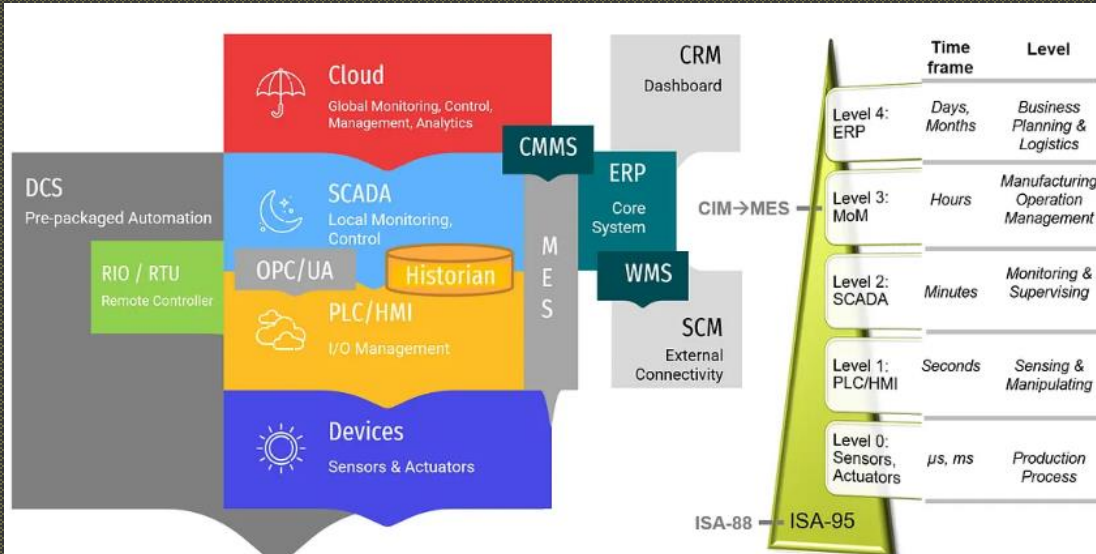


The image shows a well-lit industrial networking lab. In the foreground, there are wooden-topped workbenches with black frames and multiple rows of drawers. A black ergonomic chair is positioned in front of one of the benches. In the background, several racks of network equipment, including switches and patch panels, are visible. A projector is mounted on the ceiling, and the room has a clean, professional appearance.

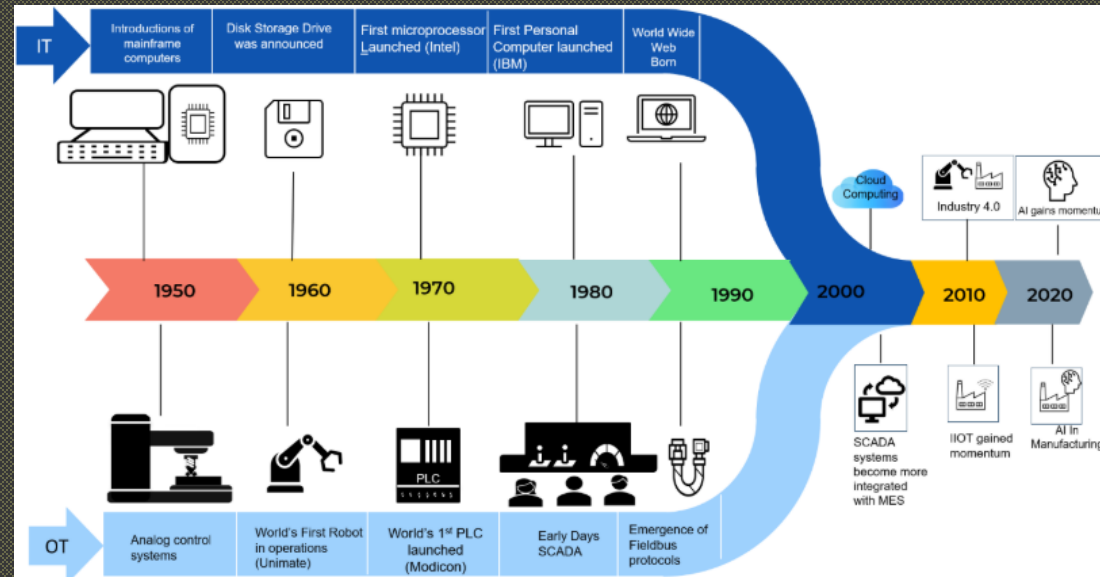
INDUSTRIAL NETWORKING

MENG 4010

INTRODUCTION TO INDUSTRY 4.0



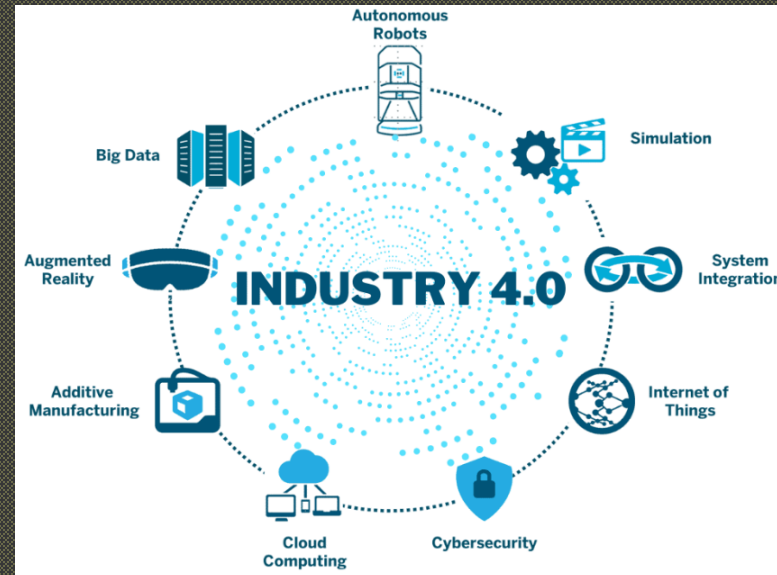
<https://andisama.medium.com/towards-industry-4-0-1-it-ot-convergence-a-digital-transformation-journey-automation-control-bdb05143ec3f>



<https://inductiveautomation.com/blog/itot-convergence-unifying-information-and-operational-technology-with-ignition>

INTRODUCTION TO INDUSTRY 4.0

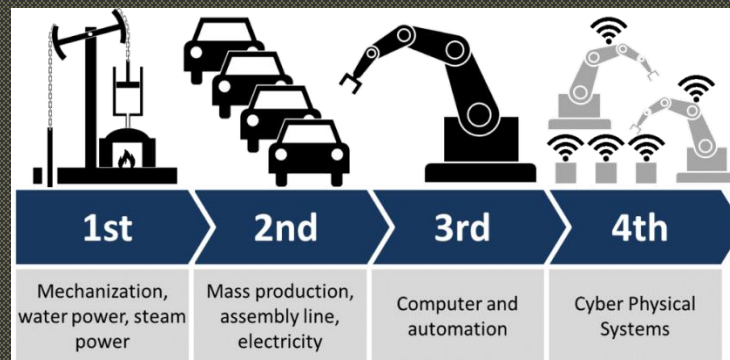
- Industry revolution is defined as the major changes and transition in manufacturing and industrial process with new innovative technologies.
- Industry 4.0 has moved beyond just a concept. It is already being adopted by many organizations. Improvements in hardware (like sensors and processors) and software tools have made real-time data collection more feasible and reliable.
- Industry 4.0 refers to the digitalization and automation of industry through technologies like IoT, cloud computing, AI, and machine learning, aiming to make systems smarter and self-diagnostic. It serves as a solid foundation for ongoing innovation and discussion in industrial automation.



<https://www.calsoft.com/what-is-industry-4-0/>

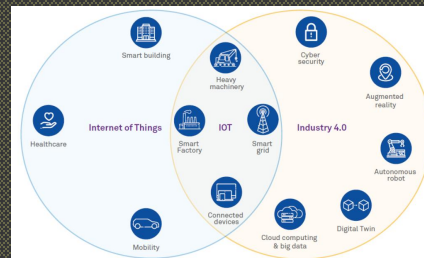
INTRODUCTION TO INDUSTRY 4.0

- First Industrial Revolution (1760s): Mechanization via water and steam power.
 - The shift from manual labor to machine-based manufacturing powered by steam and water, driven by innovations like the mechanical loom, improved steam engines, and regional advancements such as Dutch windmill technology.
- Second Industrial Revolution (1870s): Mass production with electricity and assembly lines.
 - Marked a period of rapid technological and industrial growth driven by electricity, railroads, telegraphs, and mass production methods like Ford's assembly line, laying the foundation for the modern industrialized world.
- Third Industrial Revolution (1970s): Automation using computers and electronics.
 - Manifested the shift from analog to digital technology, driven by innovations like the transistor, microprocessors, and the internet, leading to the rise of computing and the information age.
- Fourth Industrial Revolution (2010s - Present): Smart factories with AI, IoT, and real-time data analytics.
 - Builds on past advances to drive greater efficiency, sustainability, and productivity through data-driven technologies, aiming not only to improve industries but also to positively impact the environment and society, though its adoption - especially in legacy systems - requires time, effort, and thoughtful planning.



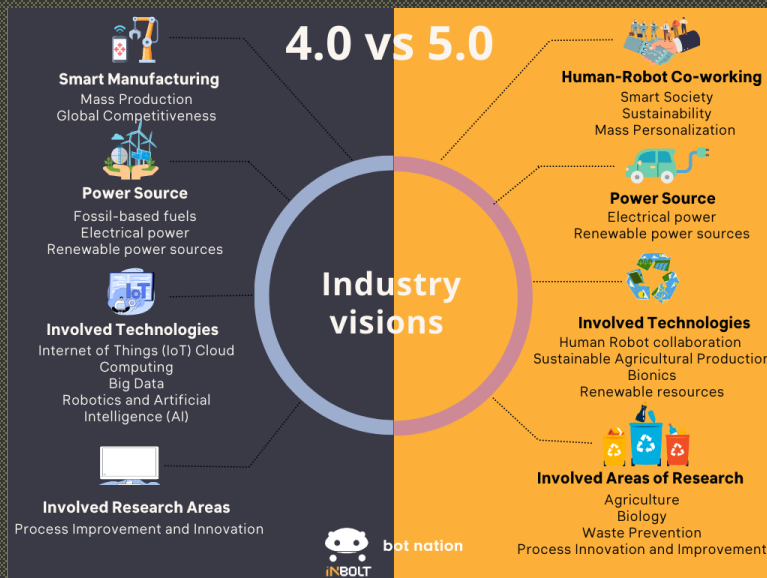
IoT as the Backbone of Industry 4.0

- Connecting “Smart” Industrial Devices
 - Industrial-strength sensors and devices - designed for harsh environments - collect real-time data and feed it into Industrial IoT (IIoT) systems.
- Scalability for Massive Deployments
 - Wireless sensors simplify large rollouts - hundreds to thousands of nodes - with less cost and hassle.
 - Scalable architectures (cloud, edge, containerized services) allow growth without overhauling systems.
- Reliability & Robustness
 - Resilient devices and protected electronics (e.g., sealed connections) ensure longevity in tough conditions.
 - Local edge processing supports resilience and continuity even if connectivity drops.
- Secure Data Handling
 - Multi-layered security (encryption, authentication, key management) is critical for safeguarding industrial data.
 - Standard practices like device identity, encrypted channels, and audits protect against threats.
- Real-Time Analytics & Predictive Insights
 - IIoT enables predictive maintenance, cutting downtime and reducing costs by 10-30%.
 - Remote monitoring - via cloud dashboards - gives usable findings into processes and assets.

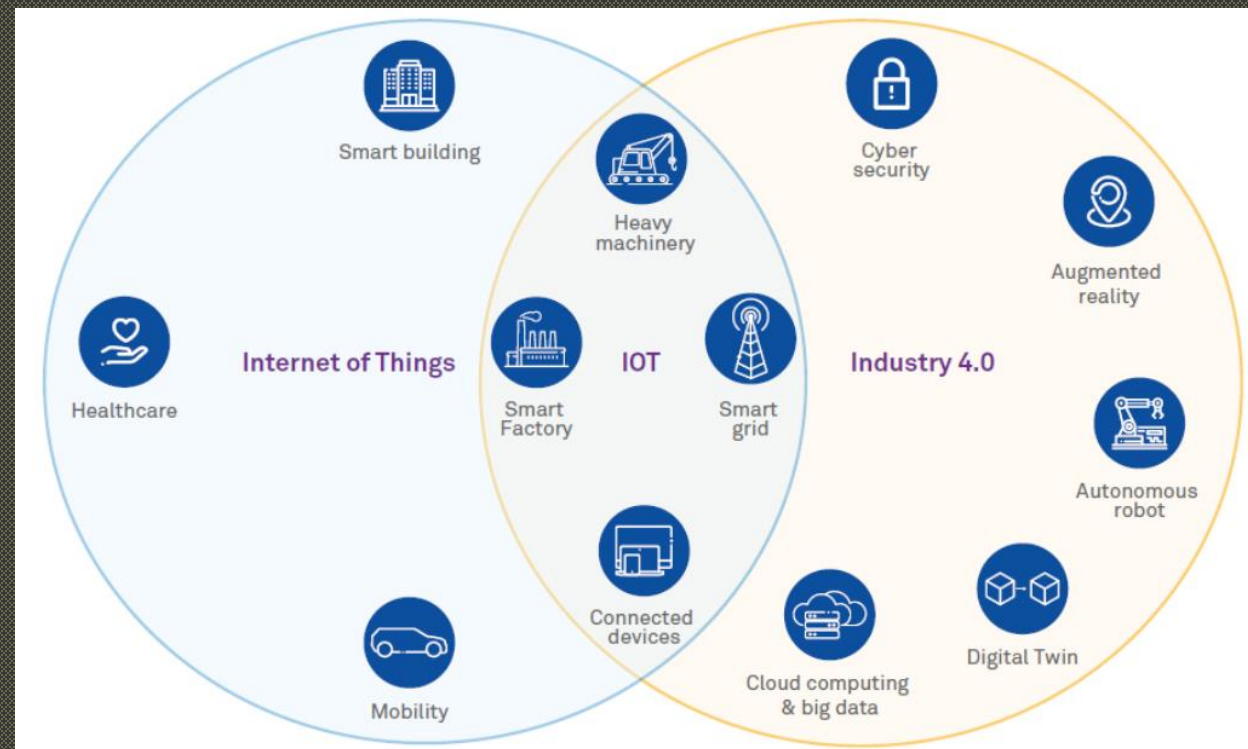


IoT as the Backbone of Industry 4.0

- **Speed:** Quick sensor deployment (even at scale) means faster insights.
- **Reliability:** Rugged hardware and local processing keep systems lit - even during connectivity issues.
- **Efficiency:** Analytics optimize resource use, cut downtime, and support smarter operations.
- **Security:** Built-in protections keep industrial systems safe from hacking or manipulation.



<https://www.inbolt.com/resources/industry-5-0>



<https://www.wipro.com/engineering/iot-in-the-manufacturing-industry-enabling-industry-4-0/>

Sensor Technology

Sensor technology includes:

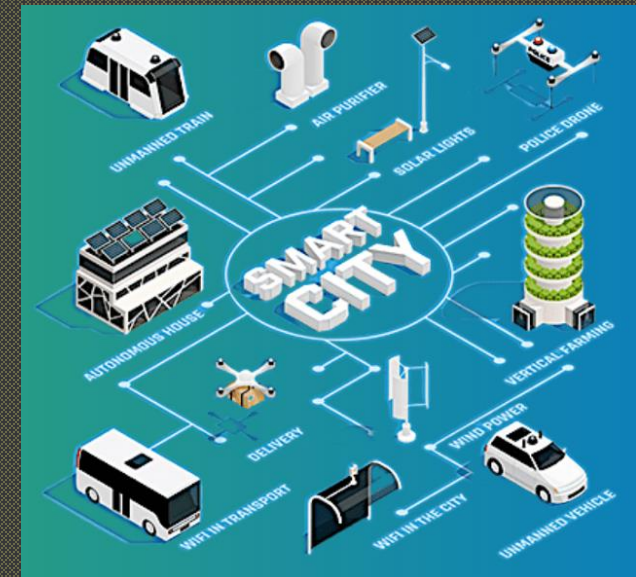
- Sensors - Devices that detect changes in the environment (like temperature, light, or motion).
- Sensor Nodes - Devices that collect data from sensors and send it to a receiver (like a computer or cloud).
- Sometimes, sensors and sensor nodes do not work well together because they use different communication methods.

Sensor technology is constantly evolving and transmission protocols, such as low-power wide-area network (LPWAN) and 5G, continue to evolve.

1. Sensors & Transmission must be compatible

- A company created a new long-distance wireless network but did not make sensors that could work with it.
- No sensors = no data, so the network failed.
- Always check if your sensors can communicate with your chosen network (like Wi-Fi, 5G, or LoRa). LoRa short for long range

5G is the fifth generation of wireless technology, It utilizes: higher frequency bands, multiple antennas MIMO (Multiple Input Multiple Output), smaller cell sites, independent networks (slices), and OFDM (Orthogonal Frequency Division Multiplexing) modulation technique.



Sensor Technology

2. Old vs. New Sensor Protocols

- SDI-12 is an old (1980s) protocol used in farming sensors. If a sensor node does not support it, the data can not be read.
- Some industries still use old standards, so your system must support them.

3. Calibrated Sensors Must Meet Standards

- Temperature sensors may need NIST calibration (a strict accuracy standard). If not available for a network, you are stuck.
- Make sure the sensors you need exist for your chosen system.

NIST - National Institute of Standards and Technology
CCCS - Canadian Centre for Cyber Security

4. Flexible Solutions Exist (Like Libelium)

- Libelium offers customizable sensor kits that work with many networks (LoRa, Wi-Fi, 4G, etc.).
- If standard sensors do not fit your needs, look for adaptable solutions.

Libelium delivers end-to-end, flexible sensor platforms that bridge physical sensors with cloud systems.

Figure 2. Smart sensor building blocks.

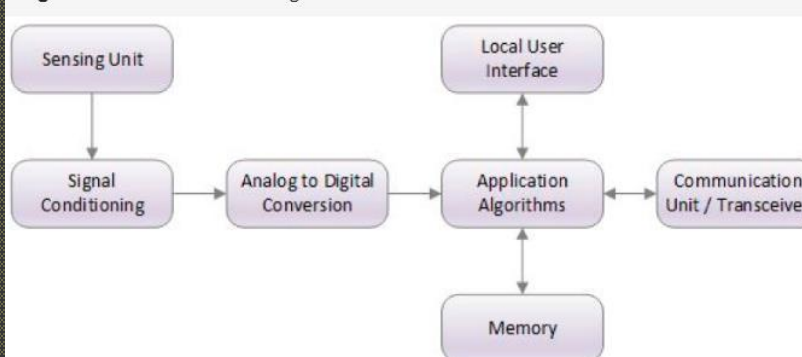
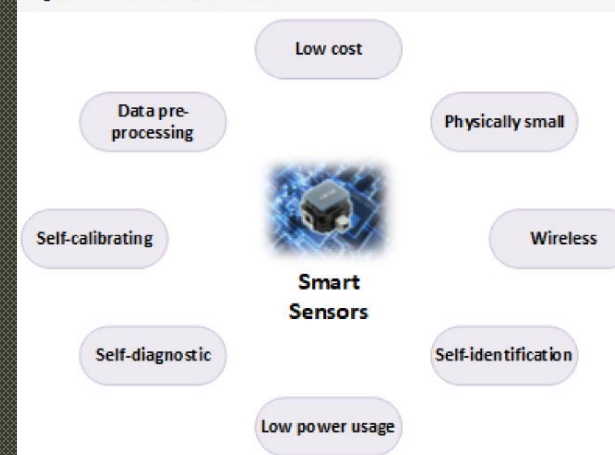


Figure 3. Features of a smart sensor.



IT / OT Convergence

- IT (Information Technology)
 - Deals with data, software, servers, and applications - like the apps on your phone or cloud servers.
 - Designed for communication, flexibility, and business processes.
- OT (Operational Technology)
 - Involves hardware and software that directly control physical machinery - like PLCs, sensors, pumps, robots, and SCADA systems.
 - Focuses on real-time monitoring and physical outcomes, prioritizing reliability and uptime.

IT/OT Convergence means bringing OT systems into the IT world - connecting machines and sensors to data networks and analytics tools.

Benefits include:

- Real-time production data from machinery (OT) can be fed into analytics and enterprise systems (IT).
- Optimizing manufacturing using combined data from IT and OT systems.
- Central dashboards monitor both machines and business KPIs in one place.

Example Use Case: A factory analyses sensor data (OT) for early fault detection while tracking orders and inventory in ERP systems (IT) - all working together

Aspect	Information Technology (IT)	Operational Technology (OT)
Focus Area	• Data management and security • Digital technology solutions	• Direct control of physical devices and processes • Reliability and continuity of operations
Devices	• Commodity Windows servers, PCs, mobile devices running standard OS and apps • Refresh cycle 3-5 years	• Customized OS devices running OEM apps, OT-proprietary embedded devices, custom production systems • Refresh cycle 20-30 years
Uptime and Redundancy	• Common redundancy and failover mechanism • Rebooting user devices with minimal impact	• Minimal redundancy • Rebooting operational devices highly disruptive due to potential impact on critical processes
Infrastructure	• Cloud-based or hybrid cloud solutions	• On-premises solutions, early steps towards cloud for certain functions
Remediation	• Simple threat remediation (e.g., system reboots) with minimal impact • Lower operational risk	• Complex threat remediation • High operational risk; incorrect actions can halt production for extended periods
Threat Identification	• Extensive public database for vulnerability identification • Lower zero-day vulnerability exposure	• Challenges in identifying domain-specific threats • Higher exposure to zero-day vulnerabilities, especially in embedded devices
Compliance	• Stringent cybersecurity compliance regulations	• Early stages of compliance adaptation, except in sectors like energy (NERC CIP)

IT / OT Convergence

- IT = Data management.
- OT = Physical control.
- Connecting IT with OT allows machines and business systems to work together, creating smarter, more responsive operations.

Example

- IT System:
 - A cloud service managing sales data and generating maintenance alerts.
- OT System:
 - On-site sensors and SCADA control machinery on a production line.
- Converged Scenario:
 - Sensor readings are sent via secure network to IT systems; analytics trigger both operator alerts and automated machine actions.

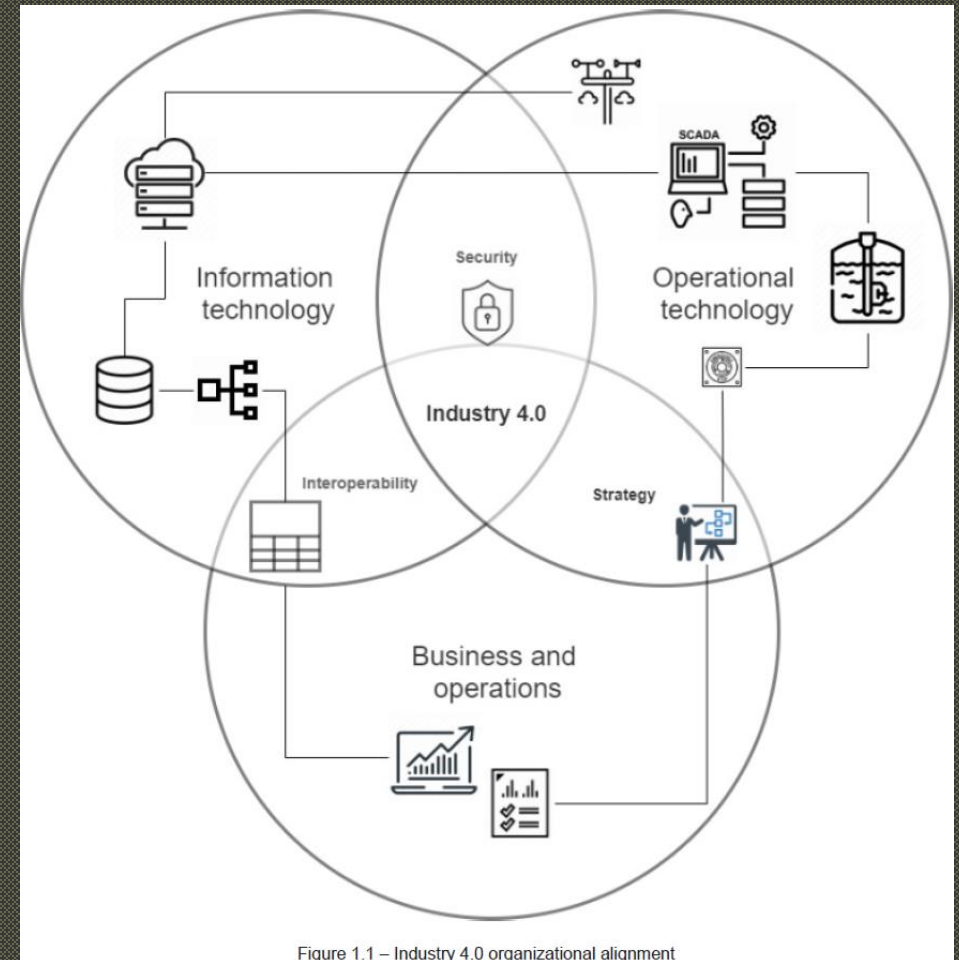


Figure 1.1 – Industry 4.0 organizational alignment

IT / OT Convergence

1. Real-Time Data and Automation

- IoT sensors collect live data on machines, energy use, or product status.
- This enables real-time process control, automation, and intelligent decision-making.
 - *Example:* Detect machine overheating and pause operations before failure.

2. Predictive Maintenance

- Continuous monitoring enables spotting issues before they become failures.
- Scheduled, proactive maintenance reduces costly downtime.
 - *Example:* A vibration sensor triggers maintenance before a motor breaks.

3. Smarter Production & Quality Control

- Sensors track production variables (e.g. temperature, humidity, pressure) in real time.
- Early detection of anomalies prevents product defects and reduces waste.
 - *Example:* Adjusting oven temperature mid-batch to meet quality specs.

4. Optimized Supply Chain and Logistics

- IoT tags and trackers offer real-time visibility for inventory and shipping.
- Enables just-in-time inventory, reduces stock waste and delays.
 - *Example:* Tracking parts across multiple facilities to auto-reorder when low.

5. Energy Efficiency & Sustainability

- Sensors monitor energy and emissions in real time, guiding efficiency improvements.
- Better reporting and compliance for sustainability goals.
 - *Example:* Fleet fuel monitoring to reduce emissions and improve logistics.

Visibility is everything.

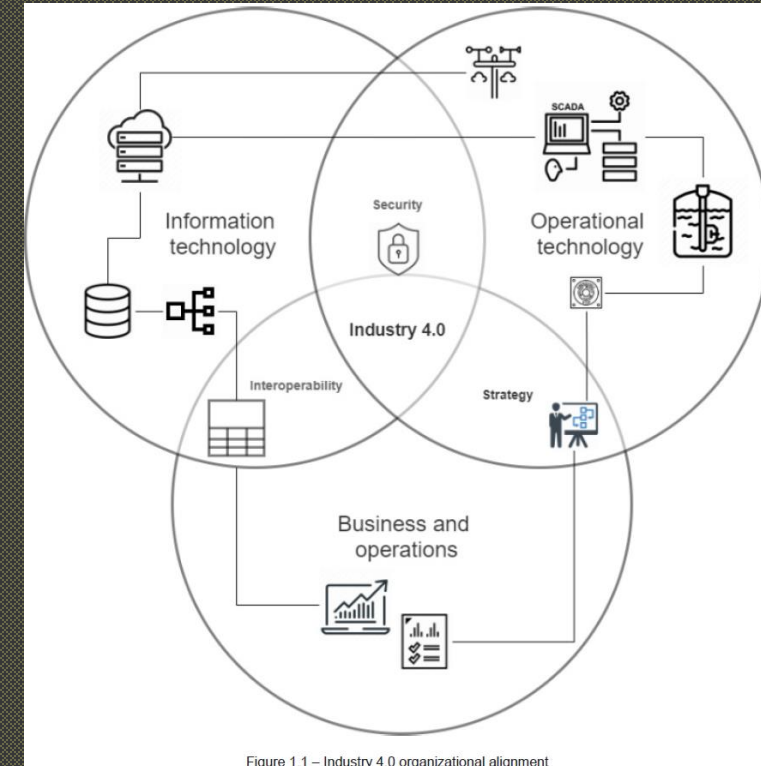
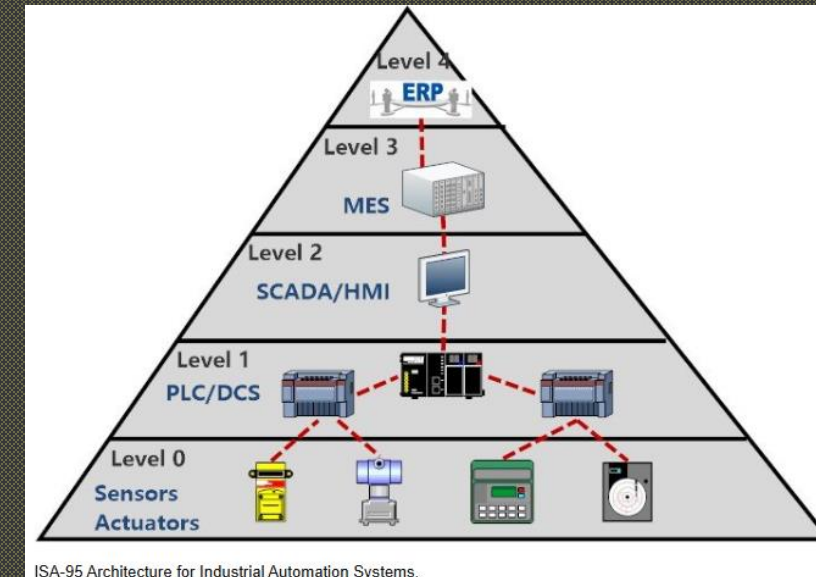


Figure 1.1 – Industry 4.0 organizational alignment

IOT Architecture

- To build a successful IoT system, use
 - Reference models as a starting point,
 - Keep plans clear and simple, and
 - Learn from proven standards like Purdue and ISA-95.
 - Create a system that meets your needs now and can grow in the future, saving time and avoiding costly mistakes.
- Level 0: Physical Process: This is where the actual manufacturing occurs, involving sensors, actuators, and other physical devices.
- Level 1: Basic Control: Includes PLCs and distributed control systems (DCS) that directly control the physical processes.
- Level 2: SCADA: Manages and monitors the equipment and processes at Level 1, often using HMI systems.
- Level 3: Manufacturing Operations Management (MOM): Focuses on activities like production scheduling, quality management, and inventory control, often involving Manufacturing Execution Systems (MES).
- Level 4: Enterprise Resource Planning (ERP): Deals with business-related activities like sales, purchasing, and financial accounting, using data from lower levels to drive business decisions.



IOT Architecture

1. On-Premise Layers (Happens Locally - Near the Machines)
 - Sensors and Actuators: Sensors collect data (e.g., temperature, pressure, speed) and actuators perform actions (e.g., open a valve, turn on a motor).
 - Device Control Systems: This includes PLCs, RTUs, or microcontrollers that directly control the machines based on sensor input.
 - Edge Servers and Applications: These are small computers or servers installed on-site that process data locally, reducing the need to send all data to the cloud. They provide fast response times and can handle real-time decisions.
2. WAN (Wide Area Network): This is the network connection (like the internet) that transfers data between on-premise systems and cloud systems.
3. Cloud Layers (Happens Remotely - Data Center or Cloud Platform)
 - Device Management: Handles the setup, updates, and health monitoring of all connected devices.
 - Cloud Data Ingestion: Responsible for collecting data from the edge devices and preparing it for processing.
 - Data Storage and Transformation: Data is stored, cleaned, and converted into a usable format.
 - Analytics and Machine Learning: Advanced tools use the data to find patterns, make predictions, or optimize operations.
 - Visualization and Reporting: Dashboards, graphs, and reports to help humans make informed decisions.

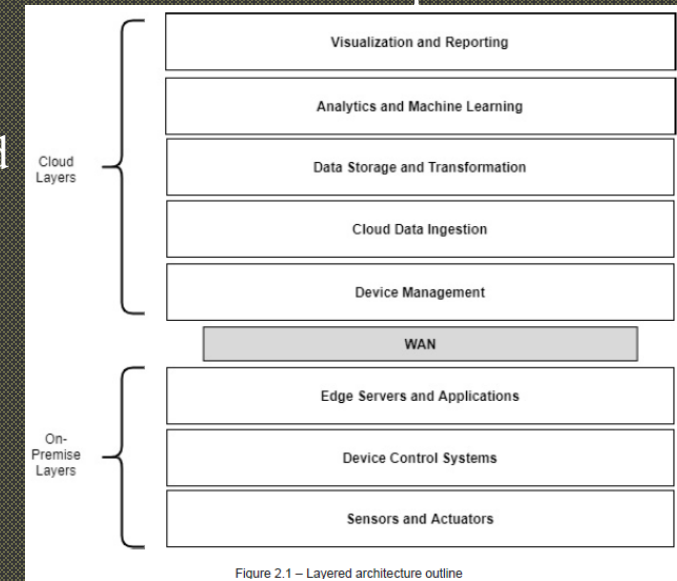


Figure 2.1 – Layered architecture outline

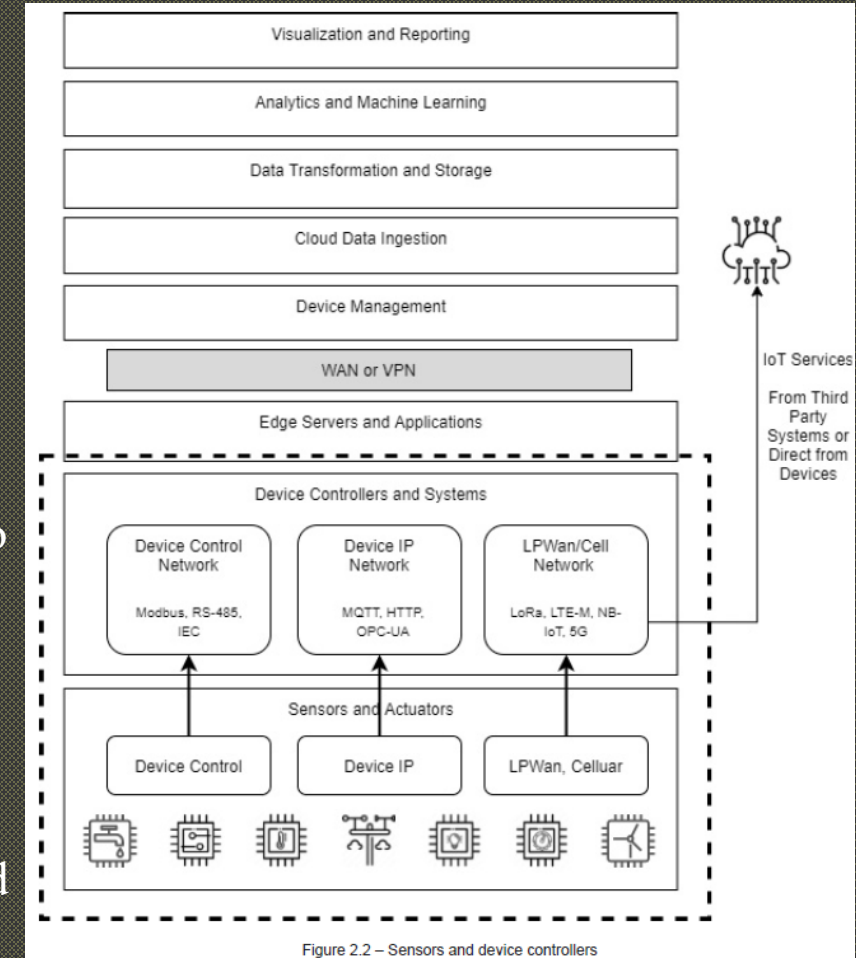
IOT Architecture

1. Sensors and Actuators (First Layer)

- Sensors are devices that measure things like temperature, pressure, or light. They take real-world information and convert it into electrical signals (analog or digital).
- Actuators are the opposite: they take electrical signals and perform physical actions. They change the system by moving parts, opening valves, or turning things on/off.

2. Device Controllers and Systems (Second Layer)

- These are devices like PLCs or Remote Telemetry Units (RTUs) that connect to sensors and actuators. They collect data from sensors and decide what to do with it. They send commands to actuators based on what they sense and any instructions from higher systems.
- A "node" is a small controller or device that manages one or more sensors. It might be small, but it plays a big role: it powers the sensors, reads the values, and sends the information to the rest of the system.
 - Example: Think of a smart greenhouse: Sensors measure temperature and humidity. A device controller reads those measurements. If it is too hot, the controller tells a fan (actuator) to turn on. All this happens automatically - even remotely - as part of a smart system.



IOT Architecture

2. Device Controllers and Systems (Second Layer continues)

- Industrial Control Systems (ICS):
 - Includes systems like SCADA, PLCs, and RTUs.
 - Uses industrial networks like Modbus, RS-485, or Profibus.
 - These long-time utilized systems often do not communicate with IT systems.
 - You may need special hardware/software to translate data and connect to the cloud.
 - The challenge is to avoid disrupting production while adding new capabilities.
- Device IP Networks:
 - These are more modern and work well with IT protocols like TCP/IP.
 - Use communication protocols like MQTT, which is simple and lightweight.
 - Easier to integrate with cloud and analytics systems.
 - Often provided by vendors who offer "plug-and-play" solutions - but you usually still have to combine different tools to get everything working.
- IoT Networks:
 - These use wireless and low-power sensors - no need to connect to existing wiring.
 - Ideal for areas where it is hard to run cables or install gateways.
 - Technologies like LoRa/LoRaWAN, 4G, or 5G allow sensors to work over long distances.
 - Useful for adding new sensors in old or hard-to-reach places.

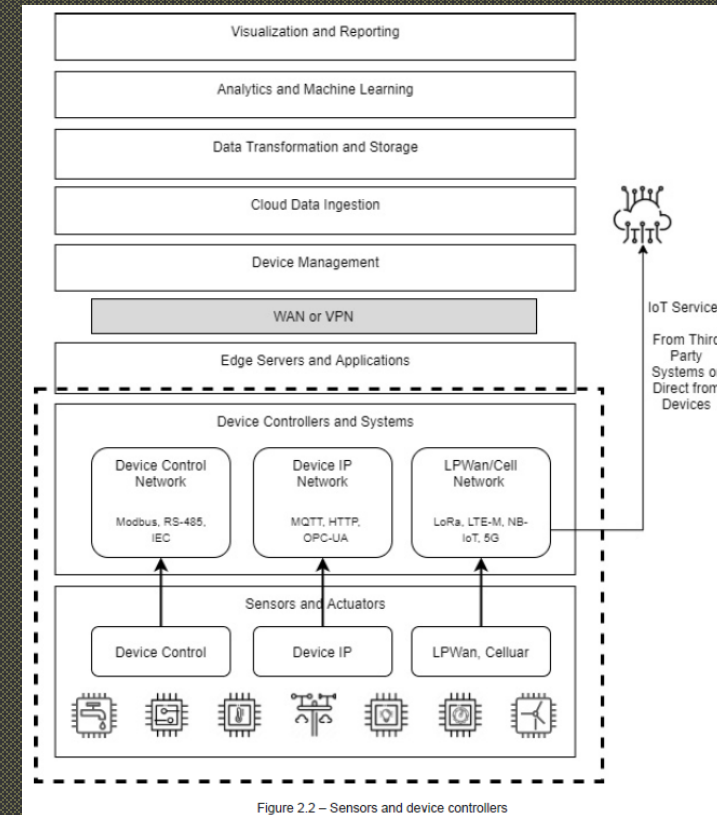


Figure 2.2 – Sensors and device controllers

IOT Architecture

3. Edge Servers and Applications (Third Layer)

- This layer is responsible for collecting, processing, and preparing data before sending it to the cloud.
- Edge devices (like edge servers, PLCs, or PACs) handle real-time data that comes from sensors or local systems (like SCADA or Modbus).
- These devices may: Convert data from factory-specific formats into IT-friendly formats (like MQTT). Filter, aggregate, or store data before sending it to the cloud. Run local analytics to make quick decisions without waiting for cloud processing. Help avoid delays (latency) caused by sending everything to the cloud immediately.
- Devices at this Layer: Can be smart (doing real-time decisions) or simple (just forwarding data). Often used to monitor and visualize operations like equipment performance or production throughput.
- WAN or VPN - Moving Data to the Cloud: A WAN (Wide Area Network) or VPN connects the factory to the cloud. It is not a layer, but a bridge between local systems and cloud services. Must be secure to protect data and prevent unauthorized access. Even if using 4G/5G or LTE, it is important to keep endpoints secure and ensure data integrity during transmission.

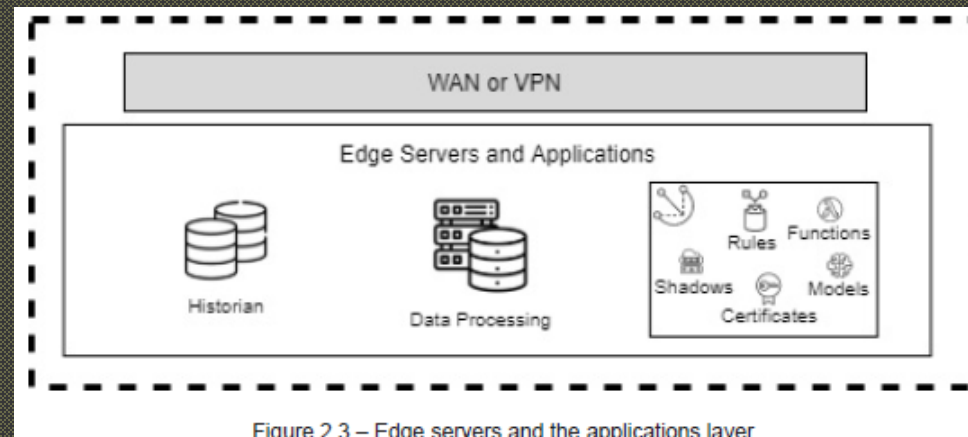
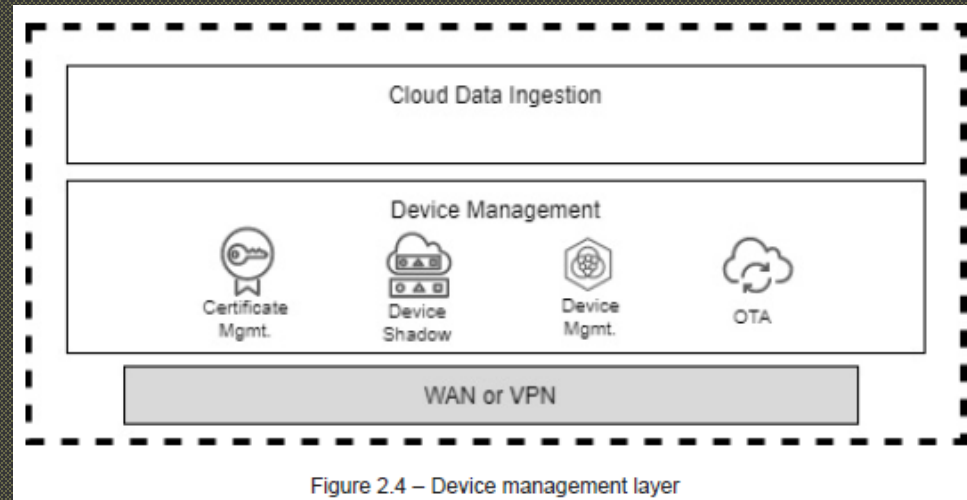


Figure 2.3 – Edge servers and the applications layer

IOT Architecture

- 4. Device Management is all about keeping track of, organizing, and securing the many devices used in an industrial or smart system - especially when you have hundreds or thousands of them.
 - Key Functions of Device Management:
 - Onboarding devices (adding them to your system)
 - Organizing and monitoring their status
 - Keeping them secure (e.g., using digital certificates)
 - Ensuring that devices behave correctly before they connect to the network
 - Device Shadows (Digital Twins) - A device shadow is a virtual copy of a real device. It keeps track of the device's latest settings and status, even if the device is offline. You can interact with the shadow instead of the real device -great for testing or managing changes. If the device goes offline, you still know its last known state.

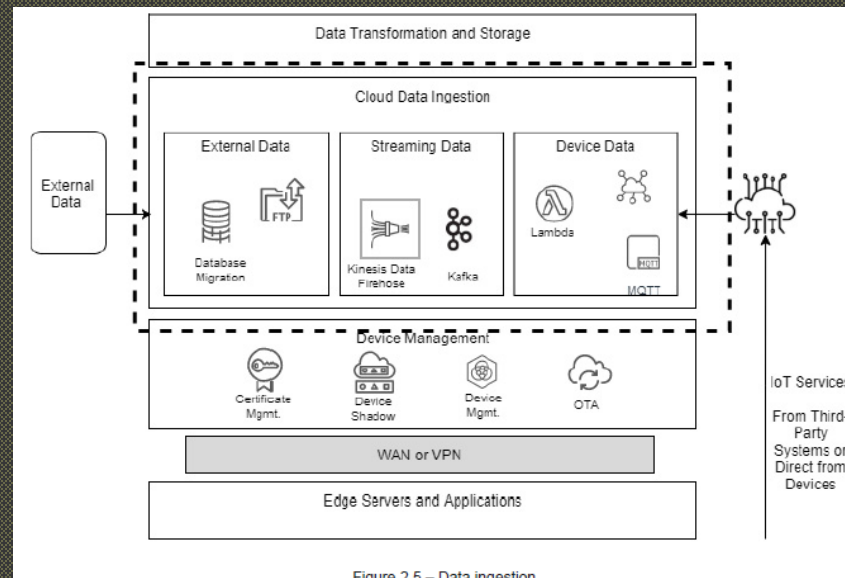
OTA - Over-the-Air allows IoT devices to receive firmware and software updates remotely via the cloud, eliminating the need for manual access



IOT Architecture

5. Cloud Data Ingestion is the process of collecting data from devices, equipment, and factories and sending it to the cloud for storage, analysis, or processing.
 - Data comes in many forms (different formats, speeds, and sizes), so organizing how to collect it is important. There is no one-size-fits-all method - each company needs a solution based on its equipment, protocols, and goals.
 - Amazon Web Services (AWS) and similar platforms offer multiple tools to help with data ingestion.
 - What to do with raw data: Store it as-is (unchanged) to trace back if needed. Or do a light transformation (like decoding binary data) before storing.
 - Why this matters: Helps maintain data accuracy and audit trails. Some regulations may require storing original (raw) data.
 - Edge vs. Cloud: Some data processing may happen at the edge (locally) before sending to the cloud to reduce data size and speed things up.

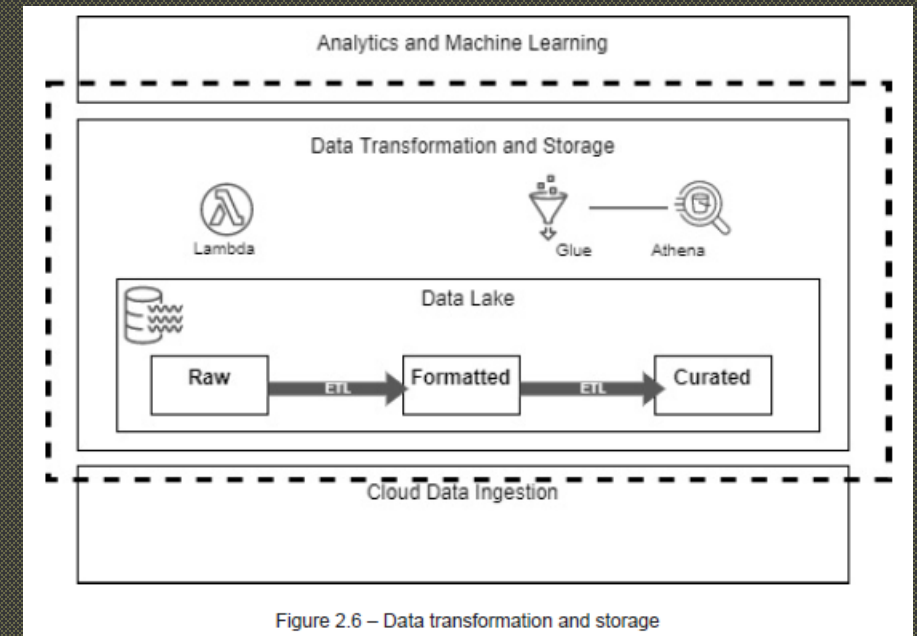
Kinesis Data Firehose is a fully managed Amazon service that simplifies the process of loading streaming data into AWS data stores and analytics services.



IOT Architecture

6. Once data is collected, it needs to be stored, organized, and transformed so it can be used for analysis, decision-making, or reporting. Main Storage Options:
- Data Lake
 - Flexible and cost-effective.
 - Stores all types of data (structured, unstructured, raw).
 - Example: AWS S3.
 - Data Warehouse
 - Stores structured, organized data (like in tables).
 - Used for relational data analysis.
 - Hybrid (Lake + Warehouse)
 - Combines flexibility of a data lake with the structure of a warehouse.

AWS S3, or Amazon Simple Storage Service, is a cloud-based object storage service offered by Amazon Web Services (AWS)

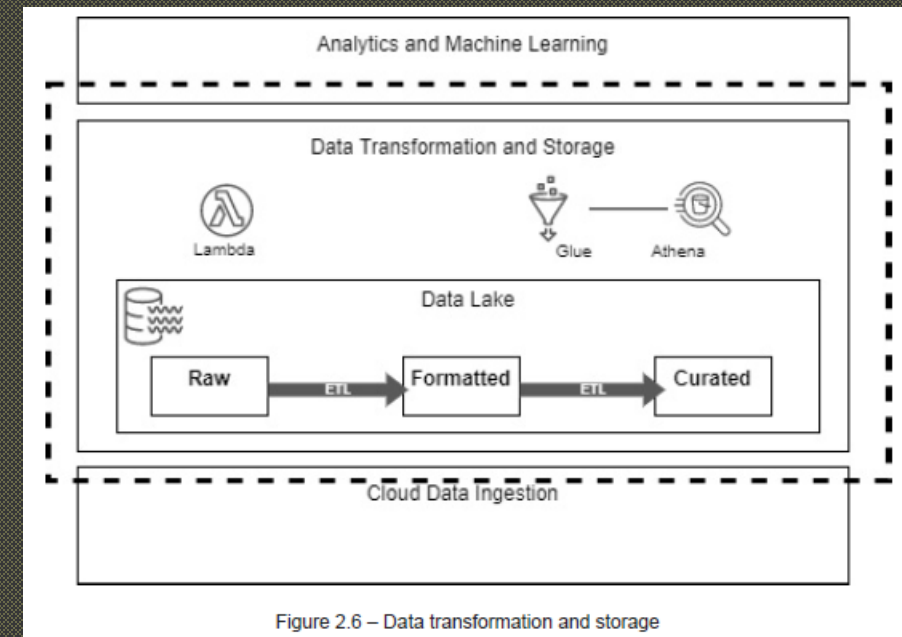


IOT Architecture

6. Three Layers in a Data Lake (like water flowing through lakes):

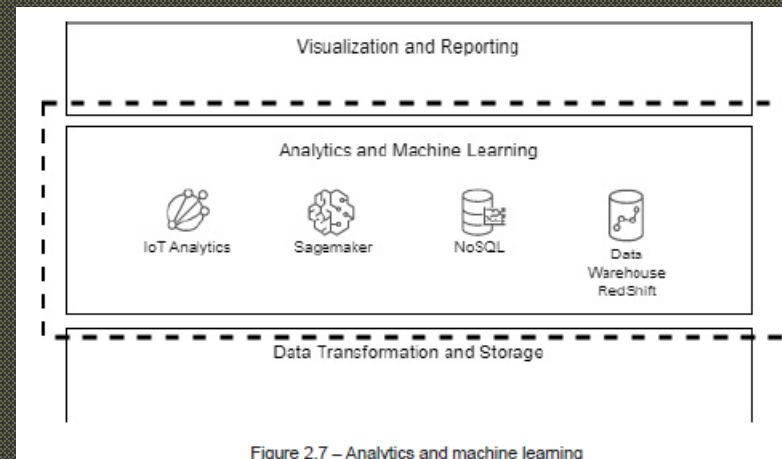
- Raw Data Layer: Stores unprocessed data exactly as it arrives. Data can not be changed and is stored securely. Useful for traceability and audits.
- Formatted Data Layer: Converts raw data into readable formats (e.g., temperature, humidity). Data becomes structured and standardized here. Example: converting binary to CSV or Parquet format.
- Curated Data Layer: Final, cleaned data ready for analysis and business use. Can combine with other sources (like financial or ERP systems). Used for analytics, reports, or machine learning.
- This three-layer approach is flexible. Some call them Bronze, Silver, and Gold layers.
 - You can add or remove layers based on your needs.

- Apache Parquet is an open-source, column-oriented data file format designed for efficient data storage and retrieval, particularly in big data environments.
- ETL - Extract, Transform, and Load
- Lambda, Glue, and Athena are serverless services that can be combined to create a powerful data processing and analysis pipeline.



IOT Architecture

7. Once we have collected data from machines and systems, the goal is to analyze it and learn from it to improve performance, predict issues, and make better decisions. Two main focus areas:
 - Machine Learning (ML) ML helps systems learn from data without being directly programmed.
 - Sensor Data Analysis: Detects unusual behavior from machines (like temperature spikes or strange vibrations) to prevent problems before they happen.
 - Image or Video Analysis: Finds defects or changes in parts using cameras - useful for quality control.
 - Advanced Use Case Example: A water utility company might predict future water needs by analyzing weather, past usage, and system data. This helps reduce costs and optimize resources.
 - Business Analytics focuses on using data to understand the past and predict the future in business operations.
 - Looks for patterns, efficiency problems, or system slowdowns.
 - Helps answer questions like: Are machines overheating? Do we need maintenance soon? Are we making enough products this month? It starts with exploring data in spreadsheets, graphs, or dashboards, then grows into using more advanced tools and predictions.

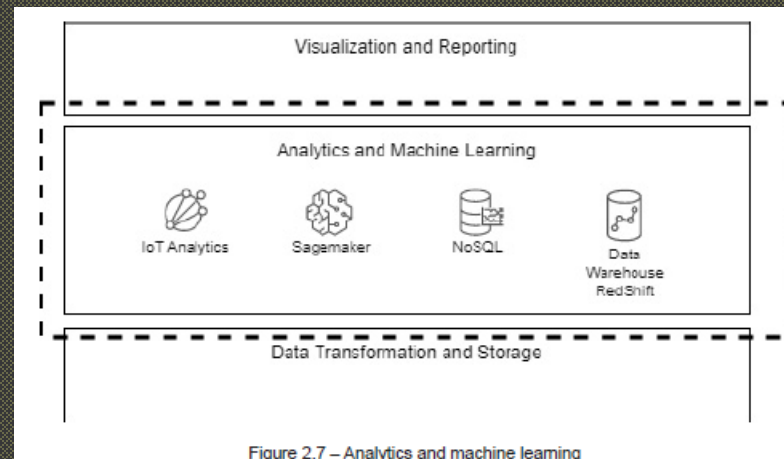


IOT Architecture

7. Edge Analytics (Quick Local Analysis)

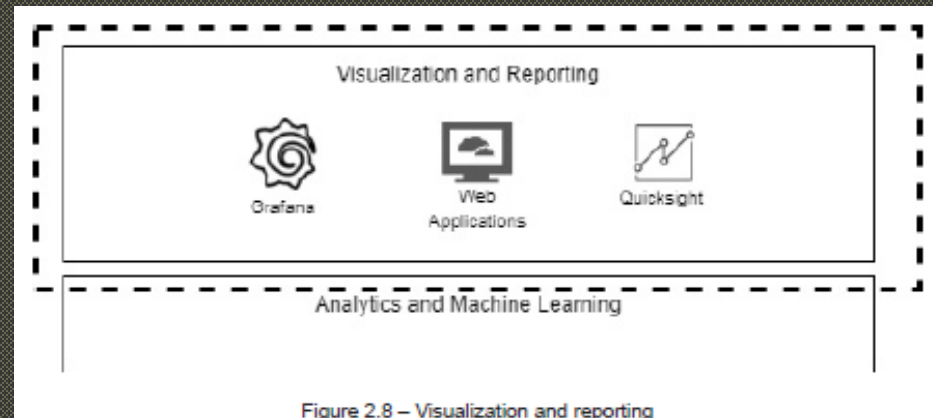
- Sometimes you cannot wait for data to go to the cloud. Edge analytics means analyzing data locally at the device or factory level to act faster.
 - Prevents delays in critical processes.
 - Saves bandwidth by filtering or processing data locally.
 - Useful when sensors collect data hundreds of times per second.
- How It All Fits Together:
 - ML helps spot patterns and predict problems using data.
 - Business analytics turns raw numbers into useful insights for managers.
 - Edge analytics lets us react fast without sending everything to the cloud.
 - Each of these areas builds on clean, organized data collected from earlier layers. When done well, they save time, reduce costs, and improve quality in industrial operations.

- Sagemaker is an Amazon service that simplifies machine learning models.
- NoSQL - Not only SQL (Structured Query Language) databases. Designed to handle unstructured data.
- Amazon redshift - fully managed, cloud-based data warehouse service



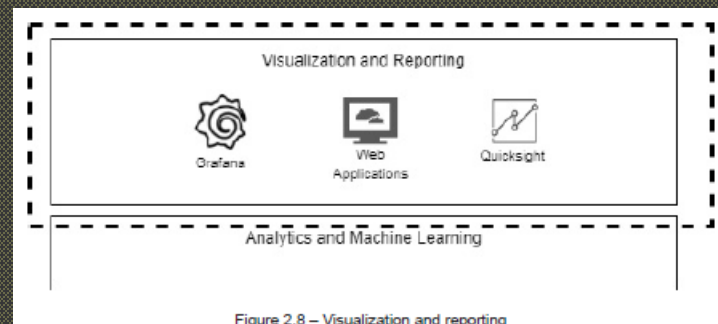
IOT Architecture

8. Visualization and reporting help people see and understand data - especially at the beginning of a project - often using graphs and dashboards.
- It helps understand what “normal” looks like in a system.
 - Operators might sense something is wrong but can’t always measure it. Visualization makes it measurable.
 - Once “normal” is known, it becomes easier to spot problems or unusual behavior (anomalies).
 - These insights can later be used in machine learning models for predictive maintenance.
 - Tools:
 - Reports: Let users explore and analyze data deeply (e.g., second-by-second performance). Useful for finding trends, issues, or areas for improvement.
 - Dashboards: Show real-time system status. Good for monitoring operations and quickly spotting issues.



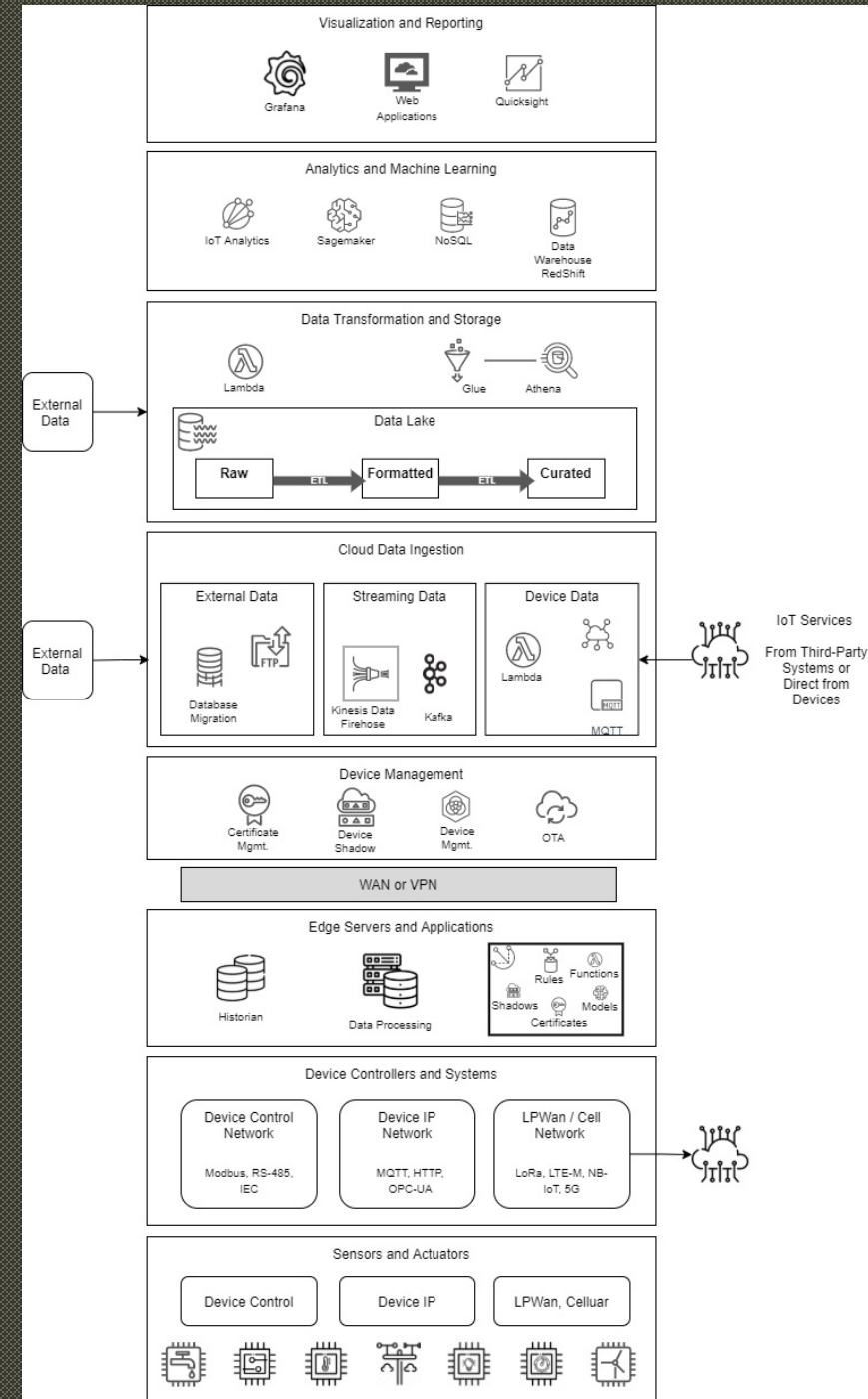
IOT Architecture

- Tool Options:
 - Out-of-the-box (OOB) tools: Already included with many systems (ERP, asset management, etc.), easy to use, but may not work well with data from other systems.
 - Custom-built tools: More flexible, can bring together multiple data sources (e.g., sensors, costs, maintenance logs), but take more time and money to develop.
 - Choose tools based on your needs. Start simple with existing tools, and expand or customize as needed. Clear and useful data visualization supports better decisions, saves costs, and improves system performance.
-
- Grafana is an open-source platform used for data visualization and monitoring.
 - Quicksight is an Amazon cloud business intelligence service for visualization, ad-hoc analysis, and quick gain business insight



IOT Architecture

- The reference model shared is not a final solution, but a good starting point for anyone beginning a digital transformation or IoT project.
- It is an overview of how different parts (like data collection, analytics, visualization, etc.) fit together and why the architecture was structured this way.
- This framework helps teams test ideas, standardize processes, and move forward with implementation.
- The full diagram shows the complete setup, it is intentionally kept at high-level, so it is easier to understand.



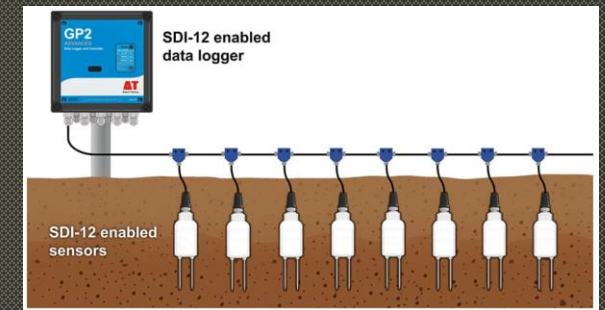
IOT Architecture

IoT Empowers Industry 4.0 at Scale

- Robust Industrial IoT: Unlike consumer-grade gadgets, industrial-grade IoT gear is designed to withstand harsh conditions, balancing cost with durability.
- Ease of Deployment: Think “plug-and-play” - sensors should be quick to set up, automatically registered, and easy to maintain in the field.
- Scalable Solutions: Systems should support hundreds or even thousands of sensors across multiple sites, with cloud-based storage and analytics ready to handle growing data volumes.
- Long-Term Reliability: Durable build quality and protective measures (like hot-glue or potting) help sensors survive in rough environments - even if it means replacing the entire unit when failures occur.
- Security by Design: Strong protection ensures data integrity from device endpoints to cloud servers, guarding against tampering or attacks.

Sensor Technology:

- The Sensor itself (e.g., temperature probe).
- The Sensor Node (the electronics that read, package, and transmit the sensor’s data).
- These often require different protocols - like an agricultural sensor speaking SDI-12 and its node transmitting via LoRa or LTE.



<https://delta-t.co.uk/an-introduction-to-sdi-12/>

IOT Architecture

This figure breaks down the key parts of a smart manufacturing system into three main categories:

- Planning & Logistics
 - Covers the "brain" of operations: engineering, finances, transportation, logistics, and working with suppliers.
- Materials & Smart Factory
 - Focuses on what goes into making products: raw materials, inventory, production, workforce, and machinery management.
- Packaging, Utilities, & Quality
 - Deals with finishing products: labeling, safety checks, quality control, utilities (like energy/water), and facility upkeep.
- Why It Matters?
- This chain shows how every step - from planning to delivery - is connected in modern, automated factories. Smart tech (like IoT/AI) helps streamline these processes for efficiency and quality.

Example:

- A smart factory might use sensors to track inventory (Materials) while AI optimizes delivery routes (Planning) and robots inspect product quality (Packaging).

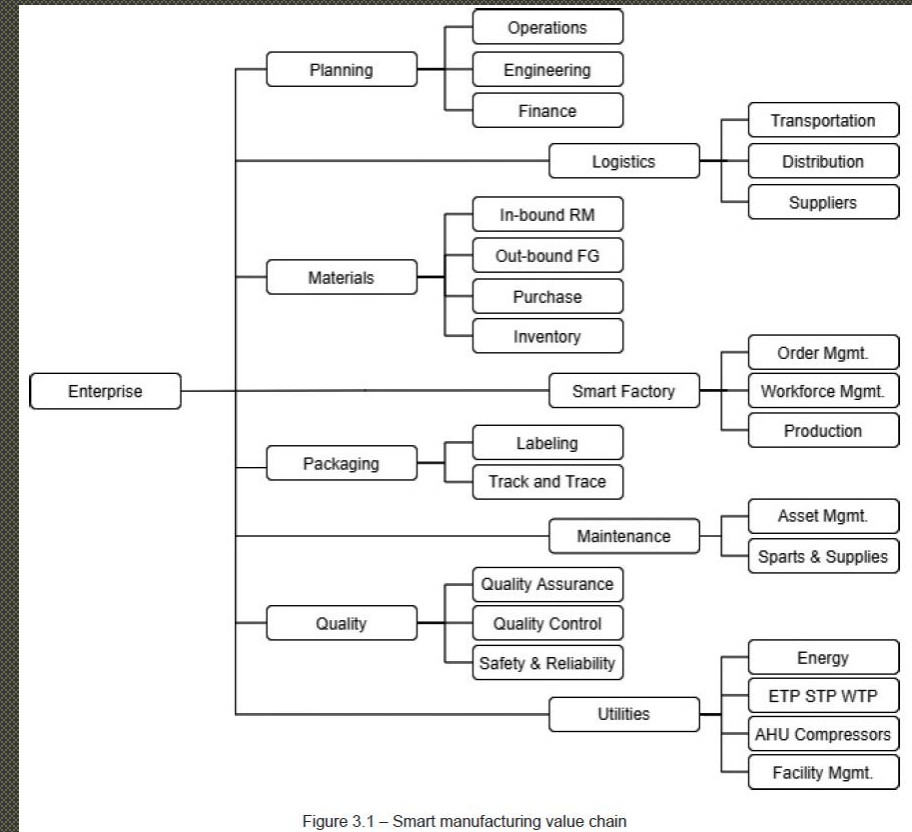


Figure 3.1 – Smart manufacturing value chain

ENVIRONMENTAL MONITORING

Environmental monitoring simply means using sensors to measure things like temperature, pressure, or water quality right where they happen. Below is an easy breakdown:

- What it is: Placing sensors in-situ (directly in their environment) to track physical changes - like a float sensor in a tank or air-quality sensors outside.
- Why it matters: Helps collect real-time data that supports decision-making and keeps manufacturing or environmental systems running smoothly.
- Types of sensors: Could be analog (smooth readings), digital (discrete steps), or binary (on/off), and often bundled together - like combining sound, vibration, and temperature into one module.

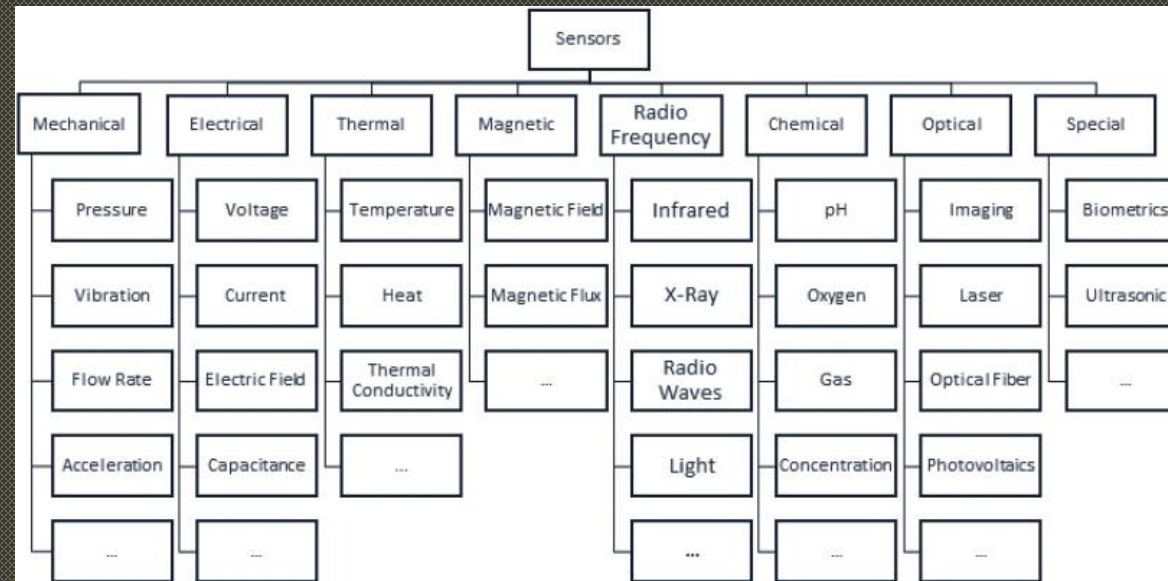


Figure 3.2 – Sensor classification based on properties

ENVIRONMENTAL MONITORING

The 6Ps framework (based on “5W1H”) guides design:

5W1H - Who, What, When, Where, Why, and How

- Premise - What problem are you solving?
- Purpose - Why do you need the data?
- Process - Which technologies, protocols, and sensitivities fit best?
- Place - Where should sensors be installed (new or retrofitted)?
- Persona - Who uses the data and what format suits them?
- Period - How often should data be collected - real-time, hourly, daily?
- There's a seventh “P” - Price, but that's considered separately for cost decisions.

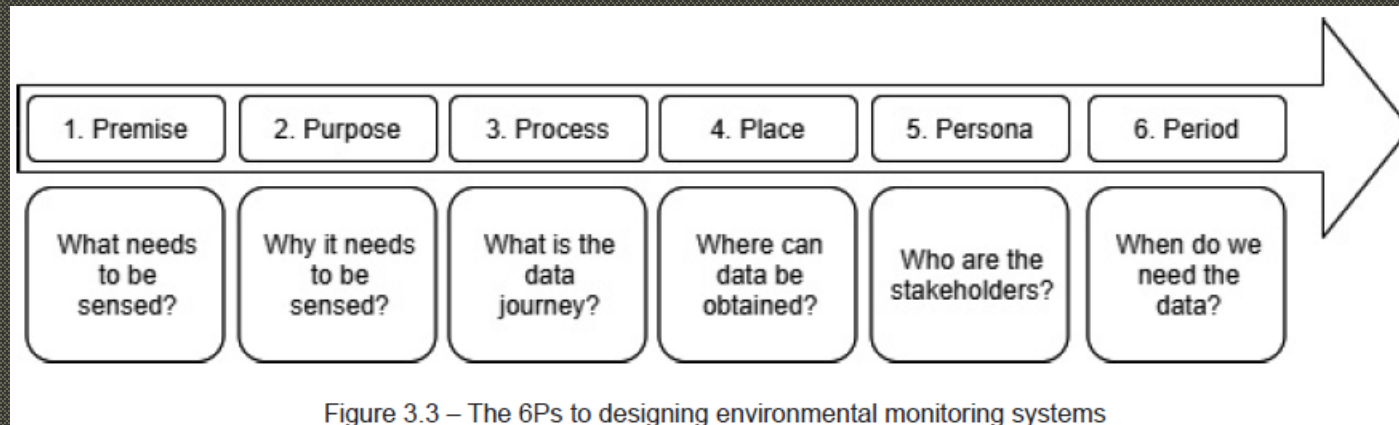


Figure 3.3 – The 6Ps to designing environmental monitoring systems

ENVIRONMENTAL MONITORING

Monitoring is essential because, as the saying goes, "you can not manage what you don't measure." In manufacturing and other systems, continuous measurement and tracking are key to:

- Identifying issues early - Real-time data lets you see problems as they arise.
- Enabling better decisions - Reliable metrics support data-driven strategies like Six Sigma.
- Driving continuous improvement - Ongoing monitoring reveals trends and areas to enhance over time.

By deploying sensors, analyzing the collected data quickly, and securely sending insights to decision systems, organizations can boost productivity, reduce waste, and ensure consistent quality.

Six Sigma Model with a fixing a leaky faucet:

1. Define - Figure out what the problem is and what needs to be fixed.
 - the faucet is leaking
2. Monitor - Keep track of the process to see how it's performing.
 - check how bad the leak is
3. Analyze - Look at the data to find the root cause of the problem.
 - find out why it's leaking
4. Improve - Make changes to fix the issue and make things better.
 - replace the broken part
5. Control - Keep the improved process running smoothly and prevent old problems from coming back.
 - check later to make sure it does not leak again

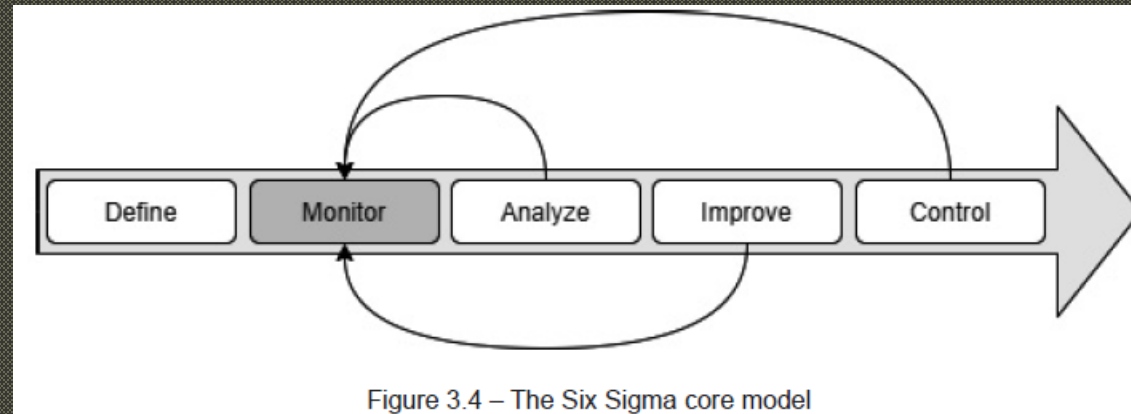


Figure 3.4 – The Six Sigma core model

IIOT Protocol Stack

1. Device (Hardware) Layer

- The physical level: sensors, actuators, controllers, PLCs
- Handles raw data collection from the real world

2. Connectivity (Network/Infrastructure) Layer

- Infrastructure layer transmits data (wired/wireless) based on speed, reliability, cost, etc.
- Includes OSI's physical to transport layers, using TCP/UDP for data transfer.

3. Management Layer

- Management layer handles governance, security, authentication, and data formatting for safe transmission.
- It defines protocols and rules to ensure secure data flow to the business layer.

4. Application (Business) Layer

- Often cloud-based, it handles data processing (ETL), storage (data lake), and integration with other systems to turn raw data into useful insights.
- Its application layer uses AI to analyze data, generate recommendations, and display findings through dashboards for decision-making.

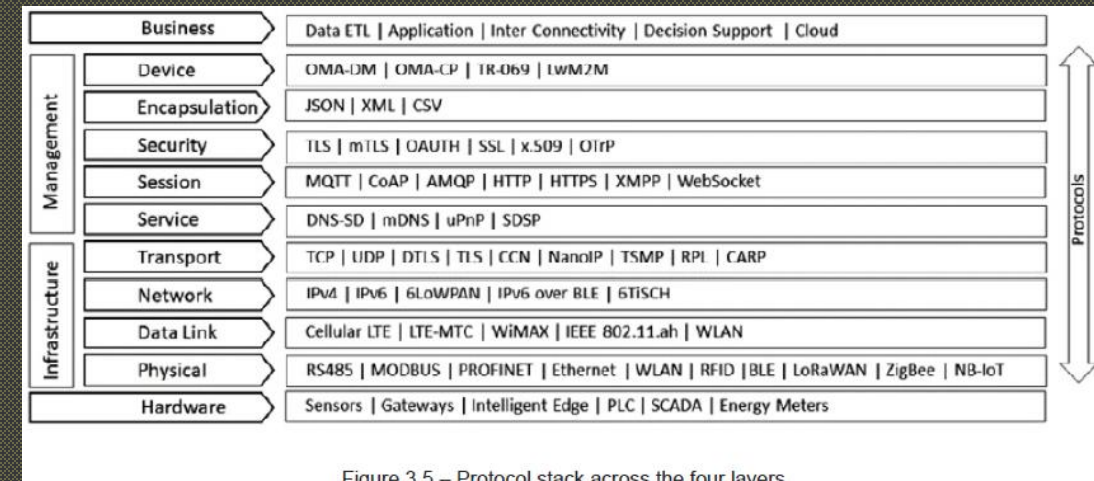


Figure 3.5 – Protocol stack across the four layers

IIoT Protocol Stack

1. Hardware & Physical Layer (Bottom)

- Soil moisture sensors and weather stations in a field use LoRaWAN/ZigBee (wireless protocols) to send raw data.
- Collects data from physical devices (sensors, gateways) and transmits it over networks like Wi-Fi, cellular (LTE), or wired connections (Ethernet).

2. Network & Transport Layer

- Data travels via IPv6/6LoWPAN (for low-power devices) and is secured with TLS/mTLS to prevent hacking.
- Ensures reliable, secure data transfer using protocols like TCP/UDP (e.g., TCP for accurate weather data, UDP for real-time alerts).

3. Management & Security Layer

- The farm's IoT gateway authenticates devices using OAuth/x.509 certificates and formats data into JSON/XML for the cloud.
- Manages device security (e.g., stopping unauthorized access) and prepares data for processing.

4. Business Layer (Top - Cloud)

- Cloud software analyzes soil data (ETL process), combines it with ERP systems (e.g., crop prices), and uses AI to recommend irrigation schedules. Results are shown on a farmer's dashboard (Decision Support).
- Turns raw data into actionable insights (e.g., "Water Field A now" or "Low yield risk detected").

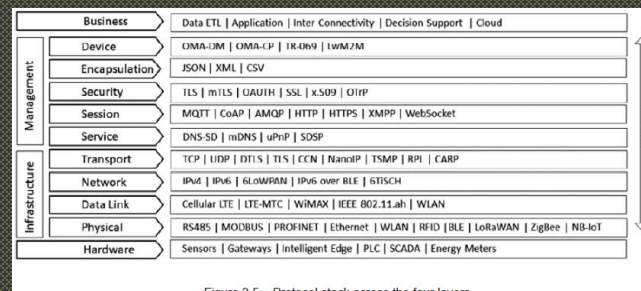


Figure 3.5 – Protocol stack across the four layers

LoRaWAN - Long Range Wide Area Network
ZigBee - Zigbee is a low-power, low-bandwidth wireless communication protocol

LTE - Long Term Evolution 4G

mTLS - Mutual Transport Layer Security

OAuth/x.509 - Methods for Authentication/Authorization

JSON/XML - JavaScript Object Notation and XML

Extensible Markup Language

IIOT Field Data

Common challenges in industrial data collection (left side) and solutions to overcome them (right side).

Challenges (Problems)

- Missing/blocked data ports - OEMs often lock access.
- Legacy systems & mixed equipment - Old and new machines do not "speak" the same language.
- Poor documentation - Hard to troubleshoot outdated systems.
- Too much scattered data - Thousands of unorganized sensor tags.

Mitigation (Solutions)

- API-based data collection - Bypass closed ports by pulling data directly.
- Unified data model - Standardize formats for analytics.
- Event-driven analytics - Focus on critical triggers (e.g., machine failure) instead of constant monitoring.

OEM - Original Equipment Manufacturer

- **Physical ports:** Hardware interfaces (e.g., USB, Ethernet, RS-485) where cables connect.
- **Virtual ports:** Software-based (e.g., TCP/IP ports like Port 80 for HTTP).
- **Protocol-Specific Ports:** Each port is linked to a communication protocol (e.g., Modbus uses Port 502, HTTPS uses Port 443).
- **Access Control:** Some ports are open for data exchange, while others are blocked by manufacturers (OEMs) for security or proprietary reasons.

API - Application Programming Interface - a set of rules and protocols that allows different software applications to communicate with each other.

Real-Life Example: Smart Factory

- **Problem:** A car factory has robots from 2005 (using Modbus) and 2020 (using Ethernet/IP). The OEM will not share data ports, and sensor tags are organized.
- **Solution:**
 - Use APIs to extract data from newer machines.
 - Retrofit gateways to translate Modbus into a modern format.
 - Build a unified dashboard to track all machines in real time.

Result: The factory now predicts maintenance needs and avoids downtime.

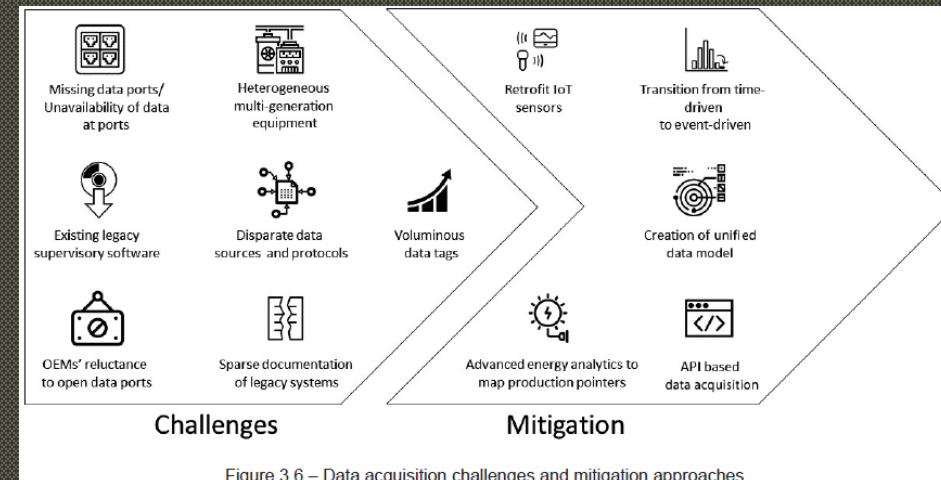


Figure 3.6 – Data acquisition challenges and mitigation approaches

IIOT Wired vs Wireless

Wired Sensors

Pros:

- Faster & more reliable (no delays, no signal loss).
- Better for critical data (e.g., factory machine monitoring).
- No battery/power issues (continuous external power).

Cons:

- Expensive & inflexible (costly to install, hard to move).
- Signal loss over long distances.

Example: A car factory uses wired sensors on its assembly line to monitor robot arms in real time - downtime is too costly for wireless risks.

- Mission-critical? → Wired (e.g., nuclear plant controls).
- Flexible/low-cost? → Wireless (e.g., farm soil sensors).

Limitation:

- Wireless struggles in harsh places (explosive gases, -40°F temps) unless specially designed.

Bottom Line:

Wired = speed & reliability; Wireless = cost & convenience.

Wireless Sensors

Pros:

- Cheaper & easier to install (no cables, place anywhere).
- Great for remote/hard-to-reach areas (e.g., agriculture, warehouses).

Cons:

- Less reliable (interference, battery life, weather impacts).
- Not always industrial-grade (a \$2 sensor won't survive a steel mill).

Example: A food factory uses LoRaWAN wireless sensors to track:

Warehouse doors (safety alerts if left open).

Freezer temps (no need to run cables through cold storage).

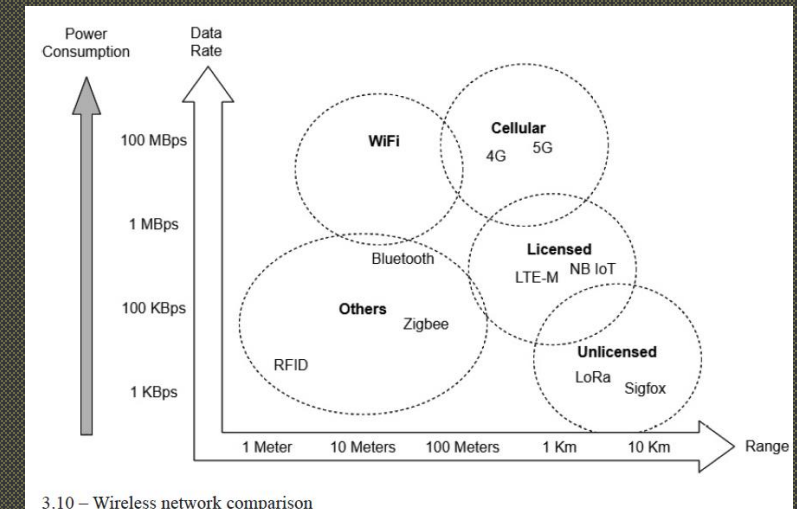
Parking spots (cheap, low-power sensors under asphalt).

IIOT - Network Evaluation

- Power
 - High data rates drain batteries faster.
 - Solar/rechargeable options help for long-term use.
 - Example: A soil moisture sensor in a farm sends data hourly on AA batteries for years, but a live video feed would need constant power.
- Range
 - Short-range (Wi-Fi, Bluetooth): Good for factories.
 - Long-range (LoRa, LTE): Best for remote areas (e.g., oil wells).
 - Example: LoRa tracks water levels in rural tanks over miles, while Wi-Fi connects machines in a workshop.
- Bandwidth
 - Low bandwidth (LoRa): For simple alerts (e.g., "tank full").
 - High bandwidth (5G/Wi-Fi): For video or real-time data.
 - Example: A security camera needs Wi-Fi, but a temperature sensor works fine on low-power LoRa.
- Availability
 - Can your sensors work with the network?
 - Example: A 4G sensor will not work in a desert with no cellular towers - use satellite instead.
- Cost
 - Millions of small data packets add up in cloud fees.
 - Example: A city's smart parking system uses cheap LoRa to avoid costly cellular data.

IIOT - Network Evaluation

- Wi-Fi
 - Best for: High-bandwidth needs (e.g., video, images).
 - Limits: High power use, short range.
 - Example: A restaurant uses Wi-Fi + Raspberry Pi cameras to count customer foot traffic and optimize marketing.
- Cellular (4G/5G)
 - Best for: Urban areas with no infrastructure.
 - Limits: Expensive for many sensors, drains batteries.
 - Example: A city uses 4G-connected trash bins to alert when full - no gateways needed.
- Licensed LPWAN (LTE-M, NB-IoT)
 - Best for: Reliable, medium-range industrial monitoring.
 - Limits: Carrier-dependent coverage.
 - Example: A utility company tracks smart meters in suburbs using NB-IoT.
- Unlicensed LPWAN (LoRa, Sigfox)
 - Best for: Cheap, long-range, low-power sensors.
 - Limits: Low bandwidth (no video/audio).
 - Example: A farm monitors soil moisture across miles with LoRaWAN.
- Others (Bluetooth, Zigbee, RFID)
 - Best for: Short-range tracking (e.g., warehouse inventory).
 - Example: A warehouse uses RFID tags to track pallets moving through docks.



IIOT - Network Evaluation

Network Type	Best For	Range	Power Use	Data Speed	Real-World Example
Wi-Fi	High-bandwidth (video, images)	Short (100m)	High	Fast (100+ Mbps)	Restaurant people-counting cameras
Cellular (4G/5G)	Urban/remote sites	Wide (km)	High	Medium (10-100 Mbps)	Smart city trash bin alerts
Licensed LPWAN (NB-IoT/LTE-M)	Reliable industrial monitoring	Medium (5-10km)	Low-Medium	Slow (100 kbps)	Utility smart meters
Unlicensed LPWAN (LoRa/Sigfox)	Cheap long-range sensors	Long (10+ km)	Very Low	Very Slow (1 kbps)	Farm soil moisture sensors
Bluetooth/Zigbee	Short-range tracking	Very Short (10m)	Low	Medium (1 Mbps)	Warehouse inventory tracking

IIOT - Wireless Gateways versus Intelligent Edge Devices

1. Wireless Gateways

- What They Do: Basic data forwarders - collect sensor data and send it to the cloud.
- Pros: Simple, low-cost, good for small-scale projects.
- Cons: No local processing - all decisions happen in the cloud (slow for urgent actions).
- Example: A factory uses a LoRa gateway to send temperature sensor data to the cloud for analysis. If a machine overheats, the alert takes seconds to reach the cloud before triggering a shutdown.

2. Intelligent Edge Devices

- What They Do: Process data locally (at the "edge" near machines) before sending to the cloud.
- Pros: Faster decisions, reduced cloud costs, works offline.
- Cons: More expensive, complex to set up.
- Example: The same factory upgrades to an edge device that detects overheating instantly and shuts down the machine before sending data to the cloud.

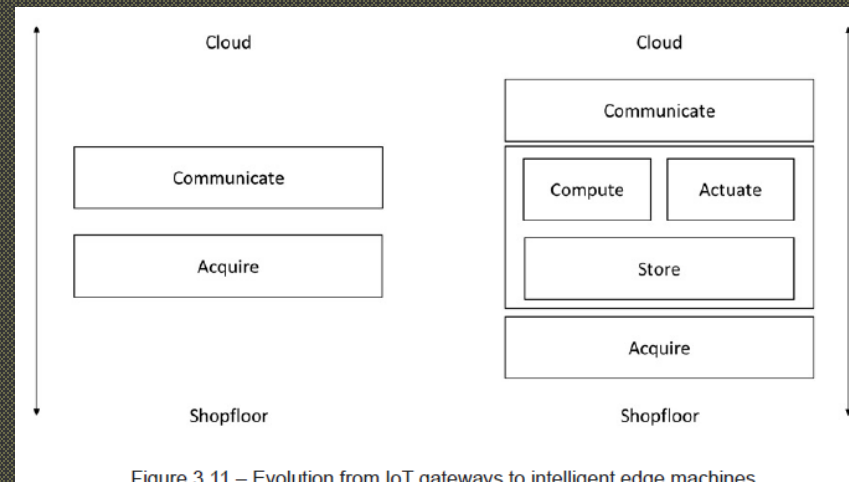


Figure 3.11 – Evolution from IoT gateways to intelligent edge machines

IIOT - Gateway and Network Setup

- Gateways act as bridges between sensors and the cloud, forwarding data with minimal processing.
- Setup is straightforward, especially with pre-configured solutions like The Things Network (TTN) or AWS IoT Core.
- Gateways are simple to set up but require careful placement for coverage (e.g., one gateway per 10km in open areas).
- Pre-configured options (TTN) save time, while commercial gateways (Milesight + AWS) offer scalability.
- Start small (e.g., one gateway + a few sensors) to test before scaling.

Real-Life Example: A farmer wants to monitor soil moisture across a large field using LoRaWAN sensors.

- Choose a Gateway:
 - TTN Indoor Gateway (plug-and-play) or Milesight UG65 (commercial, flexible).
- Configure the Gateway:
 - For TTN: Create an account, claim the gateway, connect it to Wi-Fi, and verify in the TTN console.
 - For AWS: Use the Milesight UG65, input its EUI into AWS, and upload security certificates.
- Deploy Sensors: Place LoRaWAN soil sensors in the field. They will send data to the gateway, which forwards it to the cloud for analysis.

Outcome: The farmer gets real-time soil data without running cables across the field.

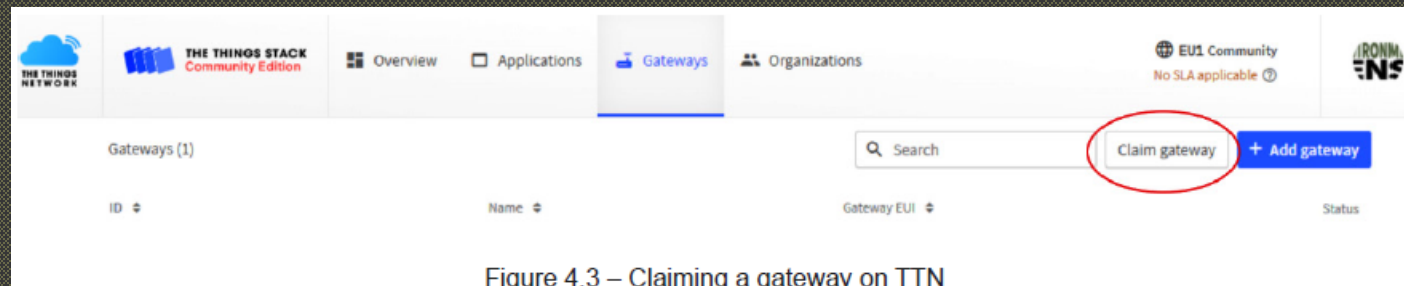
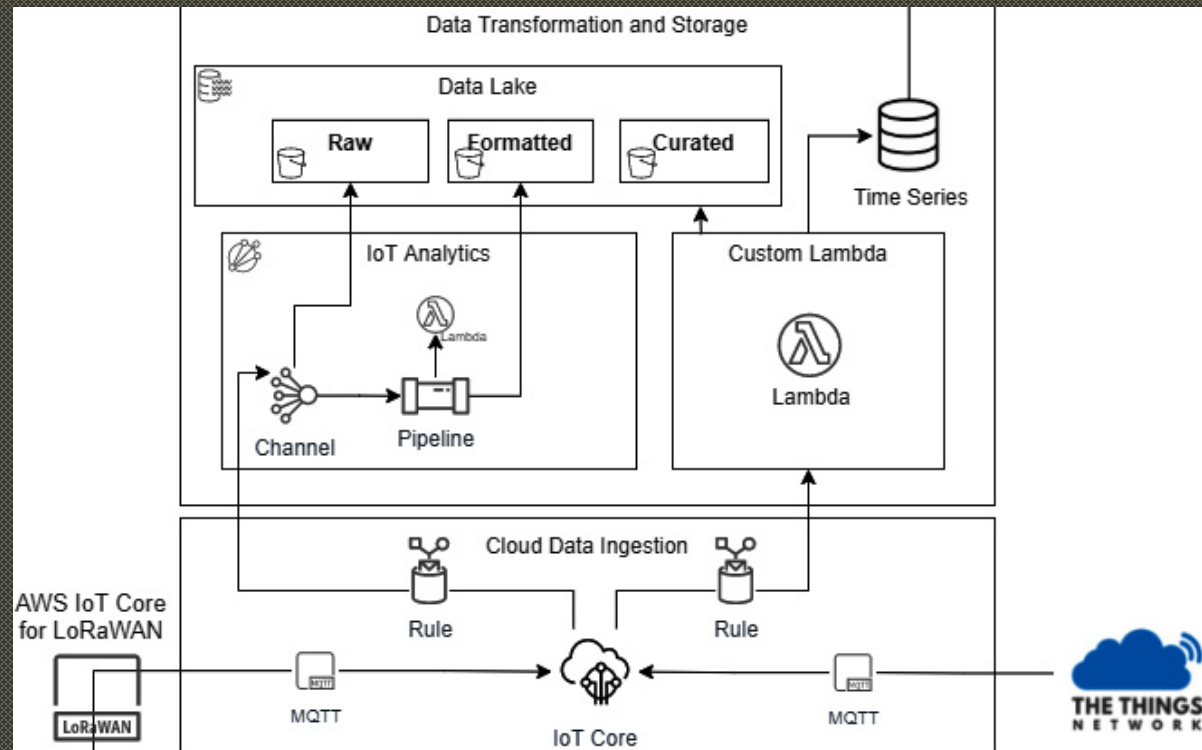


Figure 4.3 – Claiming a gateway on TTN

IIOT - Collecting Data into the Cloud

Once sensor data reaches the cloud (AWS), it needs to be organized, processed, and stored to become useful. This involves:

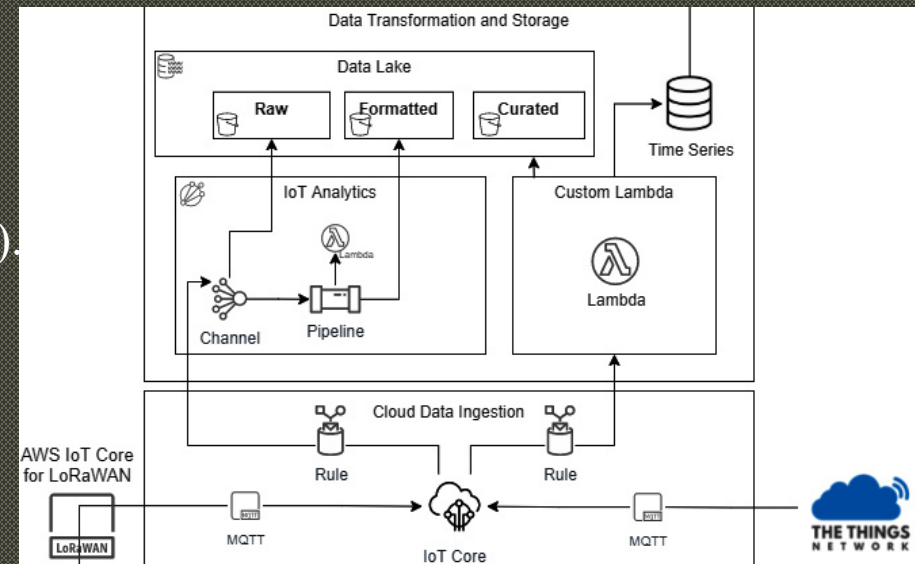
- Ingestion: Receiving raw data from sensors (e.g., via LoRaWAN gateways).
- Processing: Decoding and cleaning the data (e.g., converting encoded strings to readable numbers).
- Storage: Saving data in databases (e.g., time-series databases for trends).
- Visualization: Displaying data in dashboards or alerts.



IIOT - Collecting Data into the Cloud

Data Flow in AWS IoT (Like a Factory Assembly Line)

- Data Ingestion (Loading Dock)
 - LoRaWAN Sensors: Send raw data (e.g., temperature readings) to AWS IoT Core or The Things Network (TTN).
 - Example: Soil sensors in a farm transmit data via LoRaWAN gateways.
- Cloud Processing (Assembly Line)
 - AWS IoT Core: Acts as the "receptionist," routing data using rules (e.g., "Send all temperature data to storage").
 - MQTT: A lightweight protocol (like a conveyor belt) that moves data efficiently.
- Data Transformation (Quality Control)
 - Lambda Functions: Custom code to clean/decode data (e.g., convert "A2fX..." to {"temp": 25°C}).
 - IoT Analytics: Prepares data for analysis (e.g., averaging hourly temperatures).
- Storage (Warehouses)
 - Data Lake: Stores raw and processed data in 3 tiers:
 - Raw: Untouched data (backup).
 - Formatted: Organized data (e.g., JSON).
 - Curated: Business-ready (e.g., "Field A needs water").
 - Time-Series Database: Optimized for sensor data (e.g., tracks temperature trends over time).

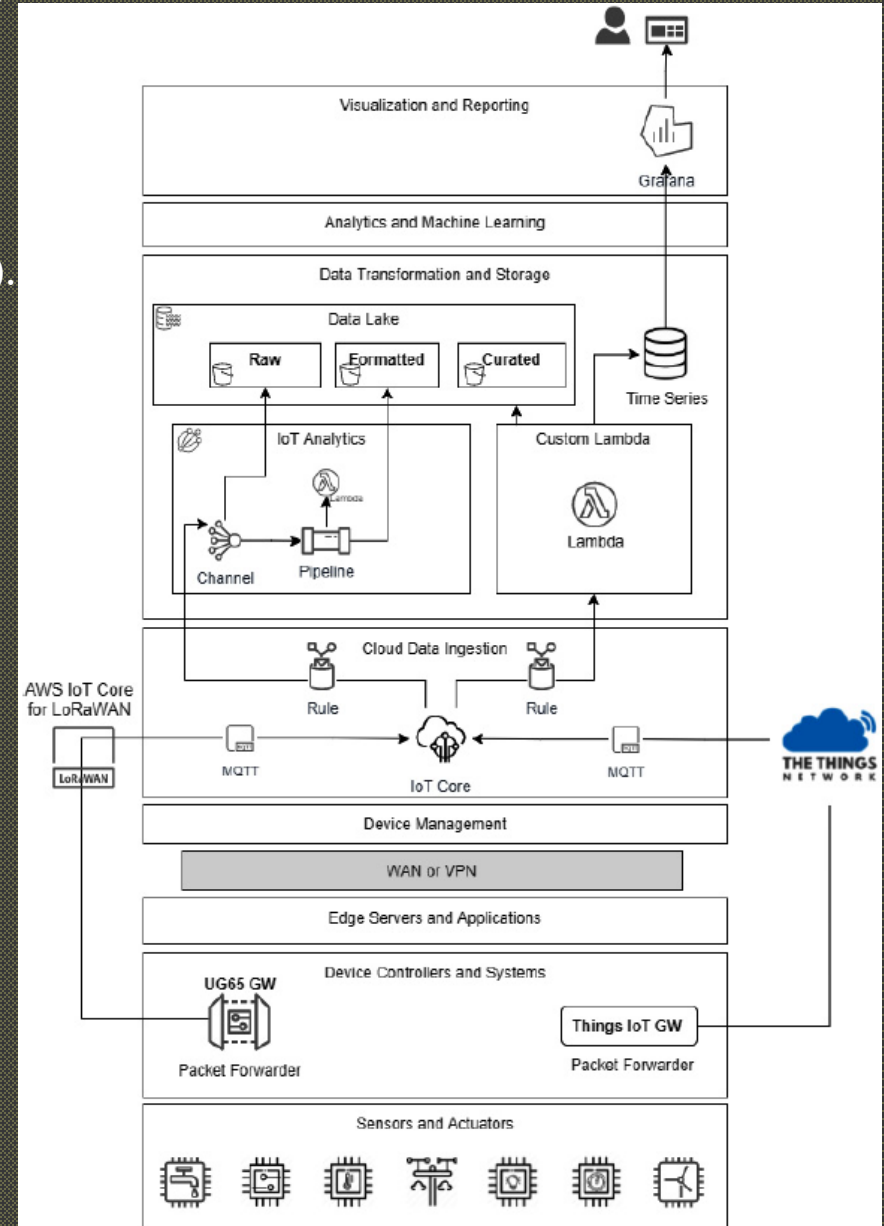


IIOT - Visualization and Recording

To turn raw sensor data into easy-to-read charts/graphs:

- AWS Grafana: Best for real-time sensor data (quick to set up).
- AWS QuickSight: Powerful for business reports (but needs setup).
- Grafana:
 - Fast for real-time data.
 - Easy to rotate multiple dashboards (e.g., switch between air quality and energy usage).
- QuickSight Pros:
 - Better for complex reports (e.g., monthly trends vs. production quality).

Start with Grafana for instant visibility, then upgrade to QuickSight later.



IIOT - Industrial Control Systems

- Industry 4.0 needs standardized, secure, and connected control systems - but legacy PLCs/SCADA were not built for the cloud era.
- Modern solutions (like edge computing) bridge this gap.

Pick a PLC if:

- You need real-time control (e.g., stopping a machine in milliseconds).
- The environment is harsh (dust, heat, vibrations).
- It is mission-critical (e.g., nuclear plant controls).

Pick a Microcontroller if:

- You are prototyping or testing ideas.
- Cost is a constraint (PLCs can cost 10× more).
- You need cloud connectivity (e.g., IoT dashboards).

Challenges

- Factories mix old/new PLCs with incompatible protocols.
- Unsecured PLCs are cyberattack targets.
- SCADA/PLCs often cannot talk to modern cloud systems.

Feature	PLCs (Industrial)	Microcontrollers (e.g., Raspberry Pi)
Purpose	Built for factory automation (mission-critical)	General-purpose, DIY projects
Hardware	Closed, rugged, plug-and-play I/O	Open-source, needs custom circuits
Reliability	Extremely reliable (24/7 operation)	Prone to crashes (OS/overload issues)
Programming	Ladder logic, real-time controls	Python/C++ (not ideal for real-time)
Cost	Expensive	Cheap (but add-ons add cost)
Best For	Factory floors, oil rigs, heavy industry	Prototyping, hobby projects, smart home

AI INSIDE PLCS - NPUS

- Neural Processing Units (NPUs) are bringing AI directly to PLCs (the "brains" of factories).
- Unlike traditional edge/cloud AI, NPUs enable real-time machine learning on the factory floor.

AI on the PLC:

- NPUs run neural networks inside PLCs, letting them make smart decisions instantly (e.g., spotting defects or predicting machine failures).
- Example: A robotic arm with an NPU can learn to handle new objects without reprogramming.

Continuous Learning:

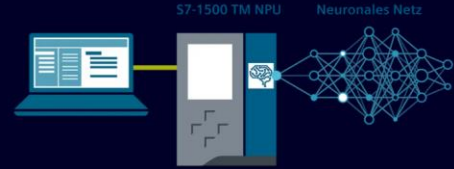
- Models improve over time by retraining with fresh data (in the cloud or at the edge).
- Example: A visual inspection system learns from human quality inspectors to reduce errors.

Key Benefits:

- Predictive maintenance: Detect anomalies before machines break.
- Adaptive automation: Robots adjust to new tasks dynamically.
- Real-time control: No lag from cloud processing.

The new SIMATIC S7-1500 TM NPU module

With the new S7-1500 TM NPU (neural processing unit) module for the SIMATIC S7-1500 controller and the ET 200MP I/O system, AI is finding its way into automation with Totally Integrated Automation. The module can be seamlessly integrated into the SIMATIC automation system, this way permitting a simple and profitable combination of AI algorithms and PLC logic. This is how inexpensive applications using SIMATIC can be implemented at the machine based on artificial intelligence. The S7-1500 TM NPU module is equipped with the AI-capable Myriad™X Vision processing unit chip from Intel Movidius™ which, among other things, enables the efficient processing of neural networks.



The diagram illustrates the integration of the SIMATIC S7-1500 TM NPU module. On the left, a laptop is connected to a PLC rack. The rack contains a SIMATIC S7-1500 controller and an ET 200MP I/O system. The NPU module is shown as a separate unit connected to the PLC. To the right, a neural network diagram is labeled 'Neuronales Netz'.

PROFINET VS OPC UA

1. PROFINET (The "Highway" for Real-Time Control)

- Industrial Ethernet protocol for ultra-fast, deterministic machine communication (e.g., robot arms synchronizing).
- Cycle times as low as 125 microseconds.
- Example: A bottling plant uses PROFINET to ensure perfect timing between conveyor belts and filling machines.

2. OPC UA (The "Universal Translator" for Data)

- Standardizes communication across machines, ERP, and cloud (no more protocol chaos).
- Provides built-in security (encryption, authentication) and a common data model for 60+ equipment types.
- Example: A pharmaceutical company uses OPC UA to connect legacy equipment to a cloud-based quality dashboard.

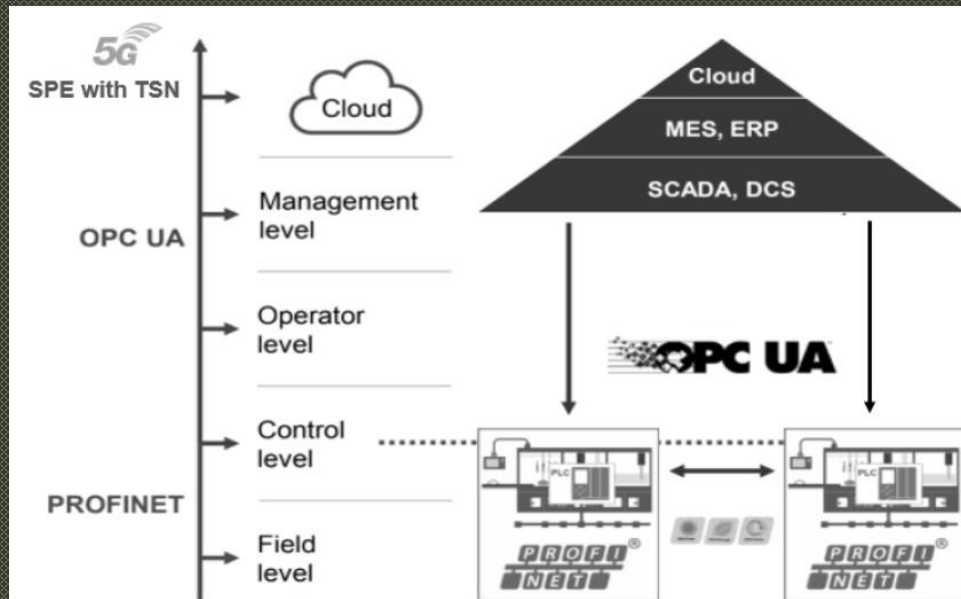


Figure 5.9 – Toward one-network infrastructure with PROFINET and OPC UA

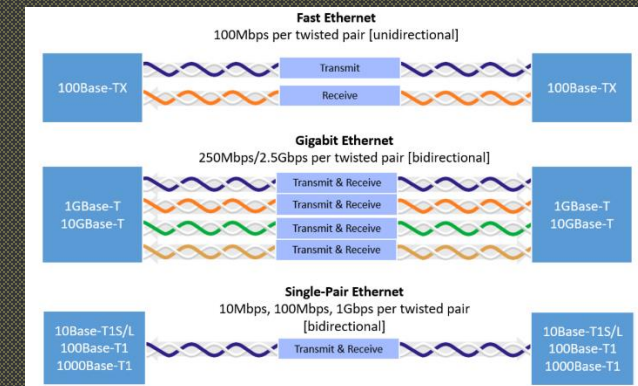
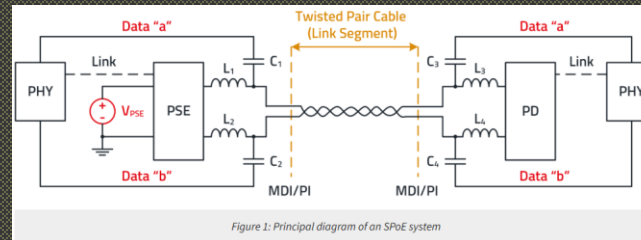
SINGLE PAIR ETHERNET (SPE)

- Ethernet communication using just one pair of wires (instead of 2 or 4 pairs in traditional Ethernet).
- Supports data + power (PoDL) over the same cable, like Power over Ethernet (PoE) but simpler.
- Speed ranges from 10 Mbps (basic sensors) to 10 Gbps (high-speed industrial apps).
- Saves Space & Cost:
 - Thinner, lighter cables vs. traditional Ethernet.
 - Easier to install in tight spaces (e.g., factory machines, car wiring).
- Works in Old Systems (Brownfield):
 - No need to rewire entire factories - retrofits existing setups.
- Power + Data in One Cable:
 - Powers devices like sensors/actuators while transmitting data (PoDL).
- Industrial & IoT Friendly:
 - Used in: Factories (connecting PLCs, sensors), Smart buildings (lighting, HVAC), and Cars (in-vehicle networks).

PoDL - Power over Data Lines
Brownfield - older networks
Greenfield - newest networks

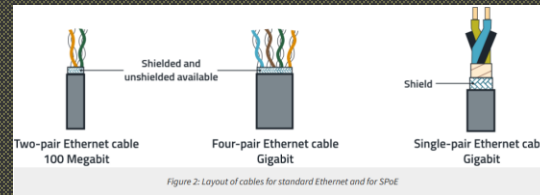
PoE is typically used with standard Ethernet (4 twisted pairs) and can transmit power up to 100 meters, while PoDL is specifically designed for SPE and has a shorter range, usually up to 40 meters.

- Power Sourcing Equipment (PSE): Provides the DC power to the network.
- Powered Device (PD): Receives the power and data.
- Coupling/Decoupling Network: Separates the power and data signals at both the PSE and PD ends.
- The interface is called “Media-Dependent interface” or “port interface” (MDI/PI).



ADVANCED PHYSICAL LAYER (APL)

- Advanced Physical Layer (APL) - A super-safe version of SPE designed for hazardous industrial areas, based on 10BASE-T1L.
- Intrinsically Safe - No sparks/fire risks (meets IEC 60079 standards), Safe for explosive environments (e.g., oil refineries, chemical plants).
- SPE is the physical data link, while SPoE (or PoDL) is the mechanism for delivering power alongside that data over the same two wires. A SPoE system inherently includes the functionality of SPE, plus the added power feature.



Since	IEEE-Standards		Speed	Max. Distance		Symbol Rate	BER	Duplex	PoDL 802.3bu
2019	802.3cg	10BASE-T1S	10 Mbit/s	25 m	UTP	12.5 MHz	1 × 10 ⁻⁹	HD	No
		10BASE-T1L		1000 m	STP	7.5 MHz			
2015	802.3bw	100BASE-T1	100 Mbit/s	15 m	UTP	66.6 MHz	1 × 10 ⁻¹⁰	FD	Yes
				25 m	STP				
2016	802.3bp	1000BASE-T1	1000 Mbit/s	15 m	Type A	750 MHz	1 × 10 ⁻¹⁰	FD	Yes
				40 m	Type B				
2020	802.3ch	2.5GBASE-T1	2.5 Gbit/s	15 m	STP	1.4 GHz	1 × 10 ⁻¹⁰	FD	Yes
		5GBASE-T1	5 Gbit/s			2.8 GHz			
		10GBASE-T1	10 Gbit/s			5.6 GHz			

Table 1: Overview of SPE/SPoE available variants and their capabilities (BER = bit error rate, the number of bit errors per unit time(s); UTP = unshielded twisted pair; STP = shielded twisted pair)

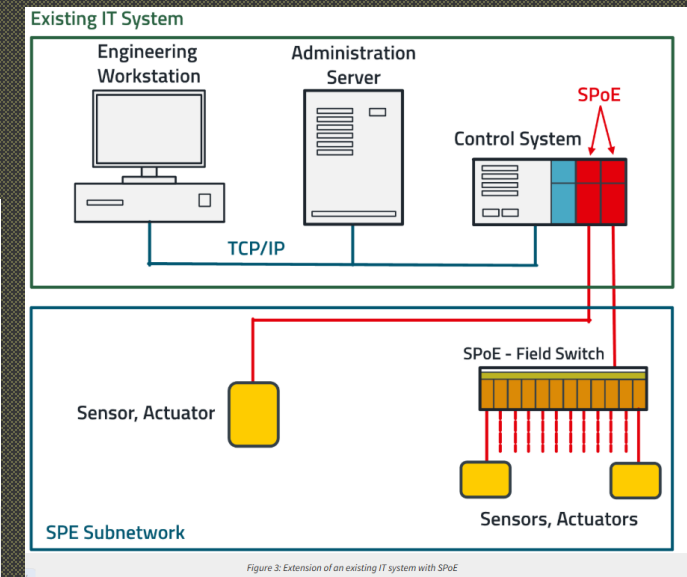
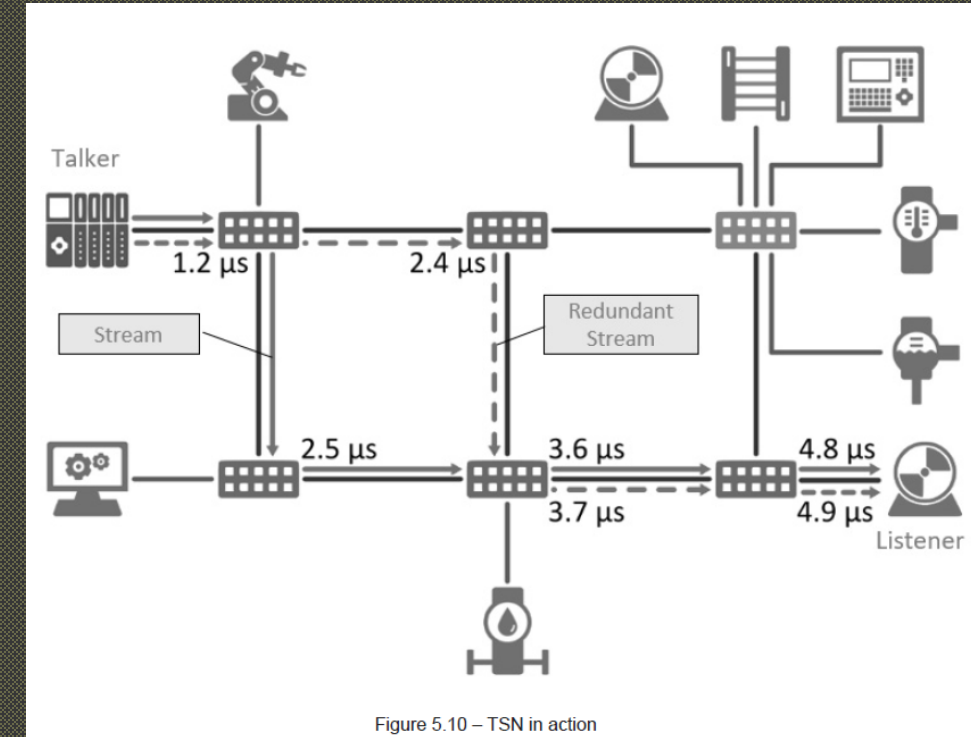


Figure 3: Extension of an existing IT system with SPoE

TIME SENSITIVE NETWORK (TSN)

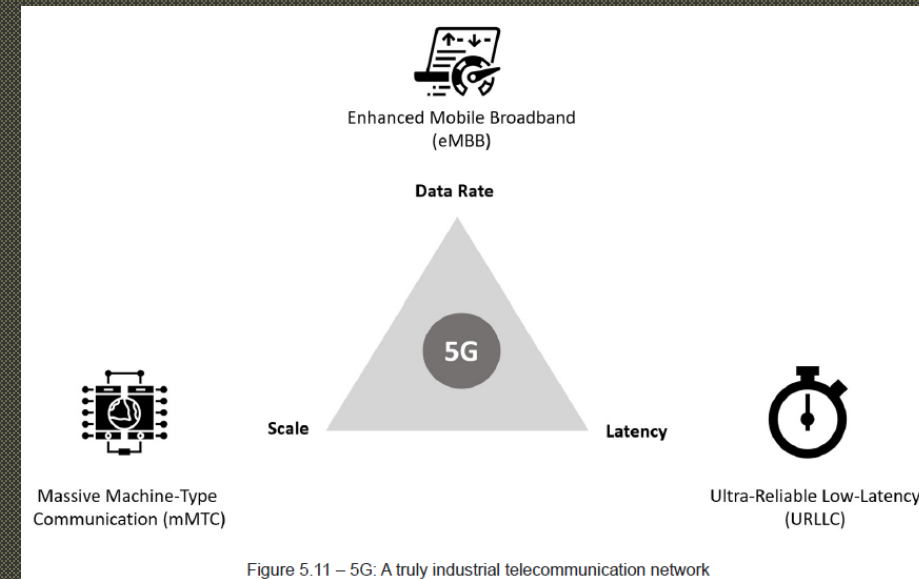
TSN is like a traffic control system for industrial networks, ensuring critical data (e.g., robot commands, safety signals) arrives on time, every time.

- Talker (Sender):
 - Sends time-critical data streams (e.g., "Robot Arm: Move NOW!").
 - Each number (1.2 μ s, 2.5 μ s, etc.) represents a precisely timed data packet.
- Listener (Receiver):
 - Receives packets in the exact same order and timing they were sent.
 - Example: A robotic arm gets commands without delays to avoid collisions.
- Features:
 - Deterministic Timing: Packets arrive predictably (e.g., always takes 1.2 microseconds).
 - Synchronization: All devices share the same clock (no "late" data).
 - Prioritization: Critical data (red) skips the queue vs. non-critical (blue).



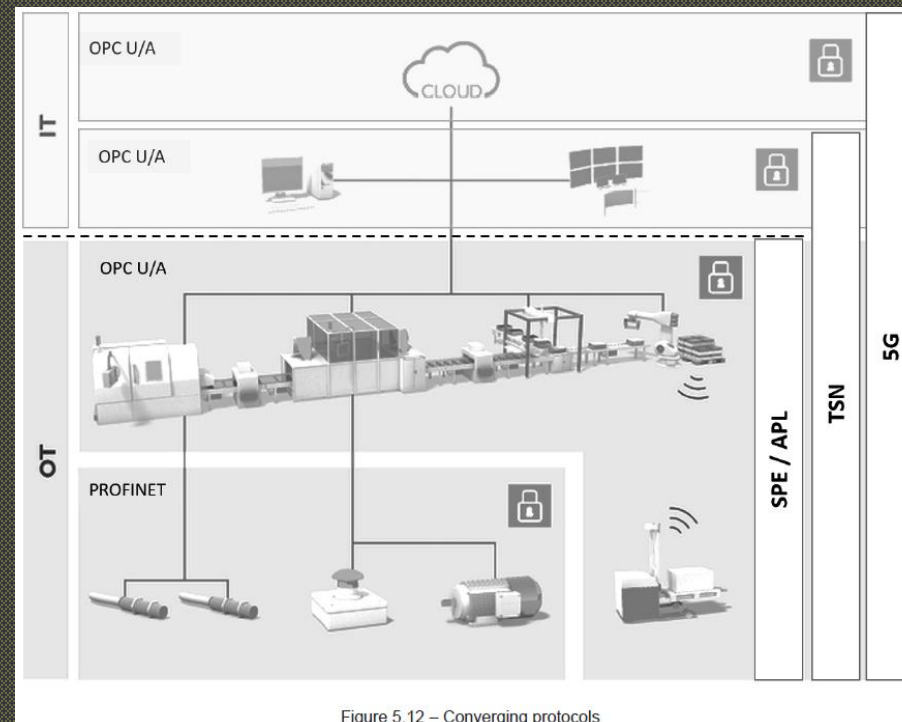
5G NETWORK

- Enhanced Mobile Broadband (eMBB)
 - Super-fast data speeds (like upgrading from a bicycle to a bullet train).
 - HD video monitoring, AR/VR for remote maintenance.
 - *Example:* A technician uses AR glasses to stream 4K instructions while fixing a turbine or AI quality check.
- Ultra-Reliable Low-Latency (URLLC)
 - Near-instant response times (1ms latency) with 99.999% reliability.
 - Robot control, safety systems.
 - *Example:* A factory uses 5G to sync 100 robots without lag, preventing collisions.
- Massive Machine-Type Communication (mMTC)
 - Connects thousands of low-power sensors (e.g., 10^6 devices per km^2).
 - Wireless - Smart agriculture, warehouse tracking.
 - *Example:* A farm monitors soil sensors across 500 acres without Wi-Fi.



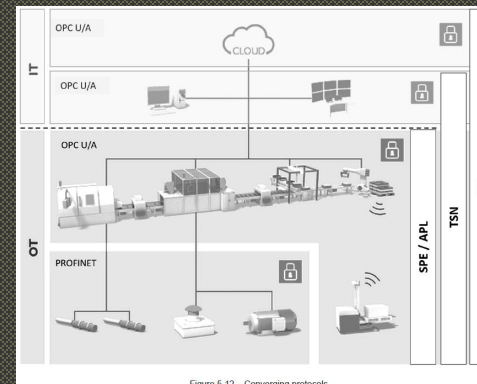
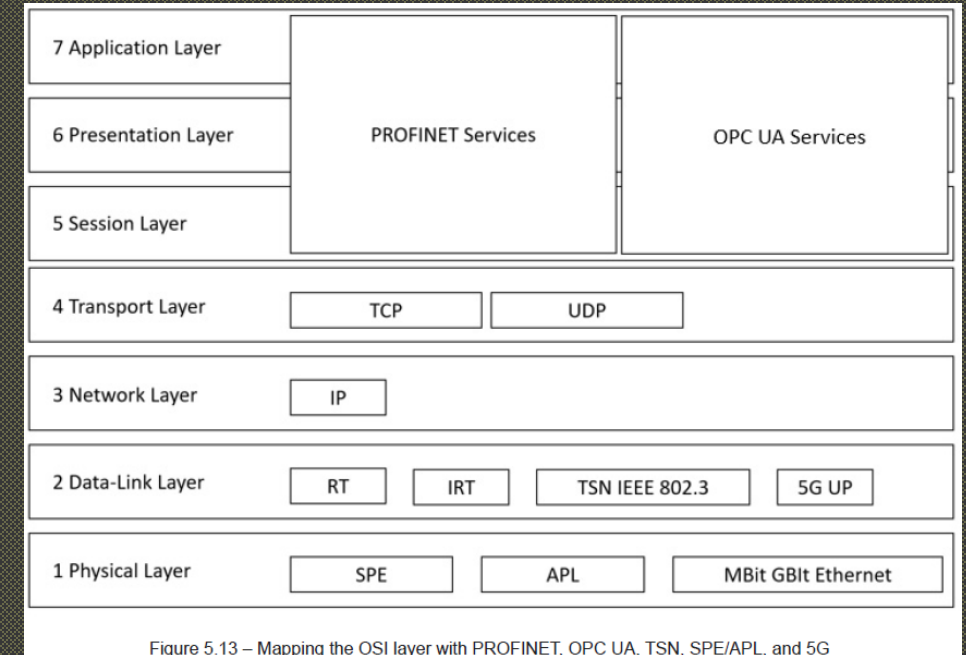
NETWORK INTEGRATION

- Field Level (Sensors → PLCs)
 - Connected via PROFINET (real-time Fieldbus)
 - Provides ultra-low latency for critical machine control
- Control Level (PLCs → SCADA/MES/ERP/Cloud)
 - Unified by OPC UA standard
 - Ensures seamless data flow from shop floor to business systems
- Physical Layer
 - Uses SPE (Single Pair Ethernet) or APL (Advanced Physical Layer)
 - Combines data + power in one cable
- Network Reliability
 - TSN (Time-Sensitive Networking) guarantees:
 - On-time message delivery
 - Correct addressing
- Wireless Backbone
 - Industrial 5G enables:
 - High-speed wireless connectivity
 - Scalability for massive IoT deployments



NETWORK INTEGRATION

Layer	Industrial Tech	What It Does	Real-World Example
7. Application	OPC UA	Talks to business apps/cloud	Sends production data to CEO's dashboard
6. Presentation	PROFINET Services	Formats machine data	Translates robot signals into numbers
5. Session	-	Manages connections	(Like a phone call handshake)
4. Transport	TCP/UDP	Data packaging	Splits video feeds into packets
3. Network	IP	Addresses devices	"Send to Robot #5 at 192.168.1.5"
2. Data Link	TSN/5G UP	Super-fast delivery lane	Ensures emergency stop signal arrives in 1ms
1. Physical	SPE/APL	Wires & wireless	Thin cables powering explosion-proof sensors



OT & IT CYBERSECURITY

OT (Operational Technology) cybersecurity is different from IT cybersecurity because OT systems run critical physical processes (like factory machines), where availability and uptime are top priorities. For example, you can not restart a machine mid-production for a software update like you might with a computer.

- OT systems often combine old and new technologies and stay in service for many years, unlike IT systems which are upgraded more often.
- Security strategies for OT must be carefully designed - not just copied from IT - because OT equipment is expensive, hard to replace, and downtime is unacceptable.
- OT cybersecurity must balance safety, availability, and reliability while protecting against threats.

1. End-to-End Data Encryption

- Scrambles data so hackers cannot read it, even if they intercept it.
- A robot's "STOP" command is encrypted so only the right machine understands it.

2. Constant Activity Monitoring

- 24/7 surveillance for suspicious behavior (like a factory security camera for data).
- Alerts if a sensor suddenly sends weird commands at 3 AM.

3. Username & Password Protection

- Strict login rules (no "admin/password123" allowed!).
- Engineers need biometric scans + one-time codes to access PLCs.

4. Industry-Standard User Access Control

- Only authorized people get access to specific machines.
- A maintenance tech can reset a conveyor but cannot alter robot programs.

5. Secure by Design

- Build security into machines from the start (not added later).
- A new CNC machine comes with built-in firewalls and encrypted ports.

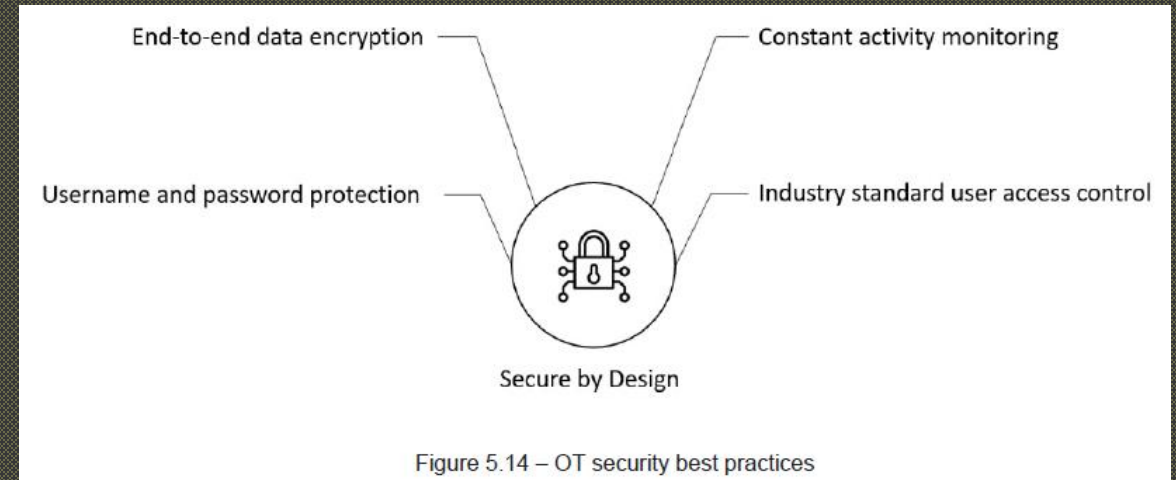
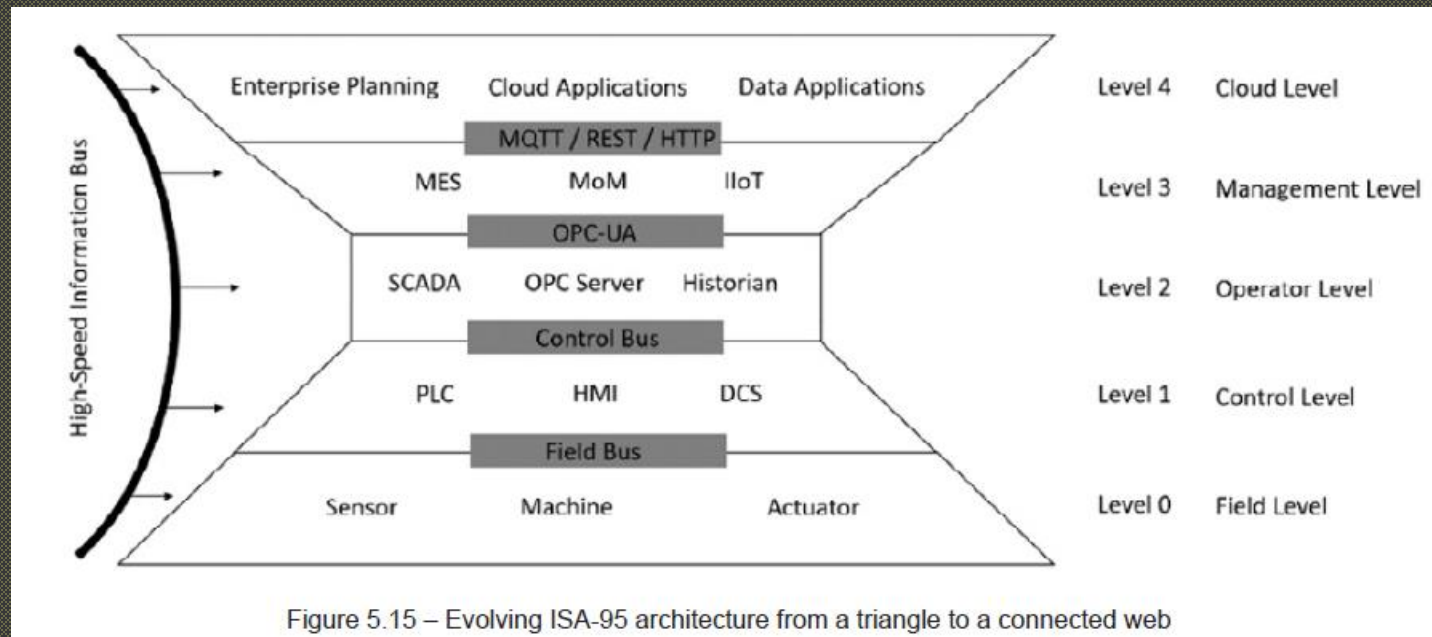


Figure 5.14 – OT security best practices

OT/IT CONVERGENCE

- Level 0: The layer where the action happens: sensors and actuators are connected via a Machine Bus.
- Level 1: This layer includes devices that manage the field equipment: PLCs, HMIs, and DCS, linked by a Field Bus.
- Level 2: Here systems like SCADA, OPC Server, and Historian help operators monitor and control the factory, connected by a Control Bus.
- Level 3: This layer handles bigger tasks with systems like MES (Manufacturing Execution Systems), MoM (Manufacturing Operations Management), and IoT (Internet of Things) platforms, using protocols like OPC-UA to share data.
- Level 4: The top layer connects to the cloud with Cloud Applications and Data Applications, using modern protocols like MQTT, REST, or HTTP, and ties into Enterprise Planning for business decisions.

MQTT - Message Queuing Telemetry Transport
REST - Representational State Transfer
API - Application Programming Interface



SMART MANUFACTURING

1. Data Layer

- Raw numbers/binary codes from machines (e.g., 1010110 = temperature reading).
- Example: A sensor reports "Motor A vibration = 4.2mm/s".

2. Information Layer

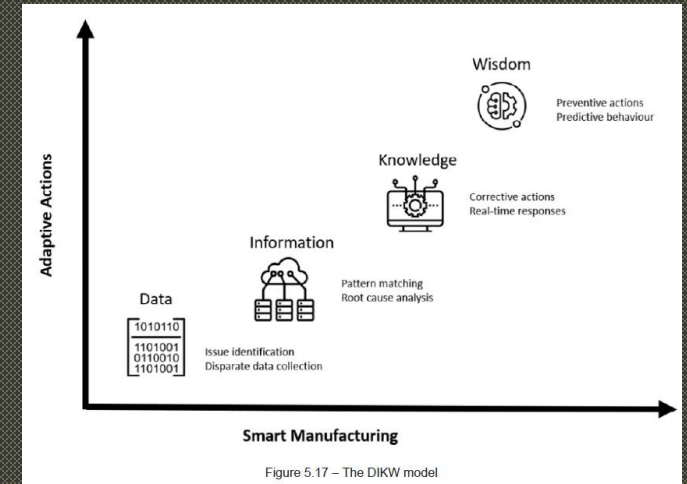
- Data is organized and labeled.
- Example: "Motor A's vibration is 4.2mm/s (above normal)".

3. Knowledge Layer

- Patterns and causes are identified.
- Example: "Motor A vibrates excessively only on night shifts → Likely due to overheating".

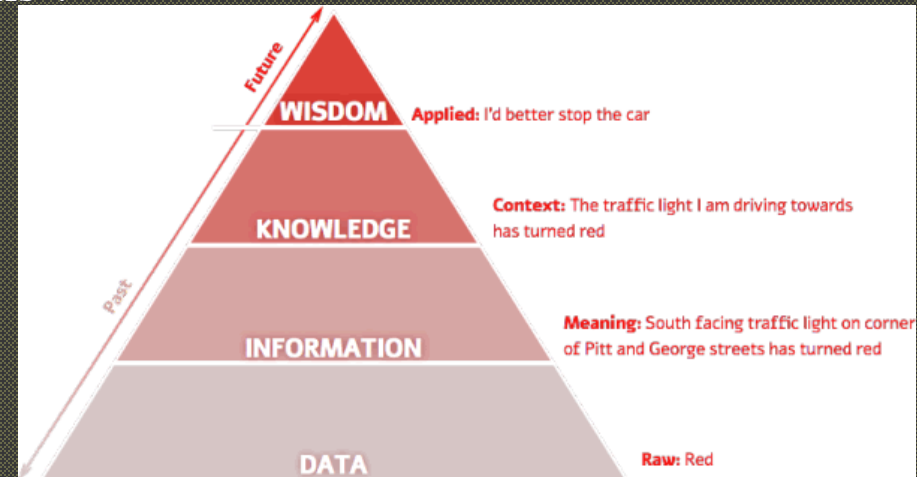
4. Wisdom Layer

- Predictive actions are taken.
- Example: "Schedule maintenance for Motor A before it fails".



Real-World Example:

- Data: Robot arm reports "Error Code 1101".
- Information: "Error 1101 = Overheating".
- Knowledge: "Overheats when room temp > 35°C".
- Wisdom: "Install AC in the robot zone next summer".



OT/IT CONVERGENCE

OT Goals:

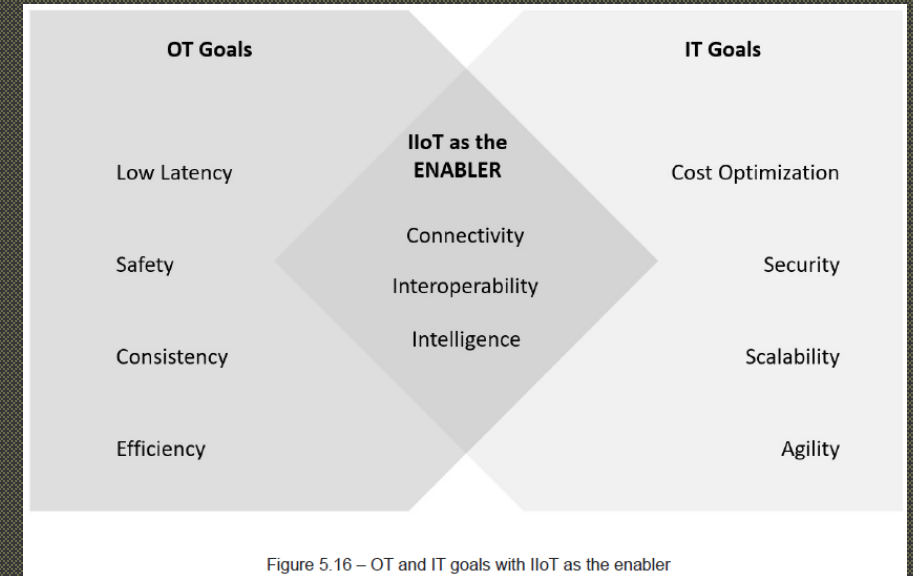
- Low Latency: Machines need instant responses (e.g., robot stops in milliseconds).
- Safety: No explosions or accidents.
- Consistency: Machines must perform identically every time.
- Efficiency: Maximize output, minimize waste.

IT Goals:

- Cost Optimization: Save money on operations.
- Security: Protect data from hackers.
- Scalability: Support growth without crashing.
- Agility: Adapt quickly to changes.

IIoT as the Enabler:

- Connectivity: Links machines to the cloud (e.g., sensors → ERP).
- Interoperability: Makes old and new systems talk (e.g., PROFINET + OPC UA).
- Intelligence: AI predicts failures or optimizes production.



DIGITAL TWIN

A digital twin is a virtual copy of a real-world machine, system, or process that updates in real-time using data from sensors.

- Sensors on a machine (like a factory robot, car engine, or power plant) collect data (speed, temperature, pressure).
- This data is sent to a digital model that mimics the real machine's behavior.
- Engineers and AI can analyze, predict, and optimize performance without touching the real machine.

Types

- Component Twin - A single part (e.g., a car's suspension).
- Asset Twin - A full machine (e.g., an entire engine).
- Process Twin - A sequence of steps (e.g., a factory production line).
- System/Network Twin - Multiple machines working together (e.g., a smart city's traffic lights).

Benefits

- Predictive Maintenance - Know when a machine will fail before it breaks.
- Real-Time Monitoring - Track performance from anywhere (no need to be on-site).
- Simulation & Testing - Try upgrades or fixes virtually before doing them in real life.
- Better Decisions - Combine sensor data with maintenance logs, specs, and AI for smarter operations.

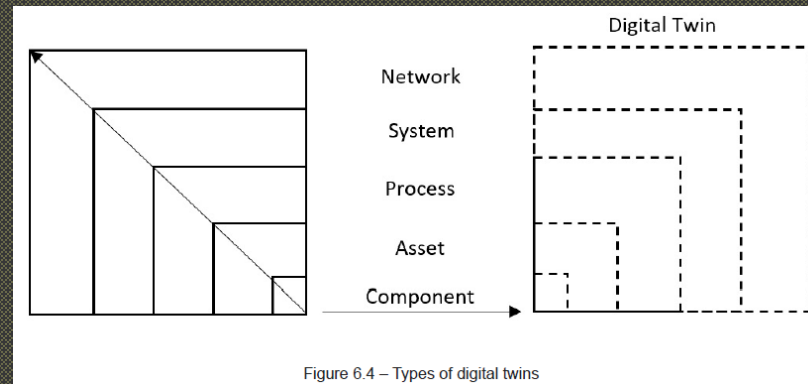


Figure 6.4 – Types of digital twins

DIGITAL TWIN

A single digital twin connects the entire car ecosystem, turning raw IoT data (like speed, temperature, or engine health) into cost savings, safety improvements, and better customer experiences.

Example

- A car's sensors send real-time data to its digital twin.
- The ride-share company uses this data to schedule maintenance.
- The insurer adjusts premiums based on actual driving habits.
- The manufacturer discovers a design flaw and releases a software update.

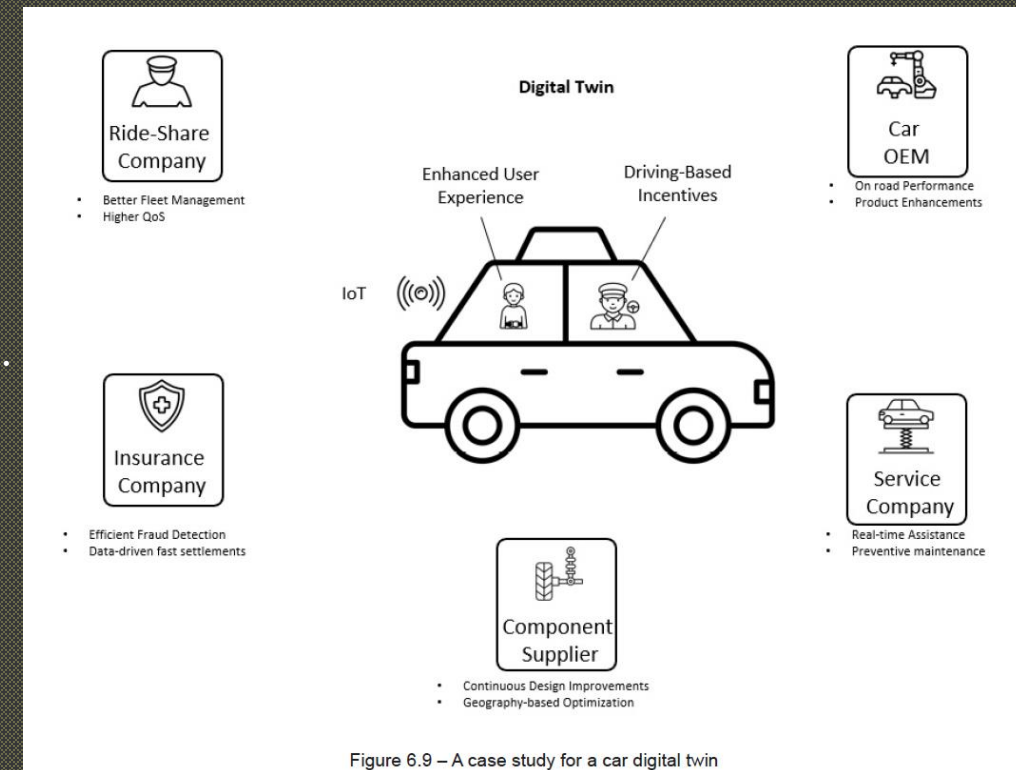


Figure 6.9 – A case study for a car digital twin

AWS IOT TWINMAKER

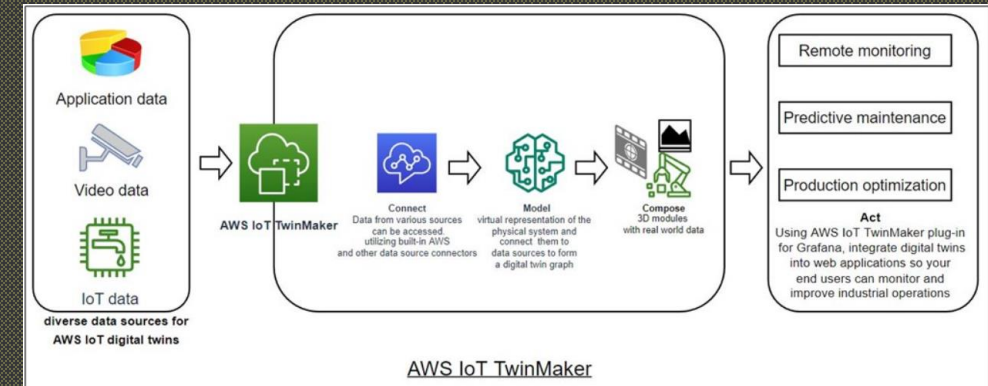
AWS IoT TwinMaker is a tool that helps you create digital twins - virtual 3D models of real-world machines, factories, or systems that update in real-time using live data.

Features

- 3D Models + Live Data
 - Upload 3D designs of your factory/machines.
 - Connect them to real-time sensor data (temperature, speed, etc.).
- Unified View
 - Combines equipment layouts, sensor feeds, and even live video into one dashboard.
- Easy Integration
 - Works with AWS services (like IoT Core) or external data sources.
- Simulate & Predict
 - Run "what-if" scenarios (e.g., "What happens if Machine X overheats?").

Example Use Case - A car factory uses TwinMaker to:

- Monitor assembly robots in 3D.
- Predict failures before they happen.
- Test layout changes virtually before rearranging the real factory.



<https://www.cloudthat.com/resources/blog/simplify-digital-twins-construction-with-aws-iot-twinmaker-2>



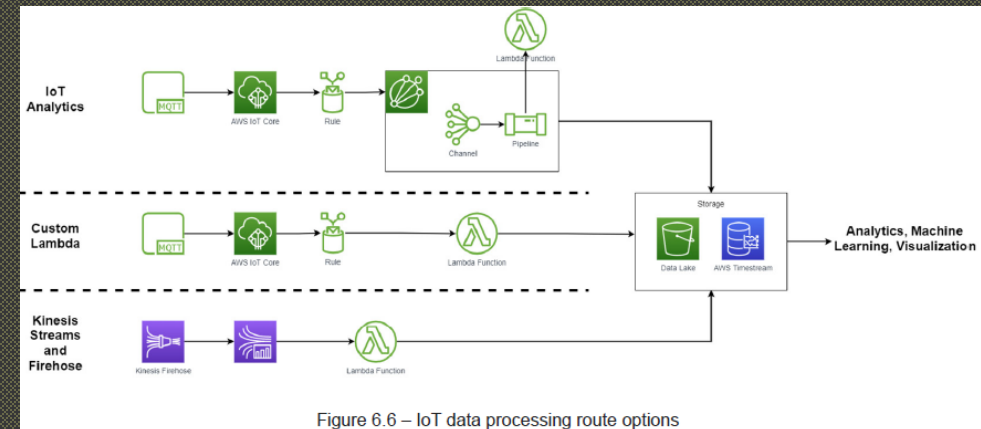
58

CLOUD DATA INGESTION

- Data Ingestion (Collecting Raw Data)
 - AWS IoT Core: The "gateway" that receives sensor data from devices (like temperature sensors or machines).
 - Kinesis Streams/Firehose: Services that collect and transport high-speed data (like a conveyor belt).
- Data Processing (Cleaning & Organizing)
 - Lambda (Custom Code): Allows writing rules to filter or transform data (e.g., "Alert if temperature > 100°C").
 - Other Tools (Rue, Lamp3d, Pipette): Likely represent custom or third-party processing tools.
- Data Storage (Saving for Later Use)
 - Storage/Data Lake: Raw data is dumped into scalable storage (like Amazon S3) for long-term use.
 - AWS Timestream: A special database for time-series data (e.g., sensor readings over time).
- Analytics & Insights (Making Data Useful)
 - Analytics: Tools like AWS IoT Analytics to spot trends (e.g., "Machine vibrations increase every Friday").
 - Machine Learning (ML): Predict failures or optimize operations (e.g., "This motor will fail in 3 days").
 - Visualization: Dashboards (like Amazon QuickSight) to show data in graphs/charts.

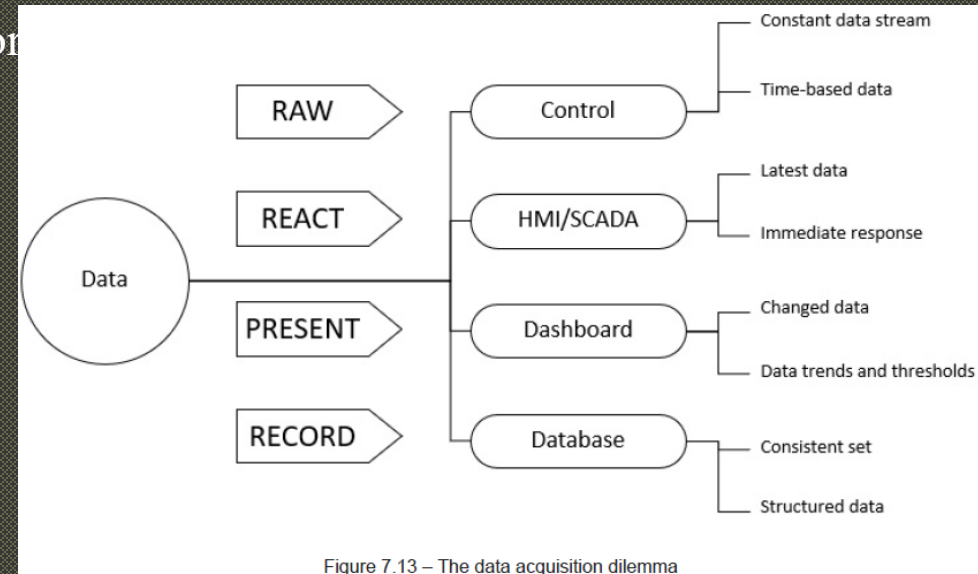
Example: A smart farm uses this pipeline to:

1. Collect soil moisture data (IoT Core).
2. Filter out faulty sensor readings (Lambda).
3. Store data (S3 Data Lake).
4. Predict irrigation needs (ML).
5. Display results on a dashboard (QuickSight).



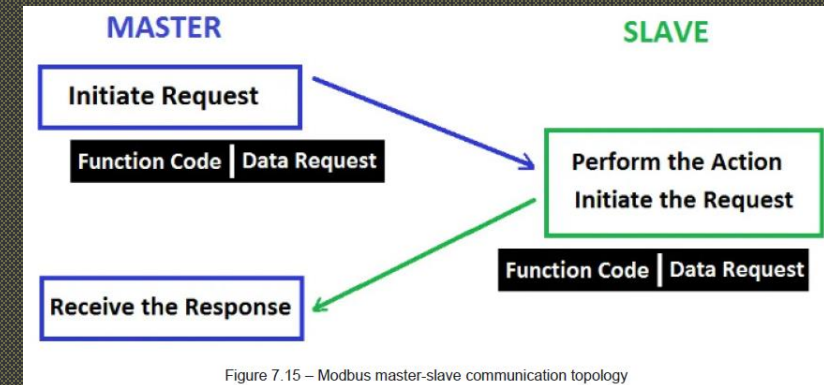
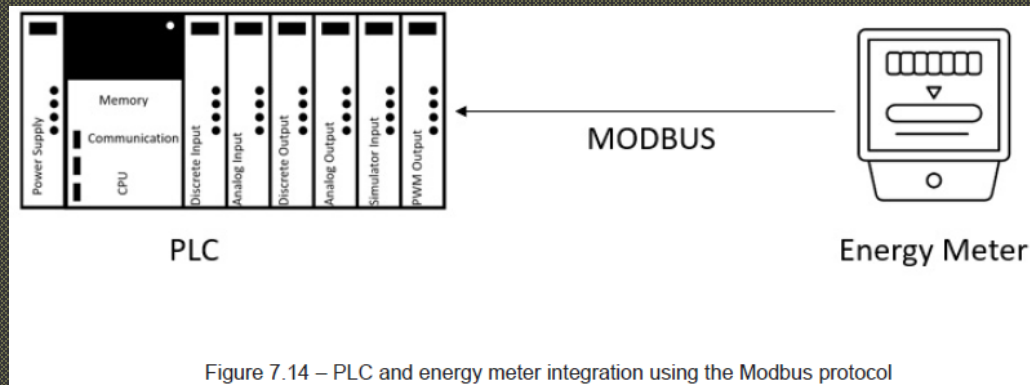
PLC DATA VISUALIZATION

- PLCs (Programmable Logic Controllers) run machines automatically, but humans still need to monitor and control them. Data visualization helps operators understand what's happening, diagnose problems, and make decisions.
 - HMI (Human-Machine Interface) - Displays real-time machine data (like alarms, errors) and lets operators adjust settings (e.g., production targets). It connects directly to a PLC.
 - SCADA (Supervisory Control and Data Acquisition) - A higher-level system that gathers data from multiple PLCs and HMIs, giving plant managers an overview of operations.
 - Uses PLC tagging to standardize data from different PLC brands.
 - Historians store historical data for trend analysis.
- Types of Data & Their Uses:
 - Real-time control data - Fast, raw data used by PLCs for instant machine actions.
 - Alarms & events - Critical issues needing immediate operator attention.
 - Dashboards & KPIs - Summarized trends and performance metrics for decision-making.
 - Stored structured data - Used for analytics, predictions, and long-term improvements.



PLC DATA VISUALIZATION

- Connecting energy meters to a central PLC (production control system) in factories can provide useful data, like tracking energy use per production line, shift, or cost. Normally, energy meters are separate and only used for utility billing, but linking them via Modbus - a simple, older communication protocol - allows better energy monitoring and planning.
- It uses a master-slave setup (PLC = master, meters = slaves).
- The master (PLC) sends requests, and slaves (meters) respond with data.
- Works over basic serial connections (RS232/RS485).
- This integration helps factories optimize energy costs and efficiency.



PLC DATA VISUALIZATION

Even though many new energy meters support wireless tech like Wi-Fi or LoRaWAN, most still use Modbus for communication. To connect an energy meter to a PLC, it is required to set up:

- Device address (like an ID number)
- Baud rate (communication speed)
- Parity bit (for error checking), etc.
- Each meter stores data (like voltage, power usage, etc.) in specific memory registers, which vary by manufacturer. The register map (a guide from the manufacturer) tells the PLC where to find each reading.
- The PLC then requests data (like current, voltage, or error codes) from these registers, allowing the factory to monitor energy usage in real time.

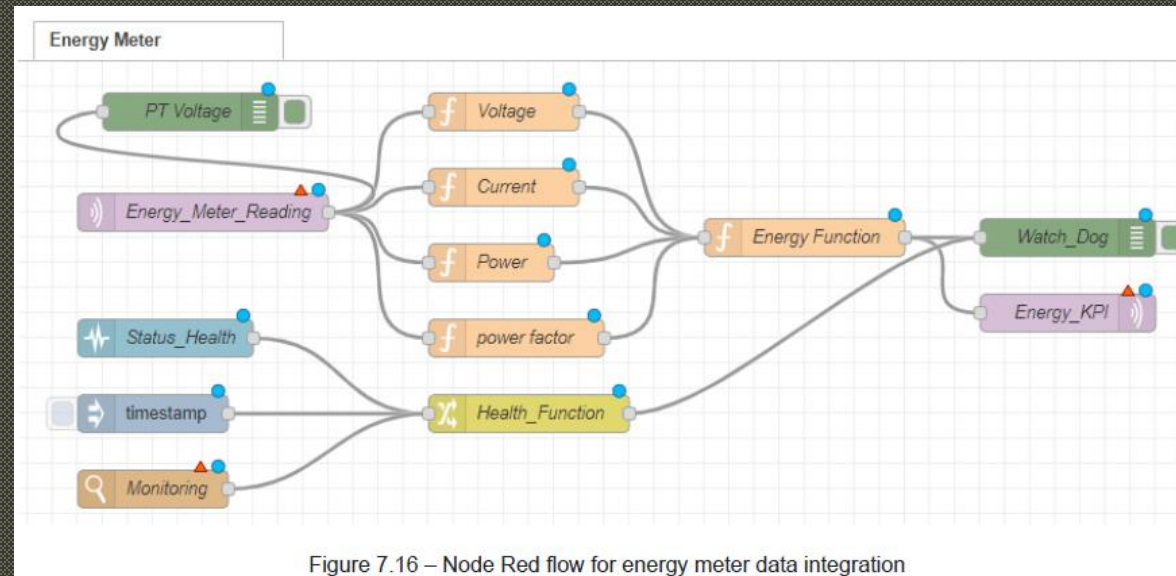


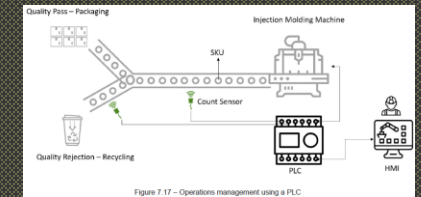
Figure 7.16 – Node Red flow for energy meter data integration

PLC DATA VISUALIZATION

A PLC (Programmable Logic Controller) and HMI (Human-Machine Interface) are used to control and monitor an injection molding production line. The PLC collects real-time data (like machine status, production speed, errors) to measure operational efficiency.

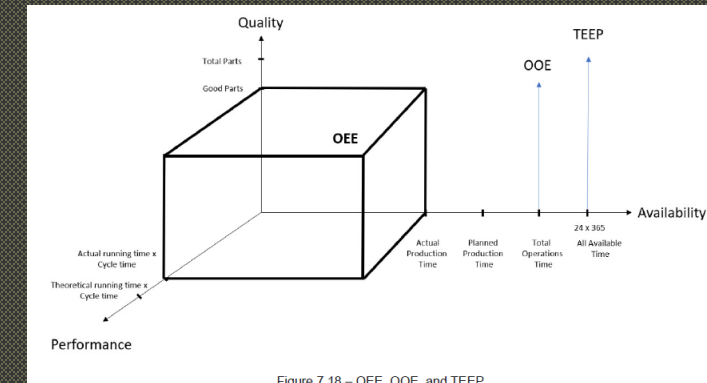
1. Key Performance Indicators (KPIs)

- OEE (Overall Equipment Effectiveness): A score (0-100%) showing how well a machine or line is performing. It combines:
 - Availability (Is the machine running or broken?).
 - Performance (Is it running at full speed or slower?).
 - Quality (Are the products good or defective?).
- OOE (Overall Operations Effectiveness): Similar to OEE but ignores unscheduled downtime (like sudden breakdowns).
- TEEP (Total Effective Equipment Performance): The "perfect score" (24/7 production with no defects). Used for benchmarking.



2. Visualizing KPIs

- Imagine a 3D box where:
 - Length (X-axis) = Time (24/7 = TEEP, less downtime = OOE).
 - Height (Z-axis) = Quality (good parts vs. total parts).
 - Depth (Y-axis) = Speed (actual vs. ideal production rate).
- OEE = Volume of the box; bigger = better efficiency



IOT ARCHITECTURAL PRINCIPLES

- The Industrial Internet of Things (IIoT) is built on three key pillars from Germany's 2030 vision:
 - Autonomy - Machines/systems make smart decisions on their own.
 - Interoperability - Different systems and devices work together seamlessly.
 - Sustainability - Eco-friendly and socially responsible manufacturing.

To make this work, IIoT designs follow five core principles (the "5 S's"):

- Simplicity
 - Keep solutions simple and modular (even for complex problems).
 - Avoid overcomplicating things - use only what's needed.
- Security
 - Build systems "secure by design" (no weak spots). Constantly monitor for threats.
 - Follow Zero-Trust Access (ZTAA) - only give minimum needed permissions.
- Serviceability
 - Ensure systems can easily connect and share data with others.
 - Make solutions plug-and-play for future reuse.
- Scalability
 - Design systems to grow (or shrink) as needed.
 - Handle changing demands without major redesigns.
- Standardization
 - Use common rules/protocols so everything works together. Avoid mismatched systems.



DATA SERVICE BUS (DSB)

The Data Service Bus (DSB) is like a central data highway for manufacturing and supply chains. It enables real-time data sharing between all stakeholders, including:

- Manufacturers (OEMs, suppliers)
- Service providers (maintenance, insurance)
- Logistics & distribution
- Customers

1. Real-Time Data Exchange - The DSB connects all systems, allowing instant updates.
 - Example: If a machine fails, the manufacturer, supplier, and service team all get alerts immediately.
2. Transparent Value Chains - Every step (from production to delivery) is tracked and visible.
3. Root Cause Analysis (RCA) - If a problem occurs (e.g., a faulty part), the DSB helps quickly trace the issue.
4. Proactive Alerts - If a defect is found, similar products in the field can be flagged for inspection.

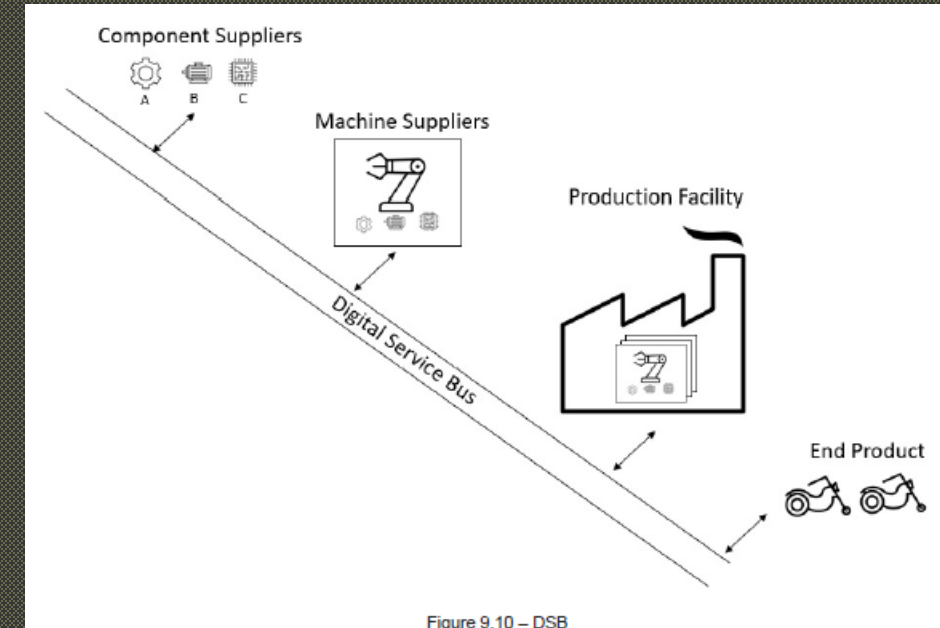


Figure 9.10 – DSB

Example: A motorcycle's brakes fail at high speed.

- The manufacturer, supplier, and insurance are alerted immediately.
- They analyze data to find the faulty part (Root Cause Analysis).
- All similar bikes get a recall warning to prevent accidents.
- Future bike designs are improved to avoid the issue.

SYSTEM CONSIDERATIONS FOR ARCHITECTURAL DESIGN

When designing an IIoT system, you need to decide where to process data (at the edge or in the cloud) based on:

1. Bandwidth & Connectivity

- Factories often have slow/no internet, so sending all raw data to the cloud isn't practical.
- Solution: Process data at the edge (near machines) and only send key info (like alerts or KPIs) to the cloud.

2. Storage Limits

- Machines generate massive data (e.g., 1 CNC machine = 14GB/year, 300 machines = 4TB/year!).
- Solution: Use smart storage rules (archive old data, delete useless data).

3. Security & Privacy

- Some factories can't send data outside (due to security or government rules).
- Solution: Keep sensitive data on local edge systems.

4. Cost & Scalability

- Cloud is great for global companies (compare plants, optimize OEE).
- But edge computing saves money on data transfers.
- Solution: Balance upfront costs (hardware) vs. long-term savings (cloud fees).

Types of Data to Consider

- System Data (machine status, errors)
- Performance Data (health checks)
- Core Data (sensor readings, camera feeds)

AI AND ML IN SMART MANUFACTURING

Leveraging AI and ML in smart manufacturing:

- AI in Factories:
 - Turns rigid processes into flexible, data-driven ones
 - Helps predict machine failures (predictive maintenance)
 - Optimizes production scheduling and resource use
 - Can improve products and create new business models
- Methods:
 - Traditional Computing: Follows fixed human-written rules
 - Machine Learning: Learns from data to make predictions
 - First trained with historical data
 - Then makes real-time decisions
 - Keeps improving over time
- Implementation Levels:
 - Ranges from human-controlled (Level 0) to fully autonomous (Level 5)
 - Current best practice: AI suggests, humans decide (assisted AI)

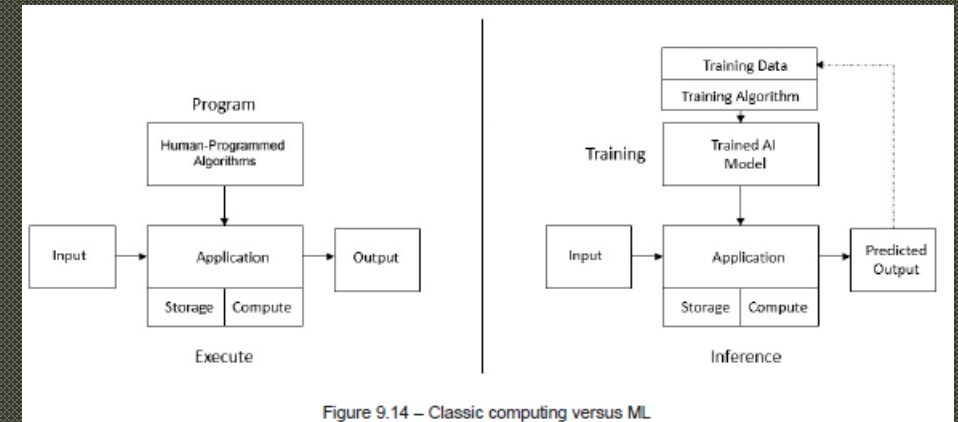


Figure 9.14 – Classic computing versus ML

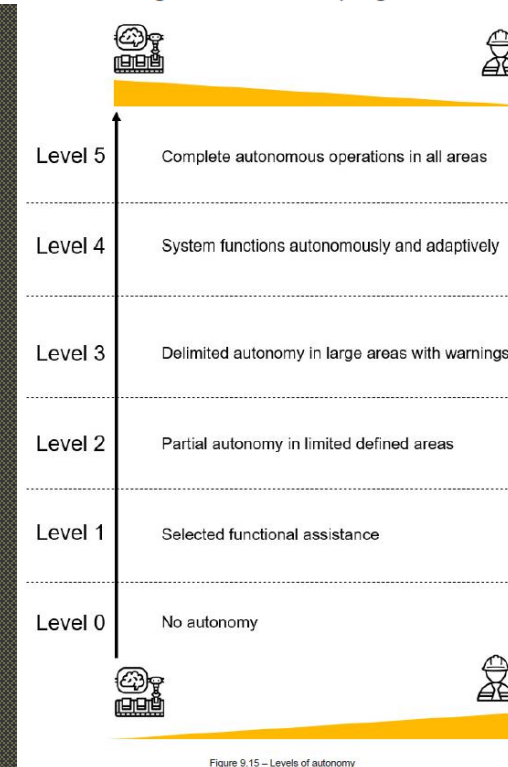
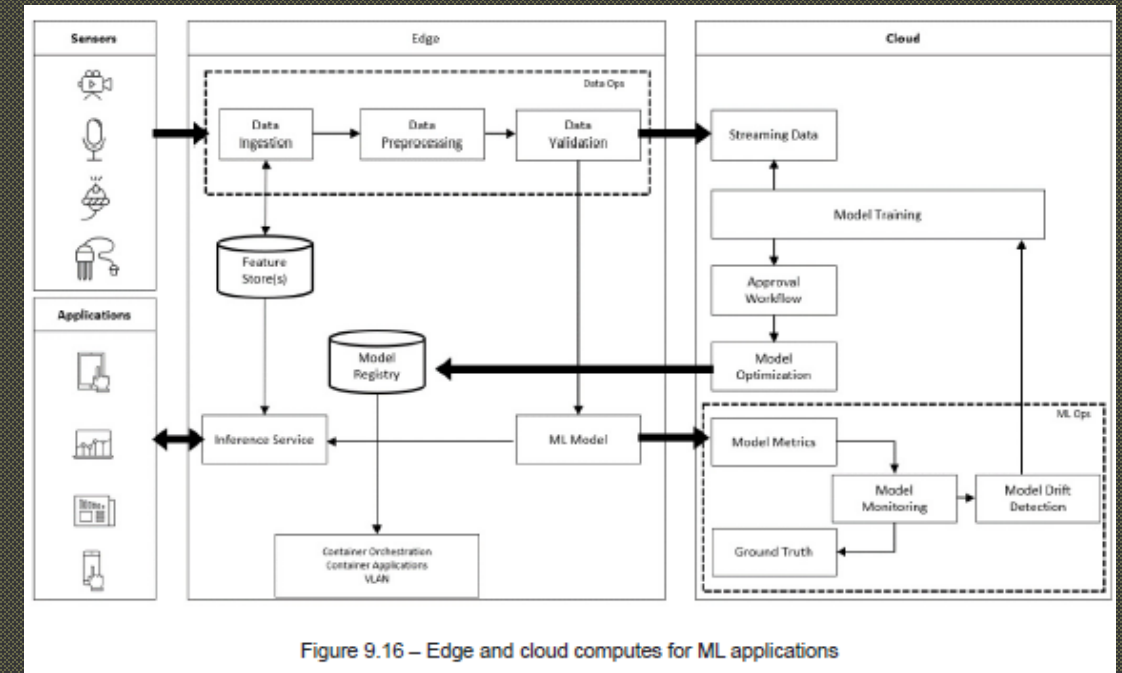


Figure 9.15 – Levels of autonomy

AI AND ML IN SMART MANUFACTURING

Leveraging AI and ML in smart manufacturing:

- Where It Runs:
 - Edge Computing (near machines):
 - Handles time-sensitive decisions
 - Uses special chips (GPUs, TPUs) for fast processing
 - Cloud Computing:
 - Handles big data analysis
 - Retrains and improves models
- Example - Quality Control:
 - Cameras scan products on assembly line
 - Edge AI checks for defects instantly
 - Cloud system improves the AI model over time
 - Humans get alerts about problems
- Challenges:
 - AI is not perfect - needs human oversight
 - Requires good quality data to work well
 - People need to trust the system



SECURITY IN MANUFACTURING

1. IT vs. OT Security Risks

- IT Problems: Apps crash, data leaks.
- OT Problems: Machines explode, people get hurt, production stops.

2. The Big Challenge

- Factories have old, unsecured systems (e.g., PLCs with no passwords).
- Hackers can attack through cables, sensors, or Wi-Fi.

3. How to Protect Factories

- A. Follow Standards
 - IEC 62443 & NIST SP 800-82: Rulebooks for securing industrial systems.
- B. Two Key Strategies
 - Defense-in-Depth
 - Like a castle with multiple walls, moats, and guards.
 - Stops hackers at many levels (but fails if they get inside).
 - Zero Trust
 - No one is trusted - even employees must prove who they are every time.
 - Hard to use with old machines (e.g., a 1990s PLC can't handle login codes).
- C. Best Solution
 - Mix both strategies (like a castle with fingerprint scanners).
 - Use SASE ("sassy") for remote access control.

Real-World Example

- A hacker attacks a power plant:
- Defense-in-Depth: Firewalls block 99% of attacks.
- Zero Trust: The hacker can not move to other systems after breaking in.
- Result: Only one machine fails (not the whole grid).
- Key Idea
- Factory security = Protecting both data AND physical safety. Start with small steps (like adding passwords to PLCs) and build up.

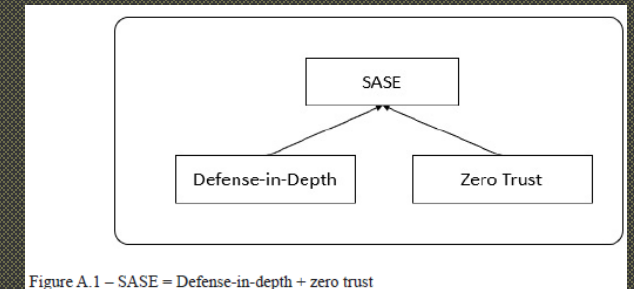


Figure A.1 – SASE = Defense-in-depth + zero trust

SASE - Secure Access Service Edge, is a cybersecurity framework that converges networking and security functionalities into a unified, cloud-delivered platform

- IEC 62443 is a series of international standards focused on cybersecurity for industrial automation and control systems (IACS)
- NIST SP 800-82 - Guide to Operational Technology (OT) Security

COMPREHENSIVE SECURITY NETWORK

1. The 5-Step Security Cycle

- Identify - Know your machines, networks, and weak spots.
- Protect - Add defenses (firewalls, passwords, encryption).
- Detect - Monitor for hackers or weird machine behavior.
- Respond - Stop attacks fast (like a fire alarm for cyber threats).
- Recover - Fix damage and learn from mistakes.

2. OT vs. IT Security

- OT (Factory Machines): Needs physical safety + uptime.
- Example: A hacked PLC can overheat and explode.
- IT (Office Systems): Needs data protection.
- Example: A hacked email leaks customer info.

3. Network Layers in a Factory

4. The iDMZ (Industrial Demilitarized Zone)

- Acts like a moat + drawbridge between factory (OT) and office (IT).
- Why? A phishing email in the office won't reach the robots.
- Uses double firewalls to filter all data crossing zones.

Real-World Example

- A hacker tries to attack a car factory:
- Phishing email tricks an office worker (IT breach).
- iDMZ blocks the hacker from reaching the assembly line robots.
- Alerts sound - IT team shuts down the attack.

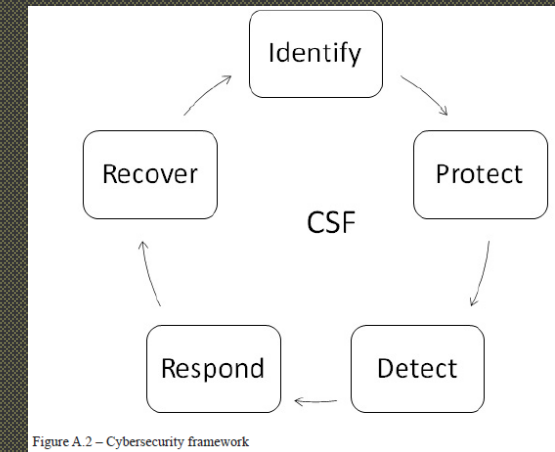


Figure A.2 - Cybersecurity framework

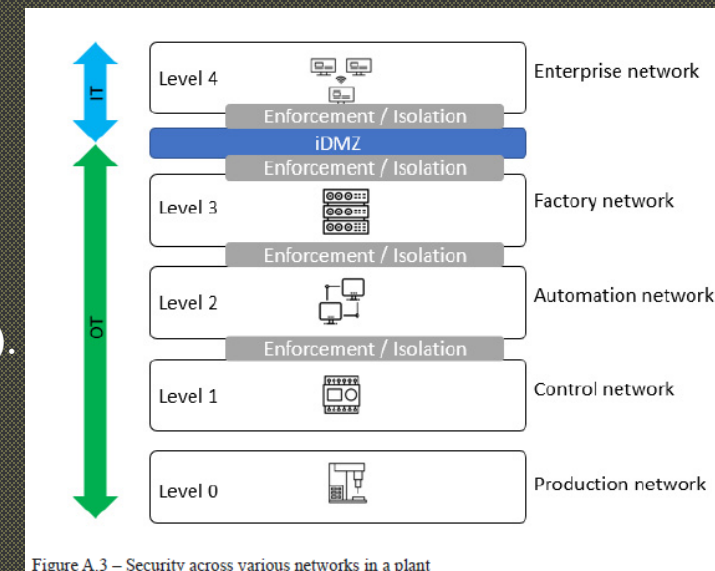


Figure A.3 - Security across various networks in a plant

DESIGNING A ROBUST SECURITY PROGRAM

A good OT security program needs:

- a leader, who is the Chief Information Security Officer (CISO),
- a Network Operations Center (NOC) team, which keeps the network running smoothly,
- a Security Operations Center (SOC), which protects against cyber threats, and
- five security pillars to protect software, hardware, networks, and physical spaces.

Five security pillars:

1. Software Security: Uses tools to protect devices and apps, with updates for antivirus, security patches, and operating systems.
2. Infra Security: Secures servers, switches, and factory equipment in the OT network.
3. OT Security Management: Oversees the SOC's work and uses tools to maintain security standards.
4. OT Network Security: Handles NOC tasks like monitoring network traffic and spotting unusual activity.
5. OT Physical Security: Involves watching over important places like server rooms and sensitive machines.

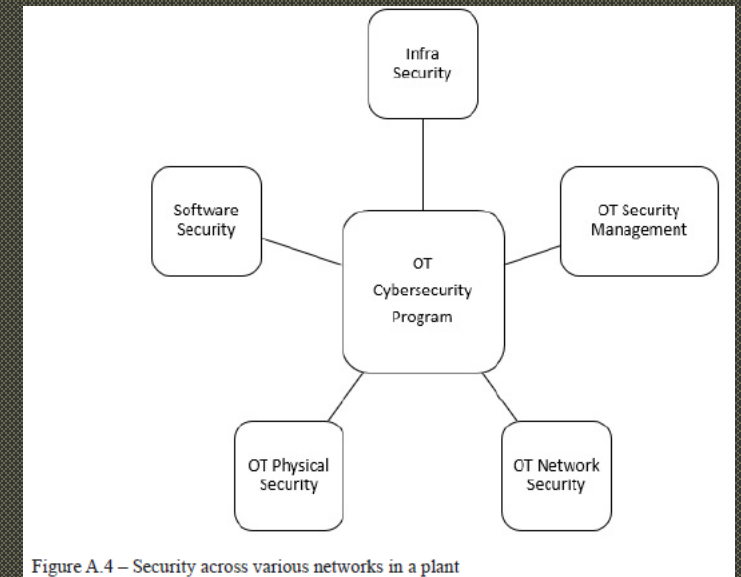
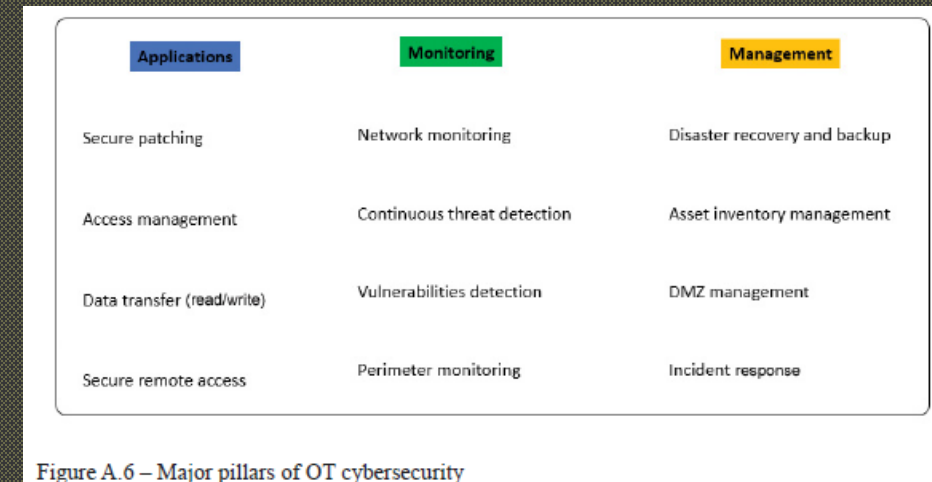


Figure A.4 – Security across various networks in a plant

DESIGNING A ROBUST SECURITY PROGRAM

OT (Operational Technology) cybersecurity is about keeping factory systems safe from cyber threats.

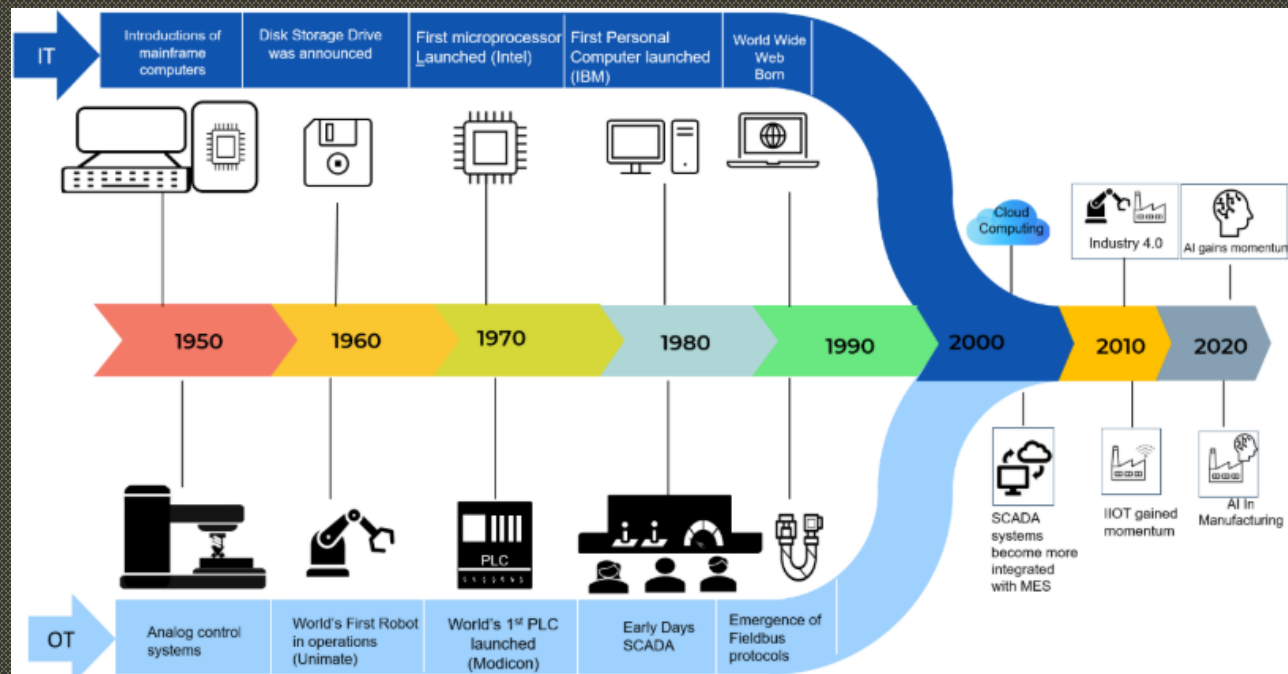
- Applications: This is about keeping the apps factory workers use (to view or control data) secure. Key tasks include:
 - Secure patching: Updating apps safely.
 - Access management: Controlling who can use them.
 - Data transfer (read/write): Safely moving data.
 - Secure remote access: Allowing safe access from outside.
- Monitoring: Watching the system to catch problems early. This includes:
 - Network monitoring: Checking the network for issues.
 - Continuous threat detection: Spotting threats all the time.
 - Vulnerabilities detection: Finding weak spots.
 - Perimeter monitoring: Watching the edges of the network.
- Management: Organizing and planning to handle problems. This involves:
 - Disaster recovery and backup: Having a plan to recover if something goes wrong.
 - Asset inventory management: Tracking all equipment.
 - DMZ management: Managing a safe zone between networks.
 - Incident response: Knowing how to react to attacks.



INTRODUCTION TO INDUSTRY 4.0

The figure is a horizontal timeline depicting the historical evolution of two interconnected domains in industrial and manufacturing contexts:

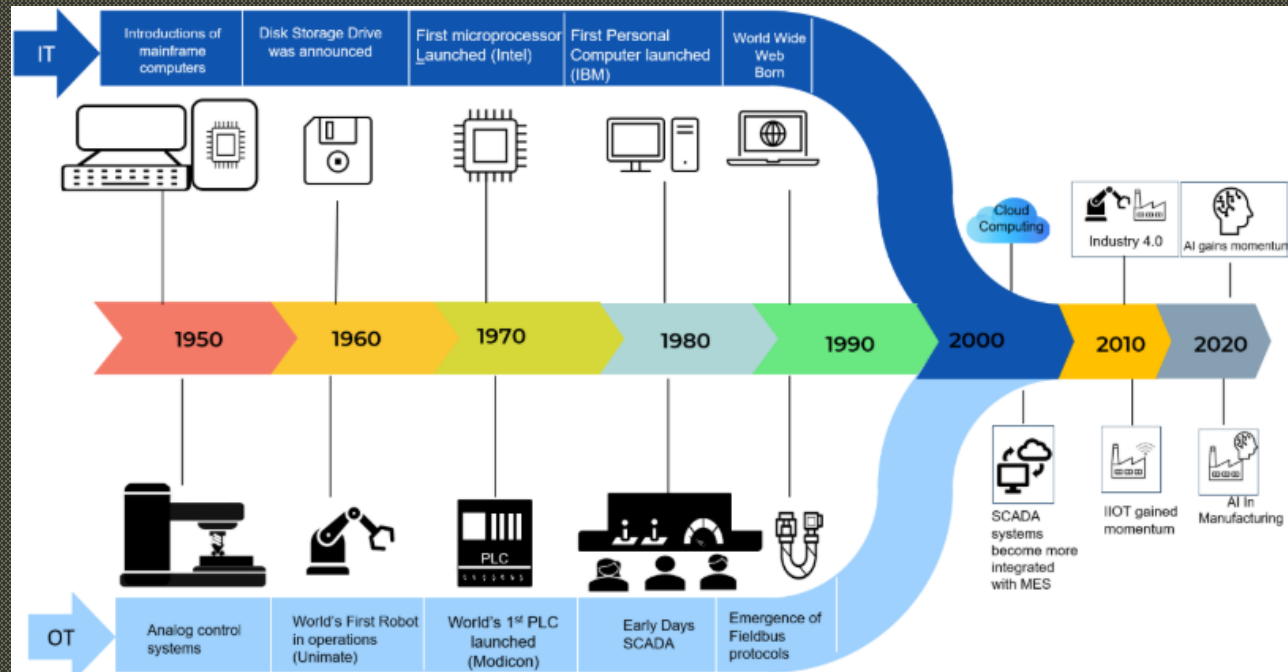
- Information Technology (IT) and Operational Technology (OT).
- It spans from 1950 to 2020, highlighting key milestones that show how these technologies have developed in parallel, eventually converging in modern concepts like Industry 4.0, IoT (Internet of Things), SCADA (Supervisory Control and Data Acquisition), MES (Manufacturing Execution Systems), and AI-driven manufacturing.



INTRODUCTION TO INDUSTRY 4.0

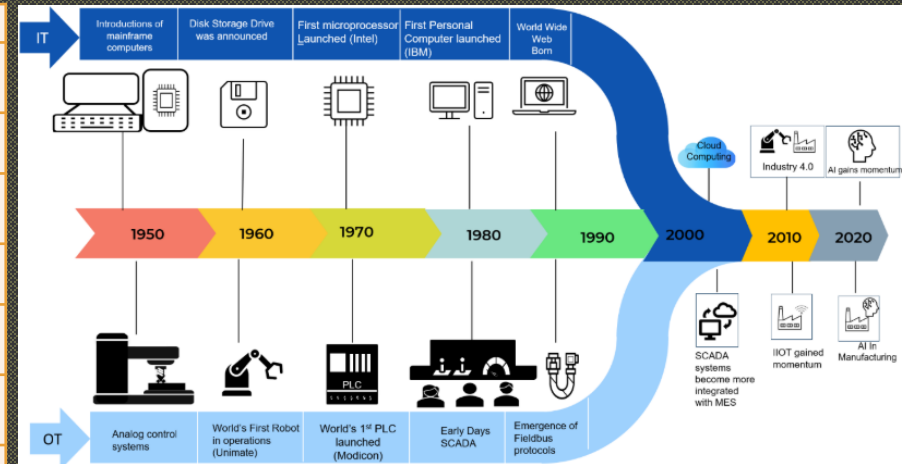
The timeline is divided into:

- IT track: Focuses on computing and data advancements that enable information processing, storage, and connectivity.
- OT track: Focuses on operational systems for controlling physical processes, automation, and industrial machinery.
- Convergence point (right side, around 2010-2020): Shows integration of IT and OT, leading to smart, connected manufacturing.
- The design uses icons (e.g., computers, robots) and color-coded arrows to visually map progress. Years are marked below the timeline (1950, 1960, 1970, 1980, 1990, 2000, 2010, 2020), with milestones placed at approximate corresponding points.



INTRODUCTION TO INDUSTRY 4.0

Year	Milestone	Description/Icon
1950	Introduction of mainframe computers	Large-scale computers for data processing in businesses and research.
1960	Disk Storage Drive was announced	First practical magnetic disk storage for reliable data retention.
1970	First microprocessor	Compact chip (e.g., Intel 4004) enabling smaller, cheaper electronics.
1980	First Personal Computer launched (IBM)	IBM PC popularized desktop computing for individuals.
1990	World Wide Web	Tim Berners-Lee's invention enabling global information sharing via internet.
2000	Cloud Computing	Shift to on-demand, scalable remote data storage/processing (e.g., early AWS).
2010-Today	SCADA more widely manufactured; Industry 4.0 AI gains momentum	Broader adoption of supervisory systems; AI integration for predictive analytics in manufacturing.



<https://inductiveautomation.com/blog/itot-convergence-unifying-information-and-operational-technology-with-ignition>

Year	Milestone	Description/Icon
1950	Analog control systems	Basic electrical/mechanical controls for processes (pre-digital era). Icon: Dial/gauge machinery.
1960	World's First Robot (Unimate)	George Devol's industrial arm for automotive assembly lines. Icon: Robotic arm.
1970	World's First PLC (Modicon)	Programmable Logic Controller by Dick Morley for flexible machine control. Icon: PLC control box.
1980	Early Days SCADA	Initial supervisory systems for monitoring industrial processes remotely. Icon: Control room with screens.
1990	Fieldbus protocols	Standardized communication networks (e.g., Profibus) for connecting factory devices. Icon: Data cables/plugs.
2000	Emergence of protocols with MES	Integration of execution systems for production tracking/scheduling. Icon: Human operators at consoles.
2010-Today	SCADA becomes more integrated with MES; IoT gained; AI in Manufacturing gains momentum.	Sensor-based connectivity (IoT) and AI for optimization; deeper IT-OT fusion. Icon: Connected factory with IoT devices and AI overlay.

INTRODUCTION TO INDUSTRY 4.0

The figure communicates four ideas:

1. OT automation pyramid

- It visually presents how devices → PLC → SCADA → MES → ERP form a hierarchical structure.

2. Data flows upward; control flows downward

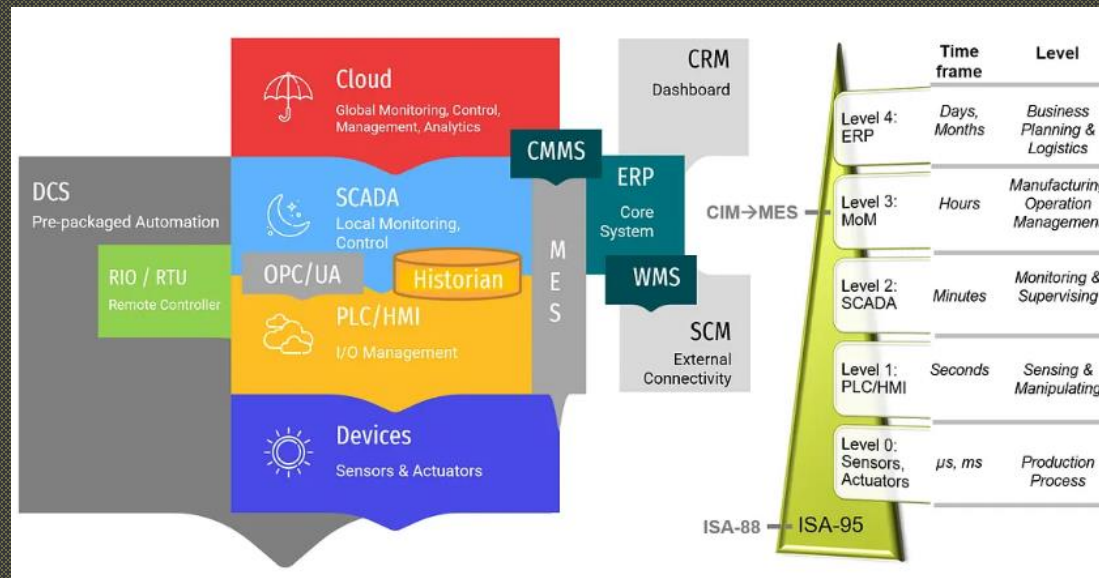
- Sensors generate the data.
- PLCs control the process.
- SCADA supervises.
- MES manages workflow.
- ERP plans business decisions.

3. Role of OPC UA and historians:

- Standardized data exchange
- Long-term storage
- Analytics
- IIoT/cloud integration

4. OT/IT convergence

- The left is OT, the right is IT.
- MES is the “bridge” between the two worlds.



<https://andisama.medium.com/towards-industry-4-0-1-it-ot-convergence-a-digital-transformation-journey-automation-control-bdb05143ec3f>

- CMMS – Computerized Maintenance Management System
- SCM - Supply Chain Management
- WMS – Warehouse Management System
- CRM - Customer Relationship Management

- A smart factory connects - from tiny sensors on machines at the bottom to big business computers at the top.
- It is based on a standard guide called ISA-95 (a blueprint for organizing factory tech), mixed with ISA-88 (for batch processes like mixing ingredients).

INTRODUCTION TO INDUSTRY 4.0

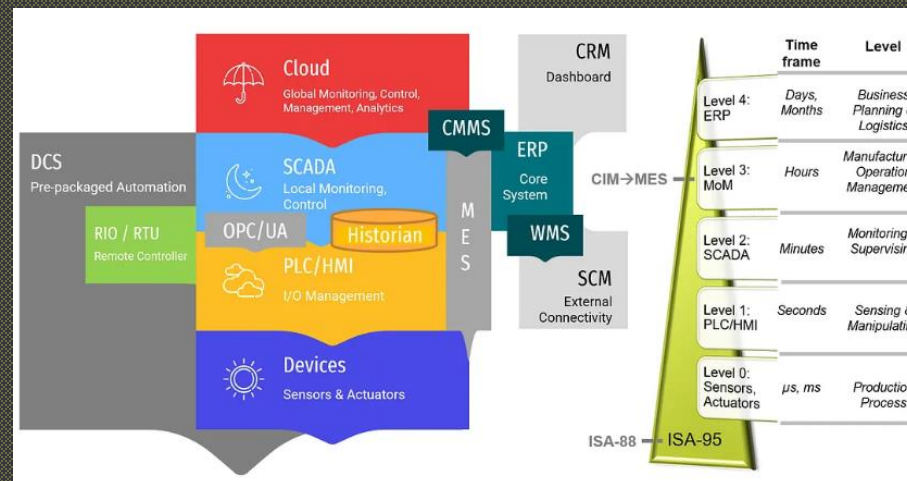
Left Side: OT Architecture Layers (Physical → Control → Supervisory → Cloud)

Level 0 - Devices (Sensors & Actuators)

- Includes temperature sensors, flow meters, pressure sensors, motors, valves, etc.
- Responsible for capturing physical measurements and performing physical actions.
- Time frame: microseconds to milliseconds.

Level 1 - PLC / HMI (I/O Management)

- PLCs (Programmable Logic Controllers) execute real-time deterministic control.
- They read data from the devices and perform logic based on ladder logic, function blocks, or structured text.
- HMIs provide local operator interfaces.
- This layer is responsible for:
 - I/O scanning
 - Real-time control decisions
 - Immediate safety interlocks
- Time frame: milliseconds to seconds.



INTRODUCTION TO INDUSTRY 4.0

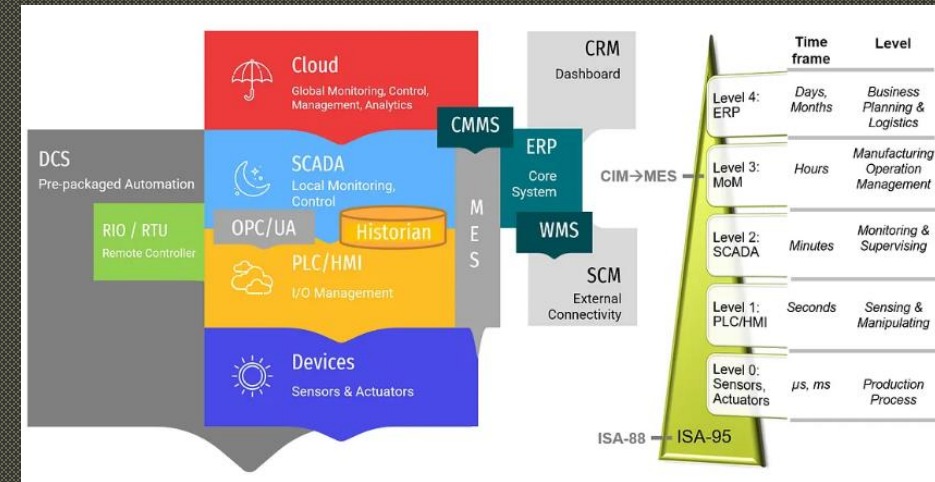
Left Side: OT Architecture Layers (Physical → Control → Supervisory → Cloud)

Level 2 - SCADA (Supervisory Control & Data Acquisition)

- SCADA systems handle:
 - Plant-wide supervisory control
 - Monitoring
 - Alarm management
 - Visualization dashboards
- They aggregate data from PLCs.
- SCADA uses protocols such as OPC UA to access data from PLCs.
- The Historian (highlighted in the figure) stores time-series process data over long periods, enabling:
 - Trend analysis
 - Performance monitoring
 - Quality analysis
 - Root cause detection

Time frame: seconds to minutes.

- Gateway & OPC UA Layer
 - Between PLC/SCADA and cloud/IT systems, OPC UA acts as a standardized communication interface, providing:
 - Security
 - Interoperability
 - Information modeling
 - Conversion of raw PLC registers into structured, contextualized data.



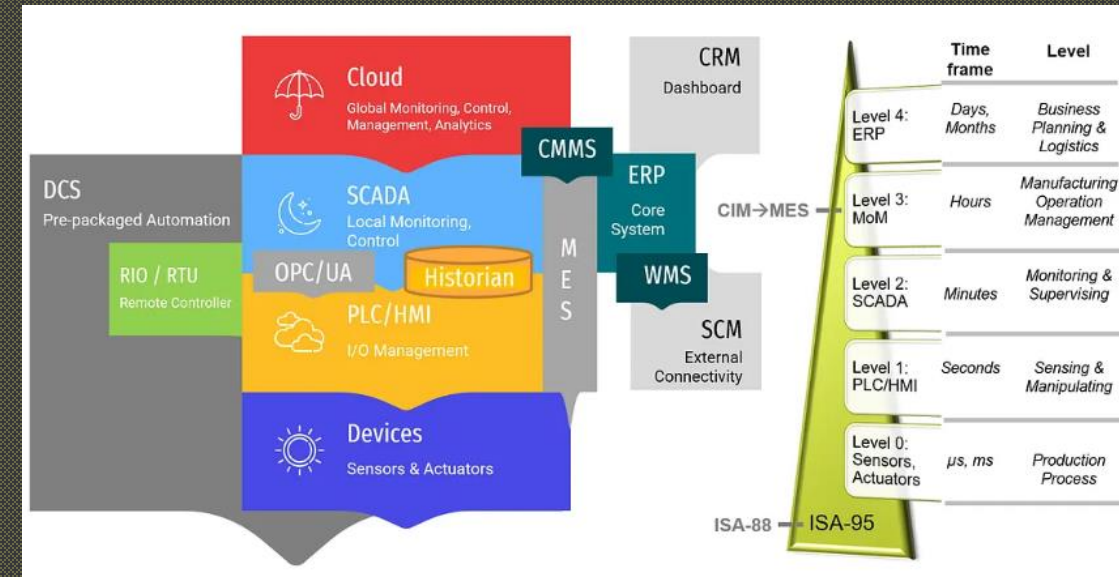
<https://andisama.medium.com/towards-industry-4-0-1-it-ot-convergence-a-digital-transformation-journey-automation-control-bdb05143ec3f>

INTRODUCTION TO INDUSTRY 4.0

Left Side: OT Architecture Layers (Physical → Control → Supervisory → Cloud)

Level 3 - MOM/MES (Manufacturing Operations Management)

- MES/MOM consumes data from SCADA and the historian.
- This level handles:
 - Production scheduling
 - Quality management
 - Work order execution
 - Maintenance coordination
- Time frame: minutes.
- This level sits between OT and IT and is the heart of OT/IT convergence.



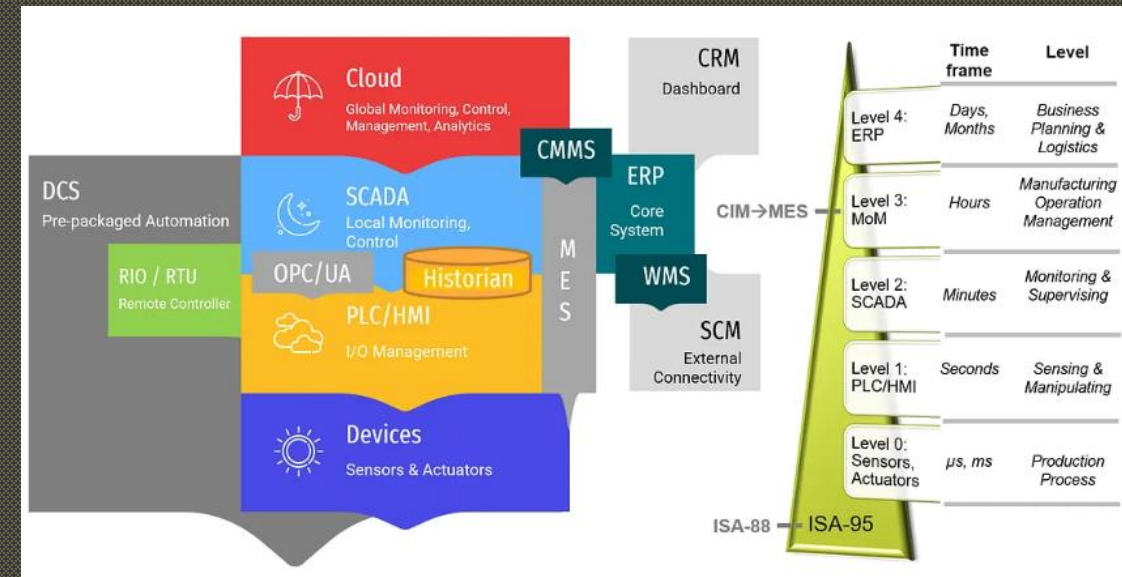
<https://andisama.medium.com/towards-industry-4-0-1-it-ot-convergence-a-digital-transformation-journey-automation-control-bdb05143ec3f>

INTRODUCTION TO INDUSTRY 4.0

Left Side: OT Architecture Layers (Physical → Control → Supervisory → Cloud)

Level 4 - Cloud / ERP (Business Planning & Logistics)

- Cloud Systems. Used for:
 - Global monitoring
 - Asset management
 - Predictive and Cross-Site analytics
 - AI/ML analysis
- Not real-time.
- ERP (Enterprise Resource Planning). Manages:
 - Finance
 - Supply chain
 - Production planning
- Connects manufacturing outcomes to business processes.
- CMMS (Computerized Maintenance Management System)
 - Tracks asset health, maintenance routines, spare parts, scheduled service tasks.
- WMS (Warehouse Management System)
 - Manages inventory, logistics, material movement.
- *Time frame*: days to months.



<https://andisama.medium.com/towards-industry-4-0-1-it-ot-convergence-a-digital-transformation-journey-automation-control-bdb05143ec3f>

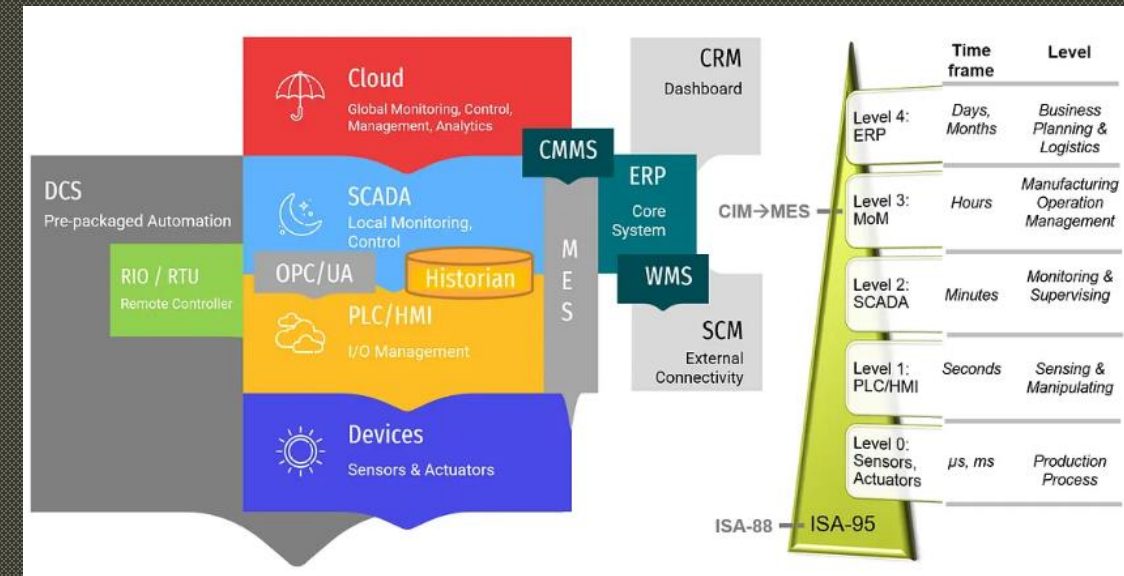
- SCM (Supply Chain Management)
- CRM (Customer Relationship Management)

INTRODUCTION TO INDUSTRY 4.0

Right Side: ISA-95 Hierarchy

The figure shows the standardized hierarchy from ISA-95, mapping the OT structure to information flow levels.

Level	Time Frame	Function
Level 4 (ERP)	Days...Months	Business planning, logistics
Level 3 (MOM)	Hours	Operations management
Level 2 (SCADA)	Minutes	Supervisory control
Level 1 (PLC/HMI)	Seconds	Sensing & manipulating
Level 0 (Sensors/Actuators)	μs...ms	Physical process



This helps classify:

- Who interacts with which systems
- What decisions are made at each level
- Time requirements for each level

<https://andisama.medium.com/towards-industry-4-0-1-it-ot-convergence-a-digital-transformation-journey-automation-control-bdb05143ec3f>

INTRODUCTION TO INDUSTRY 4.0

Middle Section: OT/IT Convergence (CIM → MES → ERP)

- The diagram in the middle shows how information flows horizontally between systems:

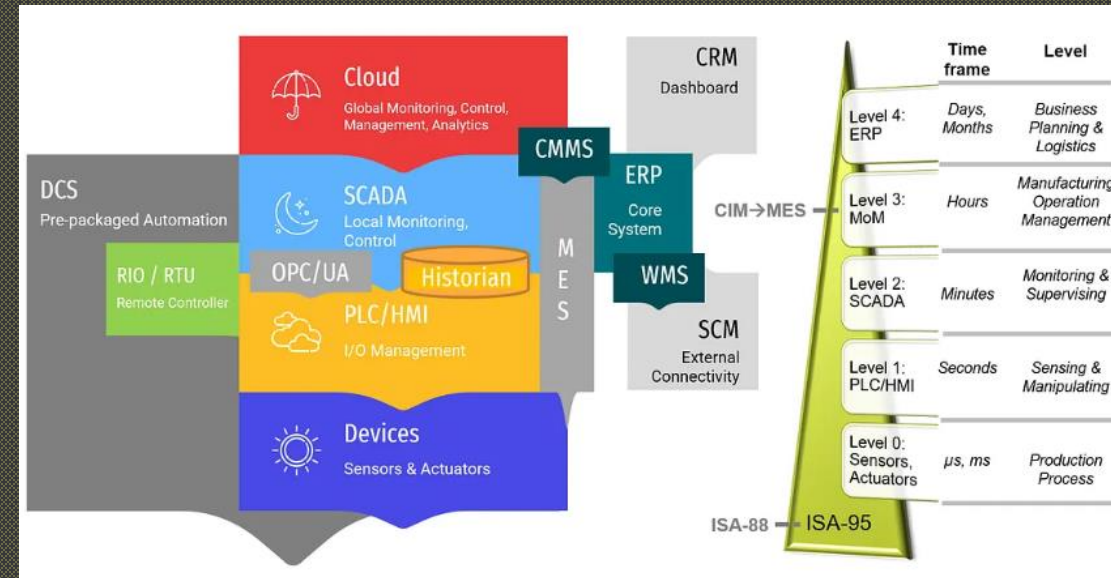
CIM → MES

- CIM (Computer Integrated Manufacturing) connects automation systems to MES.
- MES receives real-time plant data from SCADA, PLCs, historians.

MES → ERP

- MES bridges OT and IT.
 - Provides production, inventory, and performance data to ERP.
- SCM (Supply Chain Management) and CRM (Customer Relationship Management)
- SCM integrates logistics, suppliers, distributors.
 - CRM handles customer-facing dashboards, performance metrics.
 - This shows a modern, integrated manufacturing ecosystem.

- CIM – Computer Integrated Manufacturing
- CMMS – Computerized Maintenance Management System
- SCM - Supply Chain Management
- WMS – Warehouse Management System
- CRM - Customer Relationship Management



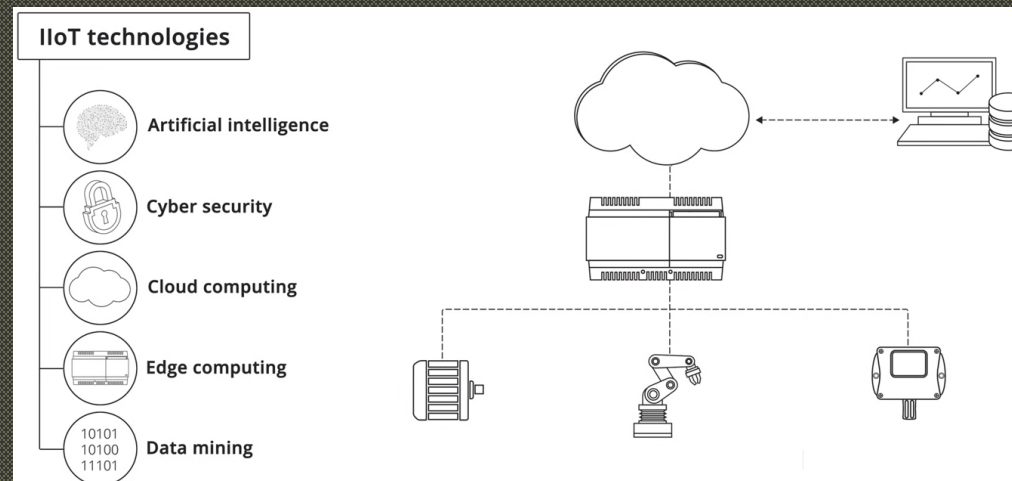
<https://andisama.medium.com/towards-industry-4-0-1-it-ot-convergence-a-digital-transformation-journey-automation-control-bdb05143ec3f>

Modern architectures no longer show CIM explicitly because its functions are increasingly distributed across:

- Edge applications
- Cloud data platforms
- IoT device management
- SCADA + historians

IIOT Technologies

- IIoT produces large and diverse datasets. Analytics tools search through this data to find trends, detect problems, and support smarter decisions, leading to better efficiency and reduced costs.
- Some decisions must be made instantly. Edge computing processes data close to the machines themselves, reducing delay and network load. Only important data is sent to the cloud, while fast responses happen locally.
- The cloud offers scalable storage and computing power for handling the massive data generated by IIoT. It supports heavy analytics, long-term storage, and complex processing that would be difficult or expensive on local systems.
- Connecting industrial systems to networks and the internet introduces cyber risks. Strong security is needed to prevent unauthorized access, protect sensitive data, and keep equipment running safely without disruption.
- AI and machine learning turn the huge amounts of data from IIoT sensors into useful information. They allow systems to predict equipment failures, optimize processes, and automate tasks by learning from patterns, reducing the need for constant human input.

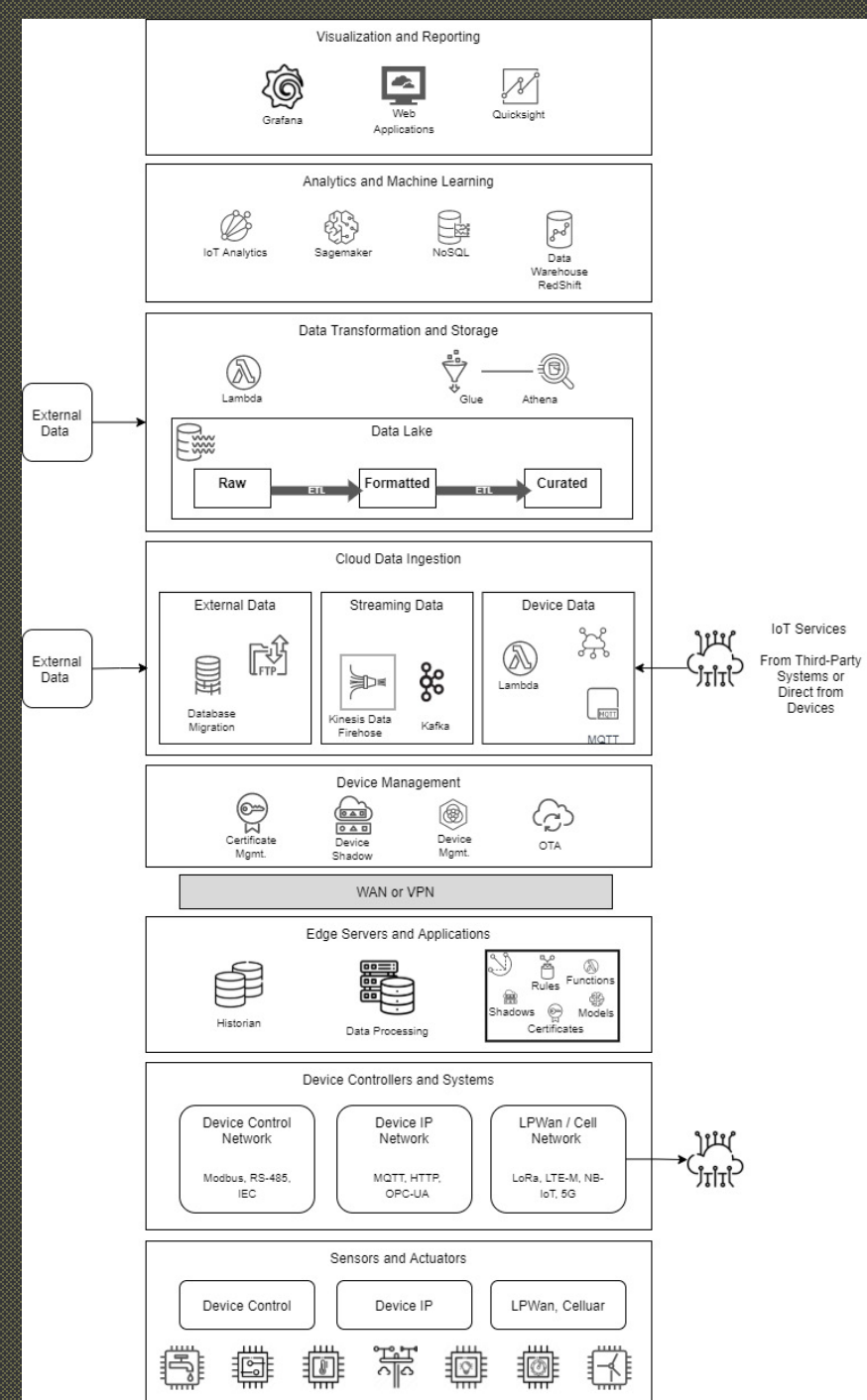


IOT Architecture

This diagram represents a complete Industrial IoT (IIoT) end-to-end architecture, showing:

1. Field Devices generate data.
2. Controllers and IP Networks transmit it.
3. Edge Servers preprocess and buffer it.
4. Device Management secures devices.
5. Cloud Ingestion Pipelines collect and route the data.
6. Data Lake + ETL transforms the data.
7. ML and Analytics Services extract intelligence.
8. Dashboards provide human-readable insights.

It is a modern, scalable, secure end-to-end digital manufacturing architecture.



IOT Architecture

1. Sensors and Actuators (Bottom Layer – Generate Data)

- These are the physical interface points to the real world.

Device Control

- Sensors connected to hardwired control I/O.
- Example: proximity switches, limit switches, 4-20 mA transmitters.

Device IP

- Sensors/actuators with IP (TCP/UDP) capability.
- Could speak MQTT, HTTP, OPC UA, or proprietary IP protocols.

LPWAN / Cellular Sensors

- Low-power wide-area sensors:

- LoRaWAN
- LTE-M
- NB-IoT
- 5G mMTC

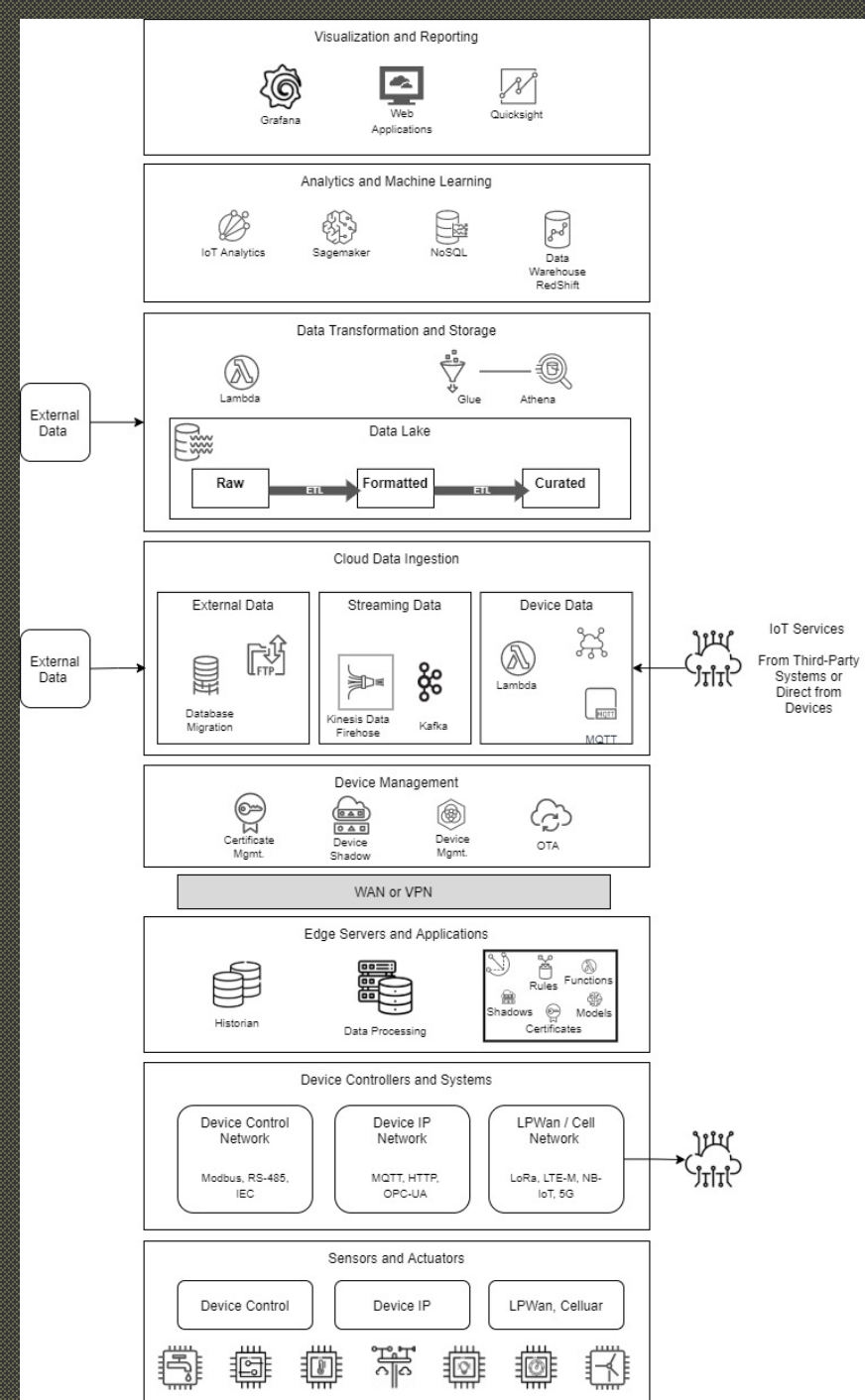
- Used for remote monitoring and long-range telemetry.

These devices create the raw operational data for the entire pipeline.

5G mMTC - Massive Machine-Type Communications

NB - Narrow Band

LTE-M - Long Term Evolution for Machines



IOT Architecture

2. Device Controllers and Systems (Transmit Data)

- This layer is where industrial control logic runs.

Device Control Network (OT Control Networks)

- Industrial fieldbuses and serial protocols:
 - Modbus
 - RS-485
- Used by PLCs, RTUs, DCS controllers.

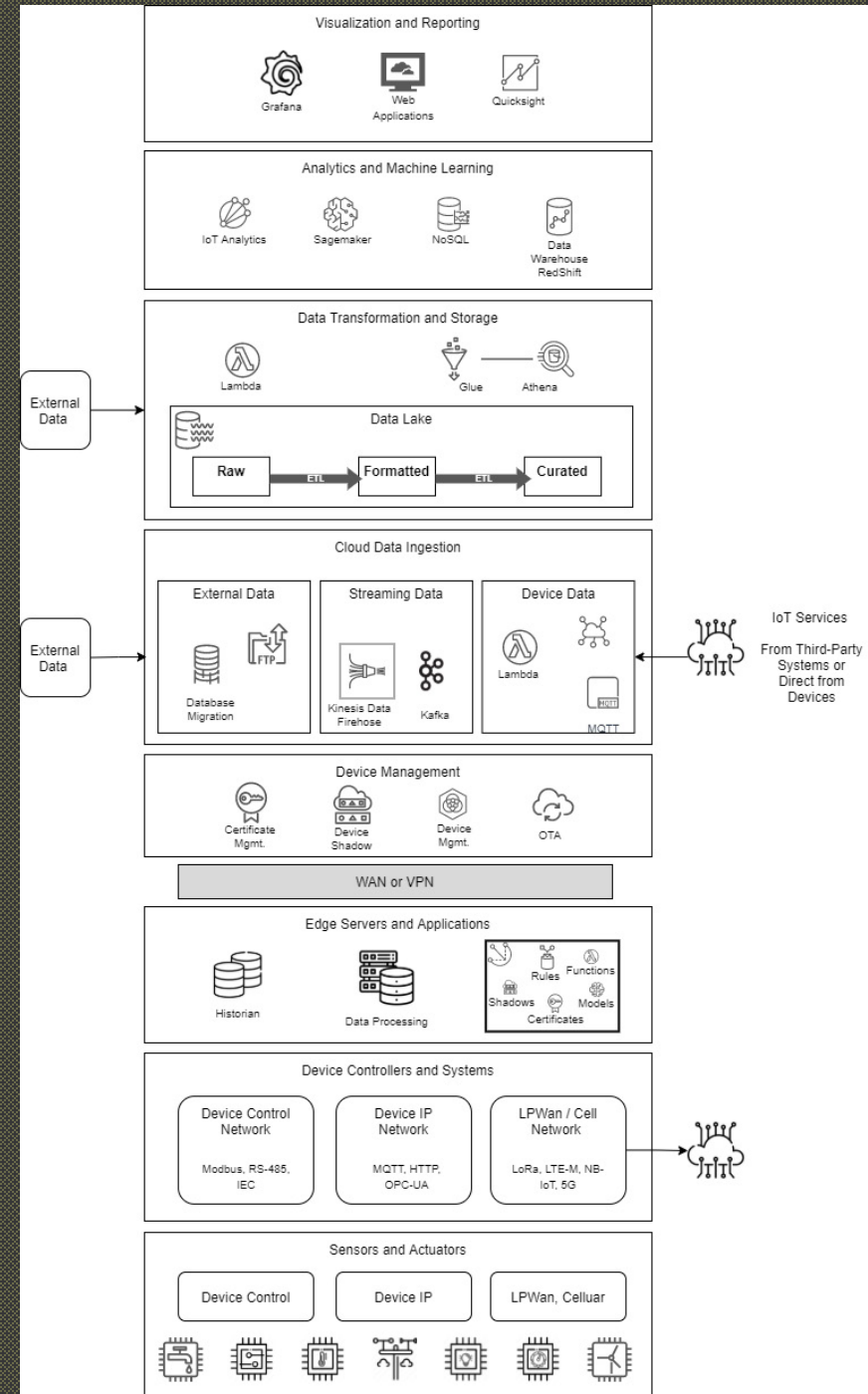
Device IP Network

- IP-based industrial device communication:
 - MQTT
 - HTTP
 - OPC-UA
- Enables device-to-cloud or device-to-edge communication.

LPWAN / Cellular Network

- Long-range communication to IoT gateways or directly to the cloud.
- Supports:
 - LoRaWAN gateways
 - NB-IoT modems
 - LTE-M modules
 - 5G IoT modules

This layer bridges raw sensors with supervisory control and communication networks.



IOT Architecture

3. Edge Servers and Applications (Preprocess Data and Buffer)

- This is the edge computing layer, placed near the factory floor.

Historian

- Local time-series database storing:
 - process values
 - alarms
 - KPIs
- Useful for local troubleshooting and buffering when the cloud is unreachable.

KPI - Key Performer Indicator

Data Processing

- Real-time data cleaning, filtering, collection.
- Running logic close to the machine to reduce latency.

Rules & Functions

- Event-driven logic (e.g., “If temperature > 80°C → trigger alarm”).

Shadows

- Digital representations - virtual copy of a physical device.
- Synchronize device state between edge and cloud.

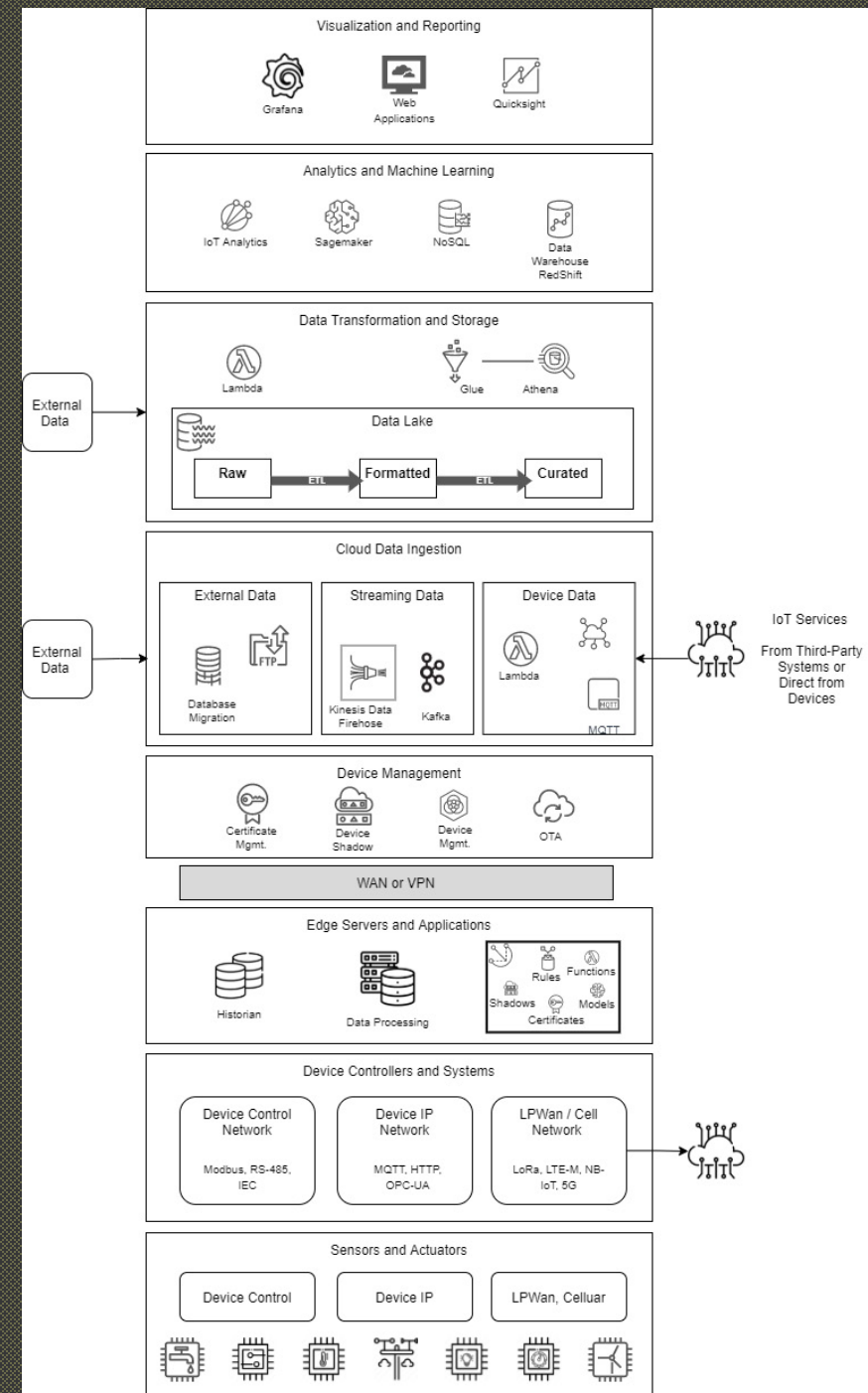
Certificates

- TLS certificates for secure communication. **TLS - Transport Layer Security**

Models

- Local ML models deployed for real-time inference.

Edge computing improves reliability, reduces bandwidth, and allows safety-critical logic to stay on-site.



IOT Architecture

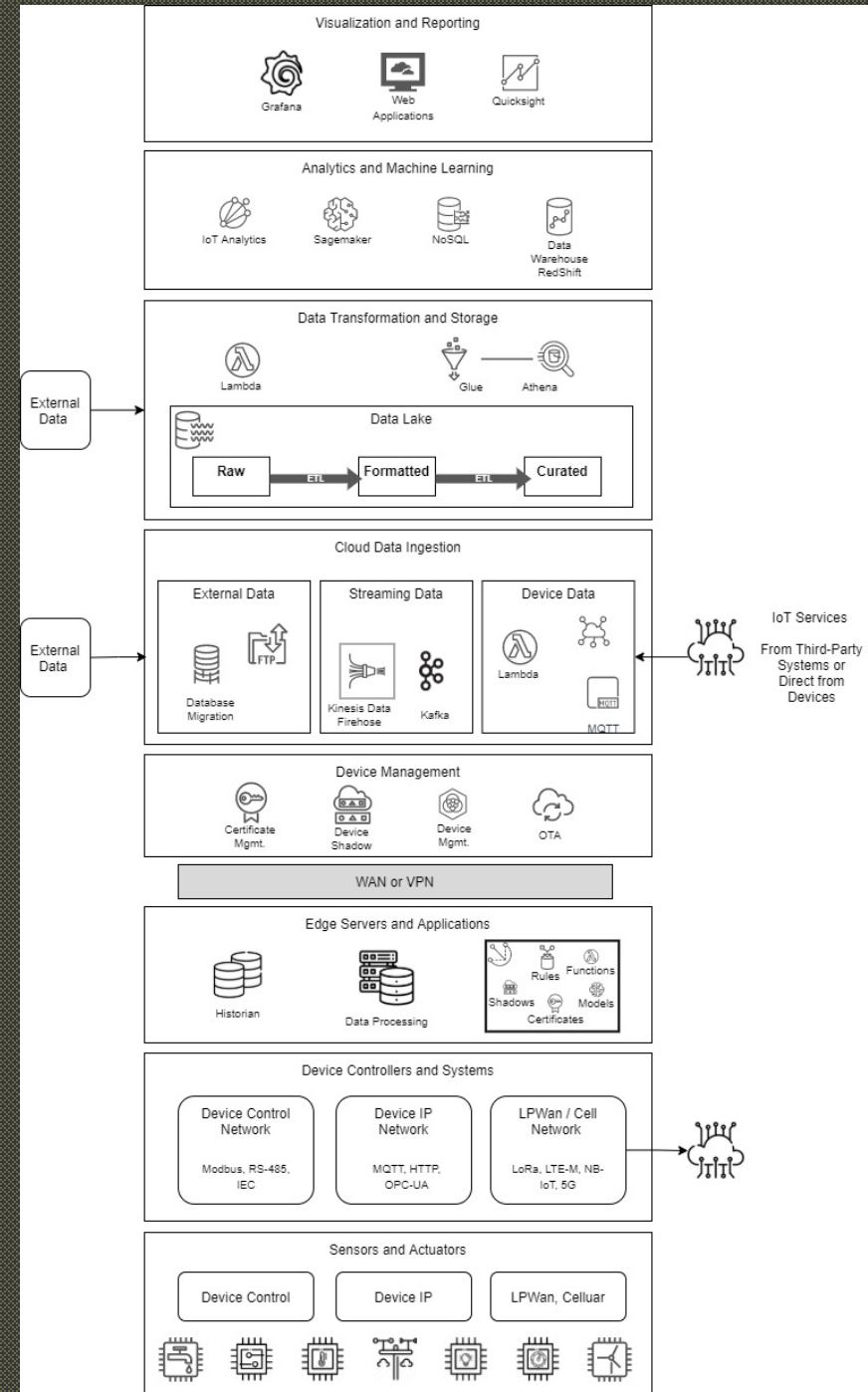
4. WAN or VPN Connectivity

Between edge and cloud, communication is secured via:

- VPN tunnels
- Private leased lines
- SD-WAN
- 5G private networks

This ensures industrial traffic is isolated, encrypted, and protected.

SD-WAN - Software-Defined Wide Area Network - a virtual network architecture that uses software to manage and optimize connections across multiple transport services (e.g., 5G LTE)



IOT Architecture

5. Device Management Layer (Secures Devices)

- A critical component for scaling IIoT.

Certificate Management

- Manages cryptographic keys to authenticate devices.
- Each device gets a unique secure identity.

Device Shadow

- Cloud-hosted virtual model of a device's state.
- A reliable digital representation of the device.

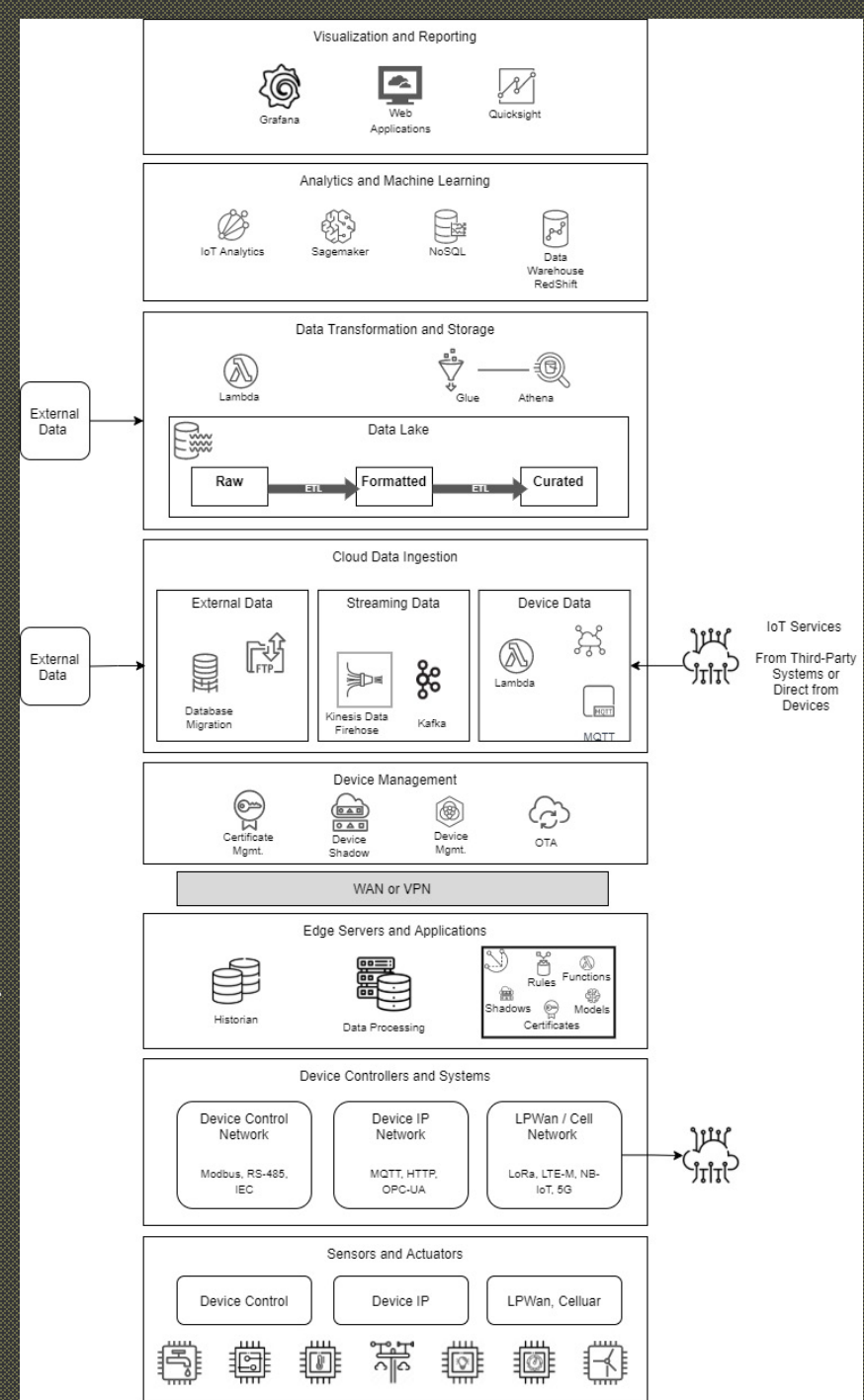
Device Management

- Provisioning - Securely onboarding new devices.
- Reset/recovery - Rebooting devices remotely.
- Status monitoring - Tracking health, connectivity, battery, etc.

OTA (Over-the-Air Updates)

- Updating firmware securely at scale.
- Adding new features.

This layer keeps thousands of field devices secure and up to date.



IOT Architecture

6. Cloud Data Ingestion Layer (Collect and Route Data)

- This is where data enters the cloud ecosystem.

External Data

- From existing databases via Database Migration Services.
- For example: ERP, MES, SCADA historian exports.

Streaming Data

- Real-time streams using services such as:
 - Kinesis Data Streams
 - Kinesis Firehose
 - Kafka
- Used for high-velocity sensor data and logs (e.g., real time data).

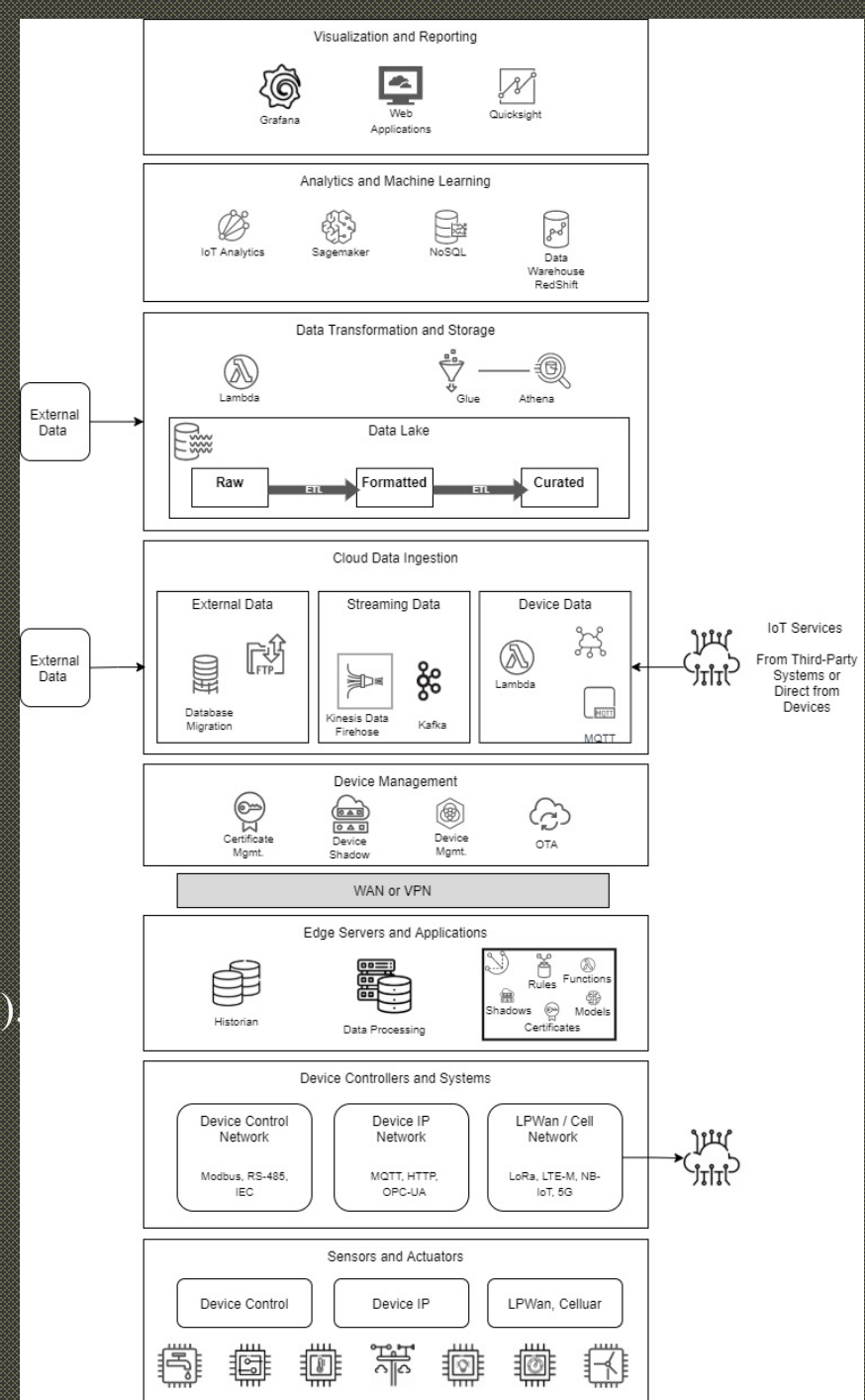
Device Data - data sent directly by field devices or edge gateways.

- Direct ingestion using:
 - Lambda - for trigger-based processing
 - MQTT brokers - for lightweight IoT messages
- Often used for IoT telemetry (temperature, vibration, energy readings).

IoT Services

- Cloud IoT Core (Azure IoT Hub, AWS IoT Core) receives MQTT traffic.
- Supports secure device connection, authentication, and routing.

This layer is the entry point for heterogeneous device data.



IOT Architecture

7. Data Transformation and Storage (Transforms Data)

- The data lake layer stores, transforms, and prepares data.

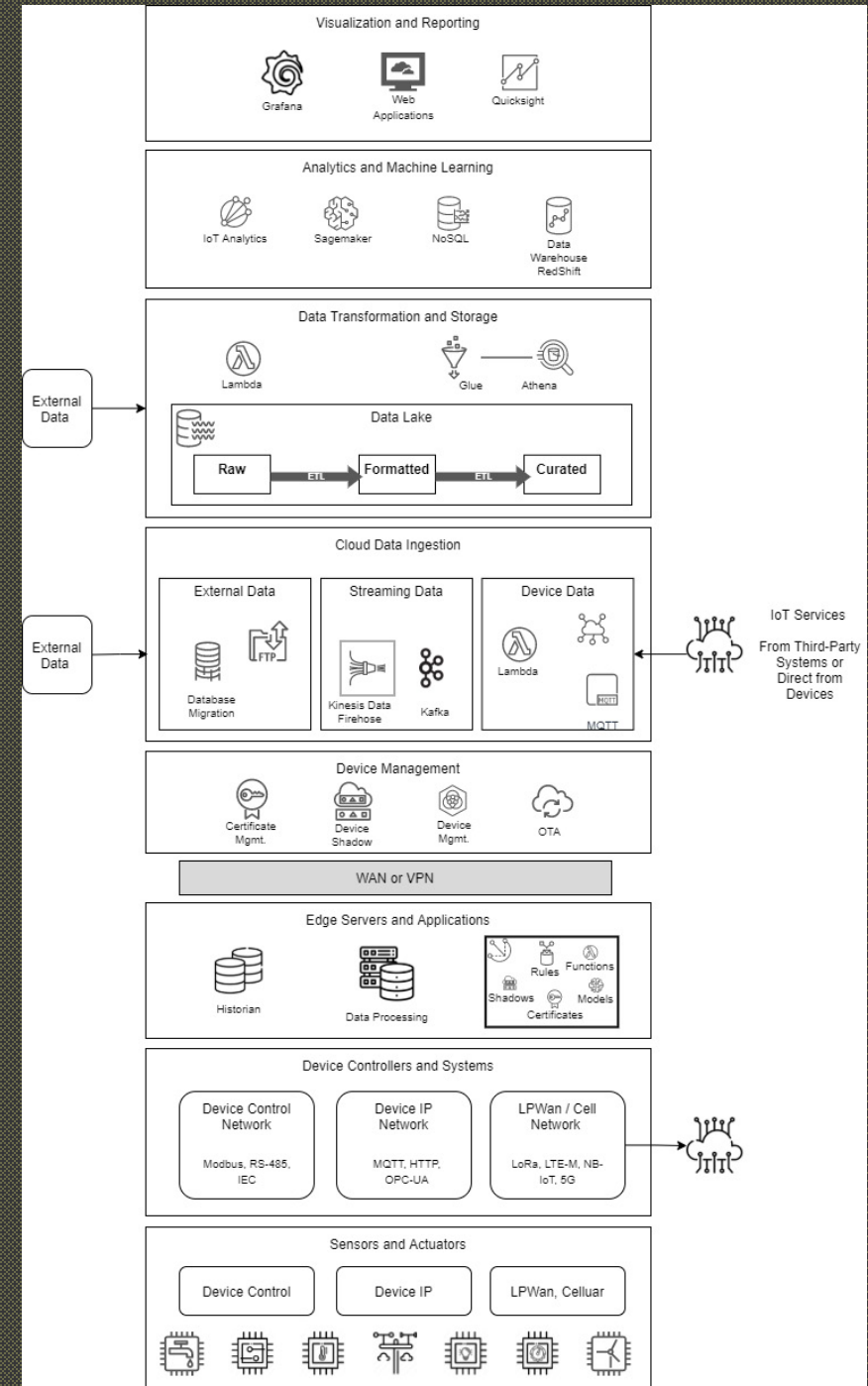
Data Lake (Raw → Formatted → Curated) - the heart of IIoT data storage.

- Raw Zone
 - Unprocessed, raw sensor data.
 - Stored exactly as ingested.
- Formatted Zone
 - Cleaned and structured data:
 - JSON normalized
 - CSV formatted **CSV - Comma Separated Values**
 - Timestamps corrected
- Curated Zone
 - Analytics-ready datasets.
 - Joined with metadata, context, and equipment models.
 - Used for dashboards and ML.

Services Supporting It

- Lambda - Event-driven processing.
- Glue - ETL pipelines, schema discovery. **ETL - Extract, Transform, Load**
- Athena - SQL queries on the data lake. **SQL - Structured Query Language**

This layer turns noisy sensor data into meaningful information.



IOT Architecture

8. Analytics and Machine Learning (Extracts Intelligence)

- Once curated, data is used for advanced analytics.

IoT Analytics

- Statistical analysis - to understand normal vs. abnormal behavior.
- Trend detection (e.g., temperature drifting upward).
- KPI generation (e.g., energy efficiency, cycle times).

SageMaker – Machine Learning Platform

- Predictive maintenance – forecasting when a machine fails.
- Anomaly detection – identifying unusual vibration patterns.
- Optimization models - Reducing energy consumption.

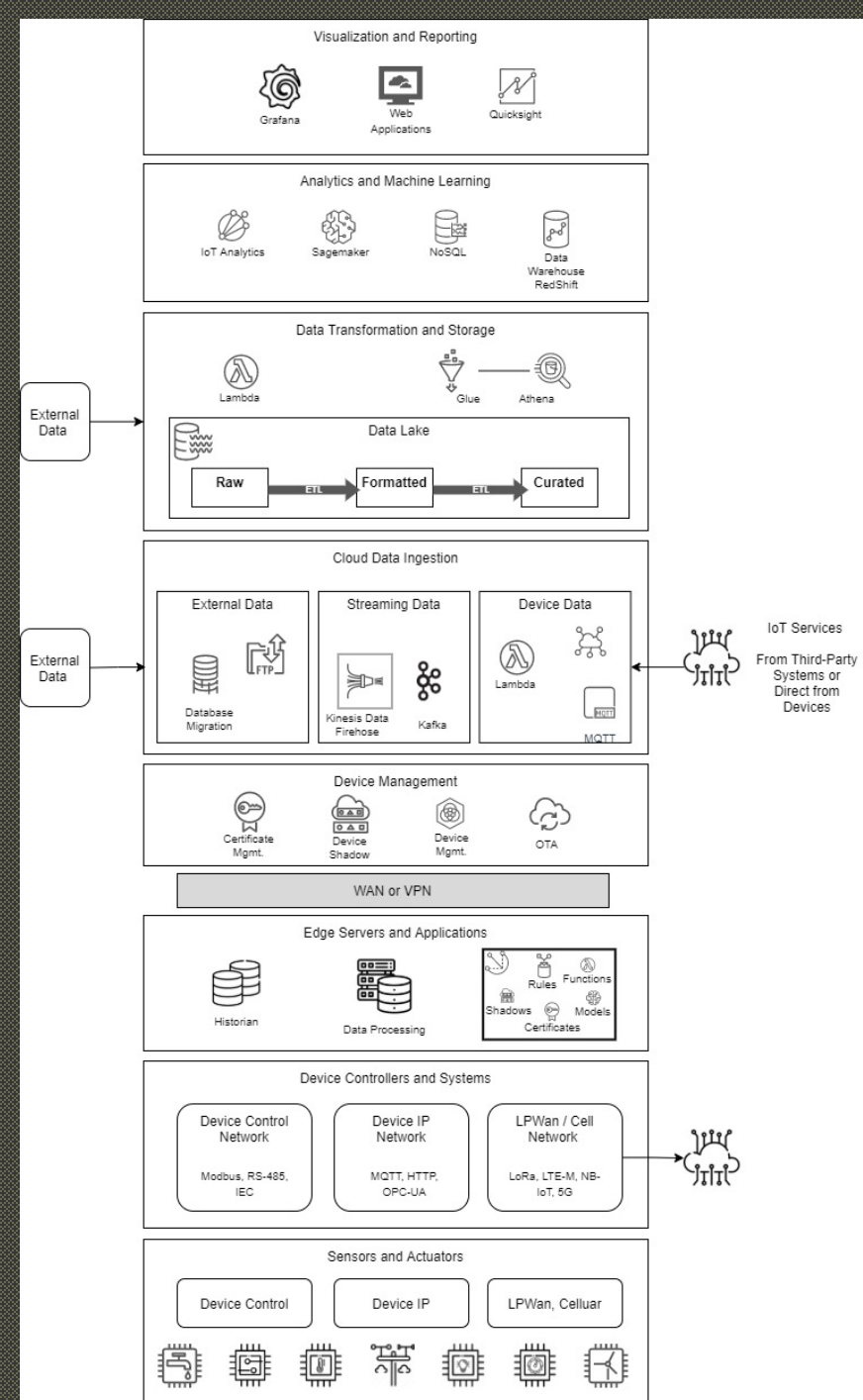
NoSQL Databases

- DynamoDB, Cassandra, MongoDB – used when data is:
 - Unstructured.
 - Needs very fast read/write

Data Warehouse (Redshift)

- Used for BI dashboards, batch analytics, historical reporting.

This layer generates insights and predictive intelligence.



IOT Architecture

9. Visualization and Reporting (Top Layer - Provides Human-Readable Insights)

- Consumed by engineers, managers, and operators.

Grafana - for live monitoring of industrial systems

- Real-time dashboard updates.
- Time-series charts for sensors and machine data.

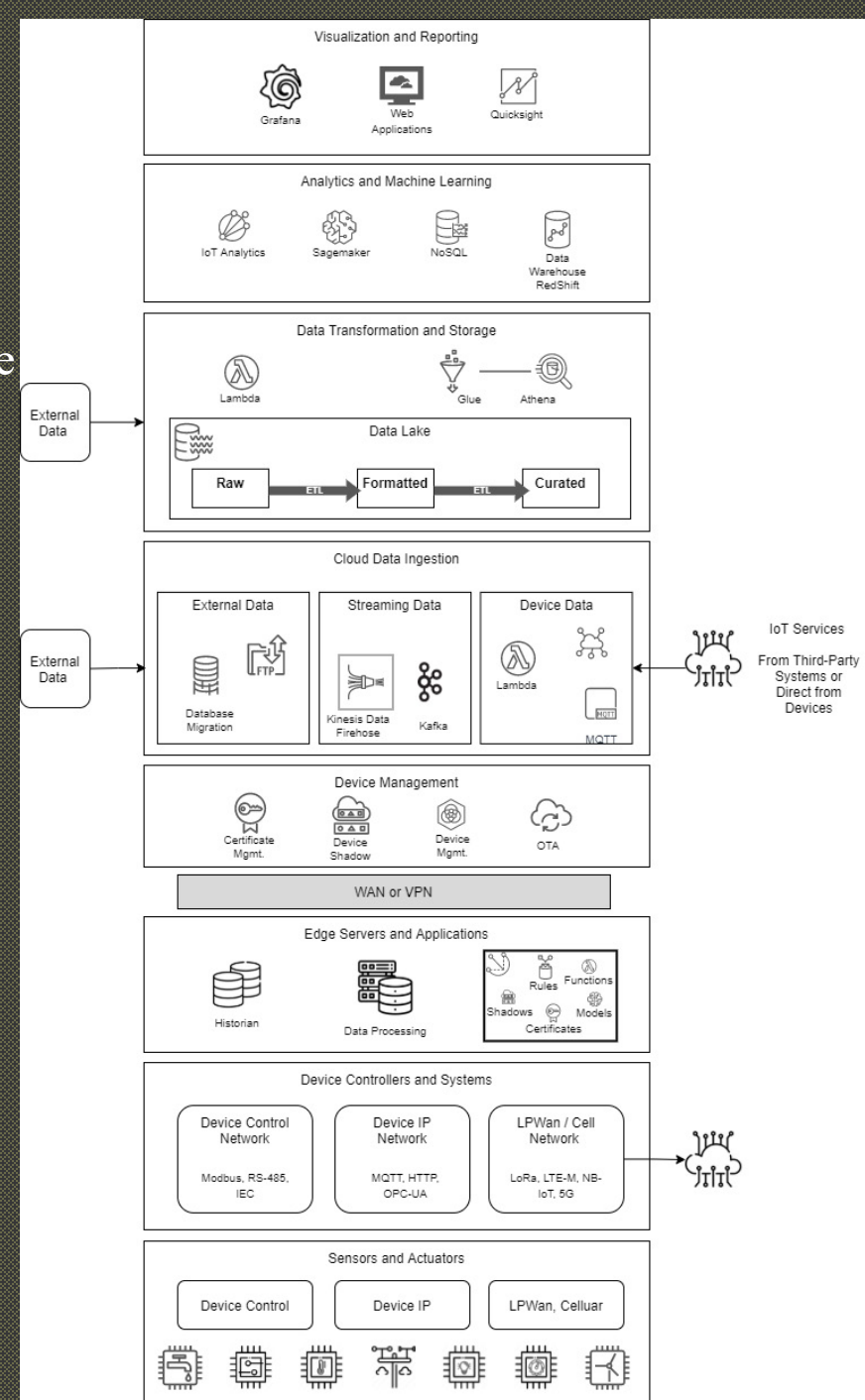
Web Applications - built for specialized industrial needs

- Custom industrial dashboards for operators.
- Cloud-hosted HMI interfaces.

QuickSight - used for higher-level business reporting, ideal for managers and executives

- KPI dashboards (energy cost, scrap rate)
- Monthly, quarterly, or yearly performance summaries
- Interactive business intelligence reports

This layer turns industrial data into actionable decision support.



REFERENCES

Industrial IoT for Architects and Engineers - 1st edition Joey Bernal, Bharath Sridhar

INDUSTRIAL NETWORKING

MENG 4010

Thank you!

Discussions?